CS1231S · Condensed Revision

Lectures 1–13, Tutorials 1–11, the AY23–25 midterms and the AY25/26 final. Definitions, theorems and worked examples are integrated by topic; Appendix A is complete. Theorem and appendix labels follow the lecture slides. Parts I–V (§1–25) are the midterm material; Parts VI–X (§26–52) cover functions, induction, cardinality, counting and probability, graphs and trees.

Find material: topic locator §63; theorem locator §65; question locators §57–58; formula sheet §56. **Final exam format (AY25/26):** 2 hours, open book, 100 marks: 23 MCQ (46) + 3 structured questions (54).

Conventions: $\mathbb{N} = \{0, 1, 2, ...\}$; $\overline{A}$ is complement; $A \setminus B$ is difference; use $\subseteq$ for subset. Largest/smallest = greatest/least. Blue boxes define terms; grey boxes contain proofs and worked examples; green boxes state results; amber boxes solve past papers.

Part I · Propositional Logic

1 · Statements, connectives, truth tables

DEFINITION Statement (2.1.1) · Statement form (2.1.5)

A **statement** (proposition) is a sentence that is true or false, but not both. A **statement form** is an expression built from statement variables and connectives that becomes a statement when statements are substituted for its variables.

Not statements (L2 s8): questions, commands, exclamations and open sentences ("x solved the question" has no truth value until x is fixed; see predicates, §5).

DEFINITION Negation (2.1.2) · Conjunction (2.1.3) · Disjunction (2.1.4)

$\sim p$: "not p"; $p \wedge q$: "p and q"; $p \vee q$: inclusive "p or q", true when at least one is true.

p	q	$\sim p$	$p \wedge q$	$p \vee q$	$p \to q$	$p \leftrightarrow q$	$p \oplus q$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	T
F	T	T	F	T	T	F	T
F	F	T	F	F	T	T	F

Order of operations. $\sim$ first; then $\wedge$ and $\vee$, which are **coequal**, so $p \wedge q \vee r$ is ambiguous and must be bracketed; then $\to, \leftrightarrow$ last. (CS2100 puts $\wedge$ before $\vee$; CS1231S does not.)

Exclusive or: $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q)$.

A $\to$ is false in exactly one row: **T** $\to$ **F**.

DEFINITION Tautology, contradiction (2.1.7–8) · Logical equivalence (2.1.6)

A **tautology** is a statement form that is true for every assignment; a **contradiction** is false for every assignment. A statement whose form is a tautology (contradiction) is a **tautological** (**contradictory**) statement. $P \equiv Q$ iff P and Q have identical truth values under every assignment; equivalently, iff $P \leftrightarrow Q$ is a tautology.

TRAP $\equiv$ is not a connective (L2 s53)

$P \leftrightarrow Q$ is a **statement form** with a truth value in each row. $P \equiv Q$ is an **assertion about** two statement forms: it has no truth table and cannot be nested inside a formula. The link is: $P \equiv Q$ iff $P \leftrightarrow Q$ is a tautology. Use $\equiv$ between the lines of a derivation and $\leftrightarrow$ inside a formula.

2 · Logical equivalences and set identities

Theorem 2.1.1 (logic) and Theorem 6.2.2 (sets) side by side (Lecture 2, slides 25–26; Lecture 5, slides 42–43). Read $\wedge$ as $\cap$, $\vee$ as $\cup$, $\sim$ as complement, **true** as U, **false** as $\varnothing$. Cite laws **by name**. **true** is any tautology and **false** any contradiction (Epp writes $\boldsymbol{t}$, $\boldsymbol{c}$).

Law	Logic (for statement variables)	Sets (subsets of U)
Commutative	$p \wedge q \equiv q \wedge p \quad p \vee q \equiv q \vee p$	$A \cap B = B \cap A \quad A \cup B = B \cup A$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(A \cap B) \cap C = A \cap (B \cap C)$
	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(A \cup B) \cup C = A \cup (B \cup C)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity	$p \wedge \mathbf{true} \equiv p \quad p \vee \mathbf{false} \equiv p$	$A \cap U = A \quad A \cup \varnothing = A$

Negation / Complement	$p \vee \sim p \equiv \mathbf{true} \quad p \wedge \sim p \equiv \mathbf{false}$	$A \cup \overline{A} = U \quad A \cap \overline{A} = \varnothing$
Double negation / complement	$\sim(\sim p) \equiv p$	$\overline{\overline{A}} = A$
Idempotent	$p \wedge p \equiv p \quad p \vee p \equiv p$	$A \cap A = A \quad A \cup A = A$
Universal bound	$p \vee \mathbf{true} \equiv \mathbf{true} \quad p \wedge \mathbf{false} \equiv \mathbf{false}$	$A \cup U = U \quad A \cap \varnothing = \varnothing$
De Morgan's	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\overline{A \cap B} = \overline{A} \cup \overline{B}$
	$\sim(p \vee q) \equiv \sim p \wedge \sim q$	$\overline{A \cup B} = \overline{A} \cap \overline{B}$
Absorption	$p \vee (p \wedge q) \equiv p \quad p \wedge (p \vee q) \equiv p$	$A \cup (A \cap B) = A \quad A \cap (A \cup B) = A$
Negations of **true**, **false**	$\sim \mathbf{true} \equiv \mathbf{false} \quad \sim \mathbf{false} \equiv \mathbf{true}$	$\overline{U} = \varnothing \quad \overline{\varnothing} = U$
Set difference	—	$A \setminus B = A \cap \overline{B}$

THEOREM **Further equivalences**

- **Implication law:** $p \to q \equiv \sim p \vee q$
- **Contrapositive:** $p \to q \equiv \sim q \to \sim p$
- **Negated conditional:** $\sim(p \to q) \equiv p \wedge \sim q$
- **Biconditional:** $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
- **Variant absorption:** $p \wedge (\sim p \vee q) \equiv p \wedge q$
- **Variant absorption:** $p \vee (\sim p \wedge q) \equiv p \vee q$
- **Exportation:** $(p \wedge q) \to r \equiv p \to (q \to r)$
- **Cases:** $(p \vee q) \to r \equiv (p \to r) \wedge (q \to r)$
- $p \to (q \vee r) \equiv (p \wedge \sim q) \to r$

In ∧/∨ form (L2 s15, s85): $p \leftrightarrow q \equiv (\sim p \vee q) \wedge (p \vee \sim q) \equiv (p \wedge q) \vee (\sim p \wedge \sim q)$; $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q) \equiv (p \vee q) \wedge (\sim p \vee \sim q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q) \equiv \sim(p \leftrightarrow q)$.

The left column is proved in Lecture 2. The right column comes from tutorials and assignments; each follows from the table in a line or two, so derive it if a question says "use the laws" (the AY24 answer key cites variant absorption as an assignment result).

PAST PAPER **AY24 Q4 · Simplify** $(((p \to q) \to (q \to r)) \to (r \to s)) \vee p) \wedge (p \to (p \to q)) \wedge p$

Call the long first conjunct $X \vee p$. Answer: $\boldsymbol{p \wedge q}$ (option C).

$(X \vee p) \wedge (p \to (p \to q)) \wedge p$
$\equiv (X \vee p) \wedge ((p \to (p \to q)) \wedge p)$ *(associative law)*
$\equiv (X \vee p) \wedge (p \wedge (p \to (p \to q)))$ *(commutative law)*
$\equiv ((X \vee p) \wedge p) \wedge (p \to (p \to q))$ *(associative law)*
$\equiv (p \wedge (p \vee X)) \wedge (p \to (p \to q))$ *(commutative law ×2)*
$\equiv p \wedge (p \to (p \to q))$ *(absorption law)*
$\equiv p \wedge (\sim p \vee (\sim p \vee q))$ *(implication law ×2)*
$\equiv p \wedge ((\sim p \vee \sim p) \vee q)$ *(associative law)*
$\equiv p \wedge (\sim p \vee q)$ *(idempotent law)*
$\equiv p \wedge q$ *(variant absorption law)*

TRAP **One law per line (L2 s27–28, T1 Q3)**

Cite a law on **every** line and never put two different laws on one line; a skipped double negative or commutative step loses credit. The **same** law may be applied several times in one step, written "×2" (L2 s28; AY24 Q4 key).

Laws match by exact shape: negation is $p \wedge \sim p$, not $\sim p \wedge p$; absorption is $p \vee (p \wedge q)$, not $p \vee (q \wedge p)$. Insert the commutative line first.

T1 Q3(a): $\sim a \wedge (\sim a \to (b \wedge a))$ takes six lines: implication, double negative, commutative ($b \wedge a$ to $a \wedge b$), absorption (leaving $\sim a \wedge a$), commutative ($a \wedge \sim a$), negation, giving **false**. Skipping straight from the implication law to absorption is not full credit.

Notation: write ≡ between lines, never =. The course writes **true** and **false** for Epp's $\boldsymbol{t}$ and $\boldsymbol{c}$, which look like statement variables (L2 s24). Bracket every mixed ∧/∨: $a \wedge \sim b \vee \sim c$ is ambiguous (T1 Q2).

METHOD **Checking tautologies and equivalences**

To test whether $\boldsymbol{P \to Q}$ is a tautology, try to make it false: set Q false and P true, and follow the consequences. If this forces a contradiction, $P \to Q$ is a tautology. **AY24 Q3(iii)**

$(p \vee q \vee \sim r) \wedge (s \vee \sim(p \vee q)) \to (s \vee \sim r)$: falsifying needs $s = F, r = T$; the premise becomes $(p \vee q) \wedge \sim(p \vee q)$, a contradiction, so the statement is a tautology. **To show $\boldsymbol{P \not\equiv Q}$,** one row is enough. AY23 Q3(i): $\sim(p \to q) \equiv p \vee (p \to \sim q)$? At $p = q = T$ the left side is F and the right side is T, so they are not equivalent.

WORKED EXAMPLE **Logic laws**

Without a truth table, show $(p \to q) \wedge (p \to \sim q) \equiv \sim p$.

$(p \to q) \wedge (p \to \sim q)$
$\equiv (\sim p \vee q) \wedge (\sim p \vee \sim q)$ *(implication law ×2)*

$\equiv \sim p \vee (q \wedge \sim q)$ *(distributive law)*
$\equiv \sim p \vee \mathbf{false}$ *(negation law)*
$\equiv \sim p$ *(identity law)*. ■

3 · Conditional statements

DEFINITION Conditional (2.2.1) and its relatives (2.2.2–4)
$p \to q$ ("if p then q", "p implies q") is false only when p is true and q is false; when p is false it is **vacuously true**. p is the **hypothesis** (antecedent) and q the **conclusion** (consequent). From $p \to q$ form the **contrapositive** $\sim q \to \sim p$, the **converse** $q \to p$, and the **inverse** $\sim p \to \sim q$.

THEOREM
A conditional ≡ its contrapositive. Its converse ≡ its inverse (they are contrapositives of each other). A conditional is **not** equivalent to its converse or its inverse.

DEFINITION Only if (2.2.5) · Biconditional (2.2.6) · Necessary/sufficient (2.2.7)
The table below translates each phrase. "p iff q" is true exactly when their truth values agree. A necessary condition must hold; a sufficient condition guarantees the conclusion.

English	Symbolic	English	Symbolic
if p then q; p implies q	$p \to q$	p only if q	$p \to q$
q if p; q whenever p	$p \to q$	p unless q	$\sim q \to p \equiv p \vee q$
p is sufficient for q	$p \to q$	p is necessary for q	$q \to p$
p if and only if q	$p \leftrightarrow q$	neither p nor q	$\sim p \wedge \sim q$

TRAP Everyday "if" is often an "iff"
"If you behave, you will get ice-cream" is usually understood as a biconditional. In mathematics it is only $b \to i$; misbehaving and still getting ice-cream does not make it false.

4 · Arguments and rules of inference

DEFINITION Argument, validity (2.3.1) · Critical row · Syllogism · Sound, unsound (2.3.2)
An **argument** (**argument form**) is a sequence of statements (statement forms); all but the last are **premises** (assumptions, hypotheses), the last (after ∴) is the **conclusion**. An argument form is **valid** iff, whatever statements are substituted for its variables, if the premises are all true then the conclusion is true. A **critical row** is a truth-table row in which all premises are T: the form is valid iff the conclusion is T in every critical row. A **syllogism** is an argument form with two premises and a conclusion (e.g. modus ponens). An argument is **sound** iff it is valid **and** all its premises are true; otherwise it is **unsound**.

THEOREM Validity as a tautology (T1 notes)
The argument $p_1, p_2, \ldots, p_k \therefore q$ is valid iff $(p_1 \wedge p_2 \wedge \cdots \wedge p_k) \to q$ is a **tautology**. So the falsifying test of §2 also tests validity: try to make every premise T and q F. Transitivity is valid because $((p \to q) \wedge (q \to r)) \to (p \to r)$ is a tautology (T1 Q6).
Critical rows by deduction (T1 Q8): do not draw the table; solve for the rows in which all premises are T.
- $p \vee (q \wedge r)$, $\sim p \therefore q \wedge r$: the premises force $p = F$, then $q = r = T$. One critical row, conclusion T: **valid**.
- $p \vee (q \wedge r)$, $\sim(p \wedge q) \therefore r$: $p = T$, $q = r = F$ makes both premises T and r F: **invalid**.
- $p \to (q \vee r)$, $q \wedge \sim r \therefore p$: $p = F$, $q = T$, $r = F$ is a critical row with conclusion F: **invalid**.
- $p \vee q$, $(p \wedge q) \to r$, $q \wedge \sim r \therefore \sim p$: premise 3 forces $q = T$, $r = F$; premise 2 then forces $p = F$. Conclusion T: **valid**.

Rule	Form	Rule	Form
Modus ponens	$p \to q, p \therefore q$	Generalization	$p \therefore p \vee q$
Modus tollens	$p \to q, \sim q \therefore \sim p$	Specialization	$p \wedge q \therefore p$
Conjunction	$p, q \therefore p \wedge q$	Elimination	$p \vee q, \sim q \therefore p$
Transitivity	$p \to q, q \to r \therefore p \to r$	Contradiction rule	$\sim p \to \mathbf{false} \therefore p$

Division into cases	$p \vee q,\ p \to r,\ q \to r \therefore r$

Fallacies (invalid): *converse error* $p \to q, q \therefore p$; *inverse error* $p \to q, \sim p \therefore \sim q$.

TRAP **Fallacies (L2 s75–78, T1 Q7)**
A **fallacy** is an error in reasoning that gives an invalid argument. Three common ones: ambiguous premises treated as unambiguous; **circular reasoning** (assuming what is to be proved); **jumping to a conclusion**. The converse error is also called **affirming the consequent**.
Validity is about form, not about the truth of the conclusion. "If at least one of two numbers is divisible by 6, so is their product; neither is divisible by 6; $\therefore$ the product is not" is $(p \vee q) \to r,\ \sim p \wedge \sim q \therefore \sim r$: an inverse error (2 and 3), and the form stays invalid even in instances with no counterexample. Conversely a valid argument with a false premise may have a false conclusion (L2 s78): valid but unsound.
Answer format (T1 Q7): define the variables, write the form, then name the rule ("valid by specialization") or the error ("invalid: converse error").

PAST PAPER **AY24 Q5 · Which conclusions follow from $p \to (q \wedge r)$, $\sim p \to \sim r$, $\sim p \wedge (r \to \sim p)$?**
1. $\sim p$ *(specialization, premise 3)* then $\sim r$ *(modus ponens with premise 2)*.
2. So every critical row has $p = r = F$ (q free). Then (i) $p \to q$ is T (vacuous); (ii) $\sim p \vee \sim r$ is T; (iii) $q \to (p \to r)$ is T since $p \to r$ is T; (iv) $\sim p \wedge \sim q \wedge r$ is F since $r = F$.

Valid: **(i), (ii), (iii)**. Answer C.

PAST PAPER **AY25 Q13 · Sound vs valid**
Sound $\Rightarrow$ valid by definition, so "$\forall x(Sound(x) \to Valid(x))$" is **true**; the converse fails (valid with a false premise), so "$\exists x(\sim Sound(x) \wedge Valid(x))$" is **true**; the other two are false. Answer: **only I and IV**.

PAST PAPER **AY23 Q4 · Knights and knaves**
Method: let A, D, N mean "is a knight". A statement S made by X gives $X \leftrightarrow S$. Here: $A \leftrightarrow D$; $D \leftrightarrow \sim N$; "Aiken is not my type" gives $N \leftrightarrow \sim(A \leftrightarrow N)$; "Aiken lies" gives $N \leftrightarrow \sim A$. From $N \leftrightarrow \sim A$, A and N differ, so $\sim(A \leftrightarrow N)$ is T, hence N is T. Then $A = F$, $D = A = F$, and $D \leftrightarrow \sim N$ checks out. **Aiken and Dueet are knaves; Noasu is a knight.** Answer B.

WORKED EXAMPLE **T1 Q9 · Knights and knaves as a written proof**
A says "Both of us are knights"; B says "A is a knave". Every native is a knight (always truthful) or a knave (always lies), not both.
1. Suppose A is a knight.
   1.1. What A says is true *(definition of knight)*, so B is a knight too *(that is what A says)*.
   1.2. So what B says is true *(definition of knight)*, so A is a knave *(that is what B says)*.
   1.3. So A is not a knight *(a native is a knight or a knave, but not both)*, contradicting line 1.
2. Hence A is not a knight, so A is a knave *(knight or knave)*.
3. So what B says is true; B cannot be a knave *(B said something true)*, so B is a knight. ■

A is a knave and B is a knight. "Knave" is not the negation of "knight": a contradiction needs $p \wedge \sim p$, so the "not both" line must come before "contradiction".
(b) C: "D is a knave". D: "C is a knave". By cases: if C is a knight, what C says is true, so D is a knave. If C is not a knight, C is a knave, what C says is false, so D is not a knave, so D is a knight. In both cases there is **one knight and one knave**.

PAST PAPER **AY25 Q16 (6 marks) · Four animals, four colours, nine clues**
Names: cat (3 letters), sheep (5), rabbit (6), penguin (7).
1. (2) six letters $\leftrightarrow$ black: **rabbit** is black; cat, sheep, penguin are not.
2. (3) an "e" $\to$ white: **sheep** and **penguin** are white.
3. (4) a repeated letter $\to$ not grey: sheep (ee), rabbit (bb), penguin (nn) are not grey. (5) brown $\to$ has an "i": cat and sheep are not brown.
4. (7) sheep has exactly one colour: **sheep = {white}**. (9) someone is grey, and only the cat can be: **cat = {grey}** (exactly one, by (7)).
5. (6) penguin has exactly two colours: white, and not black or grey, so **penguin = {white, brown}**.
6. (8) some animal is exactly {brown, black}; only the rabbit is black: **rabbit = {brown, black}**. (1) holds.

Model derivations

PROOF **Simplify $\sim(\sim p \wedge q) \wedge (p \vee q)$ using the laws (L2 s27)**
$\sim(\sim p \wedge q) \wedge (p \vee q) \equiv (\sim(\sim p) \vee \sim q) \wedge (p \vee q)$ *(De Morgan's law)*
$\equiv (p \vee \sim q) \wedge (p \vee q)$ *(double negative law)*

$\equiv p \vee (\sim q \wedge q)$ *(distributive law)*
$\equiv p \vee (q \wedge \sim q)$ *(commutative law)*
$\equiv p \vee \mathbf{false}$ *(negation law)*
$\equiv p$ ■ *(identity law)*

PROOF **Derive variant absorption $p \wedge (\sim p \vee q) \equiv p \wedge q$ from the table**
$p \wedge (\sim p \vee q)$
$\equiv (p \wedge \sim p) \vee (p \wedge q)$ *(distributive)*
$\equiv \mathbf{false} \vee (p \wedge q)$ *(negation)*
$\equiv (p \wedge q) \vee \mathbf{false}$ *(commutative)*
$\equiv p \wedge q$ *(identity).* ■

PROOF **A deduction with named rules: from $p \vee q$, $q \to r$, $(p \wedge s) \to t$, $\sim r$, $\sim q \to (u \wedge s)$, conclude t**
1. $\sim q$ *(from $q \to r$ and $\sim r$, by modus tollens)*
2. $u \wedge s$ *(from $\sim q \to (u \wedge s)$ and step 1, by modus ponens)* so s *(specialization)*
3. p *(from $p \vee q$ and step 1, by elimination)*
4. $p \wedge s$ *(steps 2, 3, by conjunction)* so t *(from $(p \wedge s) \to t$, by modus ponens).* ■

p	q	r	$p \to q$	$q \to r$	$p \to r$
T	T	T	**T**	**T**	**T**
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	**T**	**T**	**T**
F	T	F	T	F	T
F	F	T	**T**	**T**	**T**
F	F	F	**T**	**T**	**T**

METHOD **Validity by critical rows**
For $p \to q, q \to r \therefore p \to r$ (transitivity): the **critical rows** are those where every premise is T: rows 1, 5, 7 and 8 (bold). The conclusion is T in each, so the argument is **valid**. A critical row with a false conclusion would make the argument invalid; that row is a counterexample.

TRAP **$\to$ is not associative**
$(p \to q) \to r \not\equiv p \to (q \to r)$: at $p = q = r = F$ the left is $T \to F = F$, the right is T. Always bracket chains of $\to$.

WORKED EXAMPLE **Validity**
Decide validity: (a) $p \to q$, $q \to r$, $\sim r \therefore \sim p$. (b) $p \to q$, $\sim p \therefore \sim q$.
(a) **Valid:** $p \to r$ *(transitivity)* with $\sim r$, $\sim p$ *(modus tollens)*.
(b) **Invalid** (inverse error): the critical row $p = F$, $q = T$ makes both premises true and the conclusion $\sim q$ false.

WORKED EXAMPLE **Which premise is missing? (L2 s86, past midterm)**
$p \to r$, $q \to s$, (missing premises) $\therefore (p \vee q) \to (r \wedge s)$. Candidates: (I) $p \vee q$; (II) $(p \wedge q) \to (r \wedge s)$; (III) $p \to s$; (IV) $q \to r$.
Method: list the rows where the given premises are T and the conclusion is F; the added premises must be F in every such row.
Row $p = T, q = F, r = T, s = F$: only (III) is F there. Row $p = F, q = T, r = F, s = T$: only (IV) is F there. So **both (III) and (IV)** are needed, and they suffice: $p \to (r \wedge s)$ and $q \to (r \wedge s)$ give $(p \vee q) \to (r \wedge s)$ by the cases equivalence (§2). (I) and (II) are T in both rows, so they never help; (II) already follows from the two given premises.

Part II · Quantified Statements

5 · Predicates and quantifiers

DEFINITION **Predicate (3.1.1) · Truth set (3.1.2)**
A **predicate** is a sentence with finitely many variables that becomes a statement when values are substituted. The **domain** of a variable is the set of values that may be substituted for it (also called the domain of discourse, universe of discourse, universal set, or universe). The **truth set** of $P(x)$, $x \in D$, is the set of all elements of D that make $P(x)$ true: $\{x \in D : P(x)\}$ (also written $\{x \in D \mid P(x)\}$).

DEFINITION **Kinds of mathematical statements (Lecture 1)**
Universal: "all/every/any" ($\forall$); **existential:** "some/at least one" ($\exists$); **conditional:** "if … then" ($\to$).
Combined forms:
- **Universal conditional:** $\forall x\,(P(x) \to Q(x))$.
- **Universal existential:** $\forall x\,\exists y\,P(x,y)$.
- **Existential universal:** $\exists x\,\forall y\,P(x,y)$.

DEFINITION **Universal (3.1.3) and existential (3.1.4) statements · Uniqueness**
$\forall x \in D, Q(x)$ is true iff $Q(x)$ is true for **every** $x \in D$, and false iff $Q(x)$ is false for at least one x, called a **counterexample**.
$\exists x \in D$ such that $Q(x)$ is true iff $Q(x)$ is true for **at least one** $x \in D$, and false iff $Q(x)$ is false for all $x \in D$.
$\exists! x \in D, Q(x) \quad \equiv \quad \exists x \in D\,(Q(x) \wedge \forall y \in D\,(Q(y) \to y = x))$, read "there is exactly one".

METHOD **Translating English**
Use $\to$ under $\forall$ and $\wedge$ under $\exists$. "All A are B" is $\forall x(A(x) \to B(x))$; "some A is B" is $\exists x(A(x) \wedge B(x))$. Writing $\forall x(A(x) \wedge B(x))$ claims **everything** is an A; writing $\exists x(A(x) \to B(x))$ is true as soon as one non-A exists.
Implicit quantification: "If $x > 2$ then $x^2 > 4$" in a real-number context means $\forall x \in \mathbb{R}(x > 2 \to x^2 > 4)$.

DEFINITION **Bound and free variables · Scope (T2 notes; L3 s35)**
A variable is **bound** when a quantifier is applied to it and **free** otherwise: in $\forall x\,\exists y\,P(x,y)$ both are bound; in $\forall x\,P(x,y)$, y is free. The **scope** of a quantifier is the part of the formula it governs: the bracketed formula after it or, with no brackets, the **smallest** formula that follows. In $\forall x\,(\exists y\,P(x,y) \vee Q(x,y))$ the scope of $\exists y$ is only $P(x,y)$, so the y in $Q(x,y)$ is free; write $\forall x\,\exists y\,(P(x,y) \vee Q(x,y))$ to bind both.
Restricted quantifiers: $\forall x \in D\,P(x) \equiv \forall x\,(x \in D \to P(x))$ and $\exists x \in D\,P(x) \equiv \exists x\,(x \in D \wedge P(x))$. The comma in "$\forall x \in D, P(x)$" is punctuation and may be omitted.
Finite domain: if $D = \{x_1, \ldots, x_n\}$, then $\forall x \in D\,Q(x) \equiv Q(x_1) \wedge \cdots \wedge Q(x_n)$ and $\exists x \in D\,Q(x) \equiv Q(x_1) \vee \cdots \vee Q(x_n)$.

TRAP **Translation mistakes (L3 s45–47, T2 notes, T2 Q9)**
- **A predicate as an argument:** $Fly(Bird(x))$ or $Loves(x, Reindeer(y))$. A predicate evaluates to true or false, so this reads $Fly(\textbf{true})$. Join predicates with a connective: $Reindeer(y) \wedge Loves(x,y)$.
- **A comma as a connective:** $\forall x\,P(x), Q(x)$ is not a formula. The comma is not $\wedge$, $\vee$ or any connective; write $\forall x\,(P(x) \wedge Q(x))$.
- **"Not all birds can fly"** is $\exists x\,(Bird(x) \wedge \sim Fly(x))$, the negation of "all birds can fly". $\forall x\,(Bird(x) \to \sim Fly(x))$ says "all birds cannot fly"; $\exists x\,(Bird(x) \to \sim Fly(x))$ is true as soon as a non-bird exists, even with no birds at all.
- **"No A is B"** (T2 Q9d): $\forall x\,(A(x) \to \sim B(x))$, equivalently $\forall x\,(\sim A(x) \vee \sim B(x))$ or $\sim\exists x\,(A(x) \wedge B(x))$.
- **One quantifier for two witnesses** (T2 Q9e): "some who took T visited E, but some did not" is $(\exists v \in V\,(T(v) \wedge E(v))) \wedge (\exists u \in V\,(T(u) \wedge \sim E(u)))$. Merging into $\exists v \in V\,((T(v) \wedge E(v)) \wedge (T(v) \wedge \sim E(v)))$ asks for one visitor with $E(v) \wedge \sim E(v)$: always false. The letter v may be reused in the second clause, since each scope ends at its own bracket.

PAST PAPER **AY24 Q2 · "All birds cannot fly"**
$\forall x(Bird(x) \to \sim Fly(x))$, option C. Option B, $\forall x(Bird(x) \wedge \sim Fly(x))$, asserts that everything is a bird; D quantifies a second, unrelated y.

6 · Negation, vacuous truth, variants

THEOREM **Theorem 3.2.1 · Theorem 3.2.2 · Negation of quantified statements**
$\sim(\forall x \in D, P(x)) \equiv \exists x \in D, \sim P(x) \qquad \sim(\exists x \in D, P(x)) \equiv \forall x \in D, \sim P(x)$
Universal conditional: $\sim\forall x \in D(P(x) \to Q(x)) \equiv \exists x \in D(P(x) \wedge \sim Q(x))$.
Nested: push $\sim$ inward, flipping each quantifier: $\sim\forall x\exists y\,P(x,y) \equiv \exists x\forall y \sim P(x,y)$.

DEFINITION **Vacuous truth**
$\forall x \in D(P(x) \to Q(x))$ is **vacuously true** when no $x \in D$ satisfies $P(x)$. In particular, every universal statement over an **empty** domain is true, and every existential statement over an empty domain is false.

DEFINITION **Variants (3.2.1) · Necessary, sufficient, only if (3.2.2)**
For $\forall x \in D(P(x) \to Q(x))$: **contrapositive** $\forall x(\sim Q(x) \to \sim P(x))$ (equivalent), **converse** $\forall x(Q(x) \to P(x))$, **inverse** $\forall x(\sim P(x) \to \sim Q(x))$.
"$r(x)$ is sufficient for $s(x)$": $\forall x(r(x) \to s(x))$. "$r(x)$ is necessary for $s(x)$": $\forall x(s(x) \to r(x))$. "$r(x)$ only if $s(x)$": $\forall x(r(x) \to s(x))$.

PAST PAPER **AY23 Q2 · Given $\forall x \in S(SoC(x) \wedge CS1231(x))$ is true**
(i) All computing students took CS1231S: **true**. (ii) Every student is a computing student: **true**, because of the $\wedge$. (iii) $S \neq \varnothing$: **false**, since the statement is vacuously true when $S = \varnothing$. (iv) Some non-computing student took it: **false**, as there is no such student. Only (i) and (ii) hold; no option lists this, so the answer is **E**.

PAST PAPER **AY25 Q2 · $\forall x \in D(Red(x) \wedge Flying(x))$ for the objects in LT11**
A: if the statement is true, every object is red. True. B: if true, every object flies. True. C: it is true when LT11 is empty. True, vacuously. D: if it is false, there is a counterexample object, so LT11 is non-empty. True. Options A–D are all correct, so the false option is **E**.

7 · Multiple quantifiers

THEOREM **Order of quantifiers**
Adjacent quantifiers **of the same kind** commute: $\forall x \forall y \equiv \forall y \forall x$ and $\exists x \exists y \equiv \exists y \exists x$. Swapping a $\forall$ past an $\exists$ usually changes the meaning: in $\forall x \exists y\, P(x, y)$ the witness y **may depend on** x; in $\exists y \forall x\, P(x, y)$ one y must work for all x. $\quad \exists y \forall x\, P \Rightarrow \forall x \exists y\, P$, never conversely in general.

METHOD **Evaluating and proving nested statements**
Read the quantifiers from left to right. A $\forall$ variable may take any value; an $\exists$ variable is chosen afterwards and may depend on the values chosen before it.
To **prove** $\forall x \exists y\, P$: "Let x be arbitrary. Let $y = f(x)$. Then $P(x, f(x))$ because …". To **disprove** it, prove the negation $\exists x \forall y \sim P$: give one x and show that every y fails for it.

PAST PAPER **AY23 Q5 · Domain $\mathbb{R}$**
(i) $\forall x \exists y(xy = 1)$ is **false**: at $x = 0$, $0 \cdot y = 0 \neq 1$ for every y.
(ii) $\exists x \forall y(x^2 \geq y^2)$ is **false**: for any x, $y = |x| + 1$ gives $y^2 > x^2$.
(iii) $\exists x \forall y \exists z(xy^2 = yz)$ is **true**: take $x = 0$; for any y choose $z = 0$. (Or $x = 1$, $z = y$.) Answer **B**.

PAST PAPER **AY23 Q16 (6 marks) · Solving for the witnesses**
(a) $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$; smallest positive integer that can lie in C with $\forall x \in A\, \forall y \in B\, \forall z \in C\, (|x - y| \leq |y - z|)$. For each y the worst x is 1, so we need $|y - z| \geq y - 1$ for $y = 4, 5, 6$: at $z = 1$, $|y - 1| = y - 1$. So $\boldsymbol{z = 1}$.
(b) $\exists y \in \mathbb{Z}^+\, \forall x \in \{10, \ldots, 50\}\, \forall z \in \{3, 5, 7, 9\}\, (x \leq y^2 \wedge y \leq z^2)$: need $y^2 \geq 50$, so $y \geq 8$, and $y \leq 3^2 = 9$. So $\boldsymbol{y \in \{8, 9\}}$.
(c) $\forall x \in \mathbb{Z}\, \exists y \in \mathbb{R} \setminus \{0\}\, (xy < 1)$: take $\boldsymbol{y = -1}$ **if** $\boldsymbol{x \geq 0}$, $\boldsymbol{y = 1}$ **if** $\boldsymbol{x < 0}$; then $xy = -x \leq 0$ or $xy = x < 0$, both < 1. (The key lists several others, e.g. $y = 1$ at $x = 0$ and $y = -x$ otherwise.)

TRAP **Hidden domain changes**
$\forall x \in \mathbb{Z}^+\, \exists y \in \mathbb{Z}(x + y = 0)$ is true but becomes false over $y \in \mathbb{Z}^+$. Check every domain again after negating or reordering quantifiers.

WORKED EXAMPLE **Nested quantifiers**
Decide and justify: (a) $\forall x \in \mathbb{Z}\, \exists y \in \mathbb{Z}\, (y > x^2)$; $\quad$ (b) $\exists y \in \mathbb{Z}\, \forall x \in \mathbb{Z}\, (y > x^2)$.
(a) **True.** Let $x \in \mathbb{Z}$ be arbitrary; take $y = x^2 + 1 \in \mathbb{Z}$ *(closure)* then $y > x^2$.
(b) **False.** Its negation $\forall y \in \mathbb{Z}\, \exists x \in \mathbb{Z}\, (y \leq x^2)$ holds: given y, take $x = y$; then $y \leq y^2$ for every integer y (if $y \leq 0$ then $y \leq 0 \leq y^2$; if $y \geq 1$ then $y^2 = y \cdot y \geq y$).

8 · Arguments with quantified statements

DEFINITION Valid argument form (3.4.1)
Valid iff every substitution of predicates that makes all premises true also makes the conclusion true.
An argument is valid iff its form is valid.

Rule	Form	Rule	Form
Universal instantiation	$\forall x \in D\, P(x);\ a \in D \therefore P(a)$	Universal generalization	$P(a)$ for an **arbitrary** $a \in D \therefore \forall x \in D\, P(x)$
Universal modus ponens	$\forall x(P(x) \to Q(x));\ P(a) \therefore Q(a)$	Universal modus tollens	$\forall x(P(x) \to Q(x));\ {\sim}Q(a) \therefore {\sim}P(a)$
Existential instantiation	$\exists x\, P(x) \therefore P(c)$ for some **new** c	Existential generalization	$P(a);\ a \in D \therefore \exists x \in D\, P(x)$

Invalid: quantified **converse error** $\forall x(P(x) \to Q(x));\ Q(a) \therefore P(a)$ and **inverse error** $\forall x(P(x) \to Q(x));\ {\sim}P(a) \therefore {\sim}Q(a)$.
Validity with diagrams: draw "all P are Q" as the P-disc inside the Q-disc; the argument is valid iff **every** consistent drawing forces the conclusion.

METHOD **Universal transitivity · reorder the premises into a chain (L3 s94–98, T2 Q10)**
$\forall x\,(P(x) \to Q(x));\ \forall x\,(Q(x) \to R(x)) \therefore \forall x\,(P(x) \to R(x))$.
Method: (1) rewrite every premise and the conclusion as "$\forall x$, if … then …"; (2) the first premise must begin with the hypothesis of the conclusion and the last must end with its conclusion; (3) where the ends do not match, replace a premise by its **contrapositive**; (4) reorder so that each conclusion is the next hypothesis.
L3 s95–98: (1) all the triangles are blue; (2) right of all the squares → above all the circles; (3) not right of all the squares → not blue; ∴ all the triangles are above all the circles. Chain: triangle → blue *(1)* → right of all the squares *(contrapositive of 3)* → above all the circles *(2)*.
T2 Q10: (1) above all the triangles → above all the blue objects; (2) not above all the gray objects → not a square; (3) black → square; (4) above all the gray objects → above all the triangles; ∴ black → above all the blue objects. Order **3, 2 (contrapositive), 4, 1**: black → square → above all gray → above all triangles → above all blue.
In symbols each link keeps its inner quantifier inside brackets, e.g. premise 2 becomes $\forall x \in O\,(Square(x) \to \forall y \in O\,(Gray(y) \to Above(x, y)))$.

PAST PAPER **AY25 Q6 (number theory in quantifier form) · n a perfect square**
(i) "n is prime" is false ($4 = 2^2$). (ii) "n has an odd number of positive factors" is true for $n \geq 1$ (divisors pair up as $d \leftrightarrow n/d$, except $d = \sqrt{n}$), but **$n = 0$ is a perfect square with infinitely many factors**, so the key accepted both C and D. (iii) "$p \mid n \Rightarrow p^2 \mid n$" is true: if $n = m^2$ and $p \mid m^2$ then $p \mid m$ (unique factorization), so $p^2 \mid m^2$.

Model solutions with nested quantifiers

PROOF **Negating the definition of a limit (Lecture 1)**
$\lim_{n\to\infty} a_n = L$ means $\forall \varepsilon \in \mathbb{R}^+\, \exists N \in \mathbb{Z}\, \forall n \in \mathbb{Z}\,(n > N \to L - \varepsilon < a_n < L + \varepsilon)$.
Its negation, one quantifier at a time: $\exists \varepsilon \in \mathbb{R}^+\, \forall N \in \mathbb{Z}\, \exists n \in \mathbb{Z}\,(n > N \wedge (a_n \leq L - \varepsilon \vee a_n \geq L + \varepsilon))$. ■

PROOF $\forall x \in \mathbb{Z}\, \exists y \in \mathbb{Z}\,(x + y = 0)$ **is true;** $\exists y \in \mathbb{Z}\, \forall x \in \mathbb{Z}\,(x + y = 0)$ **is false**
1. Let x be an arbitrary integer and let $y = -x$. Then $y \in \mathbb{Z}$ *(closure)* and $x + y = 0$.
2. For the second, prove the negation $\forall y \in \mathbb{Z}\, \exists x \in \mathbb{Z}\,(x + y \neq 0)$: given any y, take $x = 1 - y$; then $x + y = 1 \neq 0$. ■

English	Symbolic ($L(x, y)$: x loves y)
Everybody loves somebody.	$\forall x\, \exists y\, L(x, y)$
Somebody is loved by everybody.	$\exists y\, \forall x\, L(x, y)$
Nobody loves everybody.	${\sim}\exists x\, \forall y\, L(x, y) \equiv \forall x\, \exists y\, {\sim}L(x, y)$
There is exactly one person everybody loves.	$\exists! y\, \forall x\, L(x, y)$

PROOF **Formal negation of a nested quantified conditional (Lecture 3, Tarski's world)**
"There is a square x such that for all triangles y, x is to the right of y":
$\exists x\,(Square(x) \wedge \forall y\,(Triangle(y) \to RightOf(x, y)))$. Negate step by step:

$$\begin{aligned}
&\sim\exists x\,(Square(x)\wedge\forall y\,(Triangle(y)\to RightOf(x,y)))\\
&\equiv\forall x\sim(Square(x)\wedge\forall y\,(Triangle(y)\to RightOf(x,y))) && \textit{(negation of }\exists\textit{)}\\
&\equiv\forall x\,(\sim Square(x)\vee\sim\forall y\,(Triangle(y)\to RightOf(x,y))) && \textit{(De Morgan)}\\
&\equiv\forall x\,(\sim Square(x)\vee\exists y\sim(Triangle(y)\to RightOf(x,y))) && \textit{(negation of }\forall\textit{)}\\
&\equiv\forall x\,(\sim Square(x)\vee\exists y\,(Triangle(y)\wedge\sim RightOf(x,y))) && \textit{(negated conditional)}
\end{aligned}$$

"Every object is either not a square, or has some triangle it is not to the right of." ■

DRILL **Translate, then negate (answers after the arrow)**

- "Every student has a friend": $\forall s\,\exists f\,F(s,f)$ → negation $\exists s\,\forall f\sim F(s,f)$.
- "No integer is both even and odd": $\forall n\sim(E(n)\wedge O(n))$ → negation $\exists n\,(E(n)\wedge O(n))$.
- "All primes greater than 2 are odd": $\forall p\,(P(p)\wedge p>2\to O(p))$ → $\exists p\,(P(p)\wedge p>2\wedge\sim O(p))$.
- "x is a **minimal** element": $\forall y\,(y\preccurlyeq x\to y=x)$ → $\exists y\,(y\preccurlyeq x\wedge y\neq x)$.

Part III · Number Theory and Methods of Proof

9 · Writing proofs

Term	Meaning (Lecture 1)
Definition	precise meaning of a term: all and only the required properties
Axiom / postulate	assumed true without proof (Peano, Euclid's postulates)
Theorem	a major result, proved rigorously
Lemma	a small result used to prove a theorem
Corollary	a simple deduction from a theorem
Conjecture	believed true, unproved (Goldbach)

METHOD **Proof format**

- **Number the steps;** nest sub-arguments (1.1, 1.2, ...).
- **Justify every step** in brackets: "by definition of odd", "by closure of integers under +", "by basic algebra", "by Theorem 4.3.3".
- Open a universal proof with "Let x be a **arbitrary** element of D".
- Give each existential witness a **fresh** name.
- End with the claim, restated, then ■.

DEFINITION **Basic properties of integers (Lecture 1, Appendix A)**

For all $x,y,z\in\mathbb{Z}$: **closure** ($x+y, x-y, xy\in\mathbb{Z}$), **commutativity**, **associativity**, **distributivity**, **trichotomy** (exactly one of $x<y$, $x=y$, $x>y$). Appendix A's field and order axioms for $\mathbb{R}$ may also be cited; the complete Appendix A appears in §64.

Assumption 1: every integer is even or odd, but not both. **Assumption 2:** every rational number can be reduced to lowest terms.

TRAP **Common mistakes in proofs**

Arguing from examples · reusing a letter ($m=2r$ and $n=2r$ makes $m=n$) · assuming what is to be proved · "therefore" without a reason · writing "$a\mid b=3$" ($a\mid b$ is a statement, not a number).

METHOD **Without loss of generality (Lecture 1, Ex. 1)**

"WLOG" goes before an assumption that narrows the hypothesis to one special case, and asserts that the proof of that case applies just as easily to all the others. Lecture: for two consecutive odd numbers a,b, "WLOG assume $a<b$, hence $b=a+2$"; the case $b<a$ is the same proof with the names swapped. If the remaining cases need a different argument, use division into cases instead.

TRAP **T2 Q7 · Two bad "proofs" of $\forall x \in \mathbb{R}\,(x^2 \geq 0)$**

Proof by examples disguised as cases. "Cases $x < 0$, $x = 0$, $x > 0$: if $x < 0$, for example $x = -3$, then $x^2 = 9 \geq 0$; if $x = 0$ then $x^2 = 0$; if $x > 0$, say $x = 4$, then $x^2 = 16 \geq 0$." Each case checks one value, not an arbitrary x in that case. The same "logic" proves the false claim $\forall x \in \mathbb{R}\,(x^3 = x)$ using $x = -1, 0, 1$.

Wrong negation in a proof by contradiction. "Suppose $x^2 < 0$ for all real x. Let $x = 3$: $x^2 = 9 > 0$, a contradiction." The supposition must be the negation $\exists x \in \mathbb{R} \sim (x^2 \geq 0)$; the "proof" supposes $\forall x \in \mathbb{R} \sim (x^2 \geq 0)$. The same "logic" proves $\forall x \in \mathbb{R}\,(x^3 = x)$: suppose $x^3 \neq x$ for all x; $x = 0$ contradicts it.

Checklist for "what is wrong with this proof?" Is the element arbitrary? Is the supposition the exact negation (§6)? Is the method named correctly (T1 Q11, L2 s84: contraposition is not contradiction)? Are both directions of an "iff" proved? Is every step justified? Is a diagram used in place of a proof (T2 Q8)?

WORKED EXAMPLE **T1 Q11 · Odd square iff odd integer · Proposition 4.6.4**

For $n \in \mathbb{Z}$, a proposed "contradiction proof" shows only that n even implies n^2 even. **Faults:** it uses contraposition, proves only one direction, and omits justifications. A complete proof is:

1. ($\Rightarrow$) Suppose n is even: $n = 2k$, $k \in \mathbb{Z}$. Then $n^2 = 2(2k^2)$ is even *(definition; closure)*. By contraposition, n^2 odd implies n odd.
2. ($\Leftarrow$) If $n = 2k + 1$, $k \in \mathbb{Z}$, then $n^2 = 2(2k^2 + 2k) + 1$ is odd *(algebra; closure)*. Hence n^2 is odd iff n is odd. ■

Proposition 4.6.4: n^2 even implies n even, by contraposition of line 2 and the even/odd dichotomy.

10 · Definitions and results in number theory

DEFINITION **Even, odd · Prime, composite**

$n \in \mathbb{Z}$ is **even** $\Leftrightarrow \exists k \in \mathbb{Z}\,(n = 2k)$; **odd** $\Leftrightarrow \exists k \in \mathbb{Z}\,(n = 2k + 1)$.

n is **prime** iff $n > 1$ and $\forall r, s \in \mathbb{Z}^+\,(n = rs \rightarrow (r = 1 \wedge s = n) \vee (r = n \wedge s = 1))$.

n is **composite** iff $n > 1$ and $\exists r, s \in \mathbb{Z}^+\,(n = rs \wedge 1 < r < n \wedge 1 < s < n)$. 1 is neither.

An equivalent definition of prime: $n > 1 \wedge \forall r, s \in \mathbb{Z}\,(r > 1 \wedge s > 1 \rightarrow rs \neq n)$.

DEFINITION **Rational, irrational · Lowest terms**

$r \in \mathbb{R}$ is **rational** $\Leftrightarrow \exists a, b \in \mathbb{Z}\,(r = a/b \wedge b \neq 0)$; otherwise **irrational**. A fraction a/b is in **lowest terms** if the largest integer dividing both a and b is 1.

DEFINITION **Divisibility**

For $n, d \in \mathbb{Z}$: $d \mid n \Leftrightarrow \exists k \in \mathbb{Z}\,(n = dk)$. Read: d divides n; n is a multiple of d; d is a factor/divisor of n.

Consequences: every d divides 0; 0 divides only 0; $1 \mid n$ and $n \mid n$ for all n. No division is performed: $3 \mid 12$ is **true** and $3 \mid 10$ is **false**.

DEFINITION **Absolute value (Lecture 1) · Colorful (Lecture 1, CS1231S only)**

For $x \in \mathbb{R}$, the **absolute value** of x is $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.

An integer n is **colorful** iff there exists some integer k such that $n = 3k$. (Non-standard term, used only in the lecture: $-1353 = 3 \cdot (-451)$ and $0 = 3 \cdot 0$ are colorful; 7 is not.)

THEOREM **Theorem 4.4.1 · Quotient–Remainder · div and mod**

For $n \in \mathbb{Z}$ and $d \in \mathbb{Z}^+$ there exist **unique** $q, r \in \mathbb{Z}$ with $n = dq + r$ and $0 \leq r < d$. We write n div $d = q$ and $n \bmod d = r$. Hence every integer is of exactly one form $dq, dq + 1, \ldots, dq + (d - 1)$. This is the basis for **division into cases**.

Slide-numbered results and proofs: §12; **theorem locator:** §65; **Appendix A:** §64.

11 · Choosing the method

To show	Method	Skeleton
$\exists x \in D\,P(x)$	Construction	Name a witness $x = \ldots$; verify $x \in D$ and $P(x)$. There is no need to explain how it was found.
$\sim\forall x \in D(P(x) \rightarrow Q(x))$	Counterexample	One x with $P(x)$ true and $Q(x)$ false. One suffices.
$\forall x \in D\,P(x)$, D finite	Exhaustion	Check every case explicitly.

$\forall x \in D(P(x) \to Q(x))$	Direct (generic particular)	Let $x \in D$ be arbitrary with $P(x)$. Unpack definitions; rebuild $Q(x)$.
same	Division into cases	Split by parity / by $x \bmod d$ (QR theorem) / by sign; prove each case.
same	Contraposition	Prove $\forall x(\sim Q(x) \to \sim P(x))$ directly.
any statement S	Contradiction	Suppose $\sim S$. Derive a contradiction. Conclude S.
$P \Leftrightarrow Q$	Two directions	Prove $P \Rightarrow Q$ and $Q \Rightarrow P$ separately.
$\exists! x\, P(x)$	Existence + uniqueness	Give x with $P(x)$; then if $P(y)$, show $y = x$.

METHOD **Contradiction or contraposition?**
Use **contraposition** when the negated conclusion is easier to work with; "n is odd" is easier to use than "n^2 is not odd". Use **contradiction** for statements such as "there is no …", "there are infinitely many …" and "… is irrational", whose negations assert that some object exists.

WORKED EXAMPLE **T2 Q8 · Cases inside a conditional**
Prove $\forall x \in \mathbb{R}\,((x^2 > x) \to ((x < 0) \vee (x > 1)))$.
1. Let $r \in \mathbb{R}$ with $r^2 > r$. Then $r(r-1) > 0$ *(algebra)*.
2. By Appendix A, Theorem T25, r and $r-1$ have the same sign. If both are positive, $r - 1 > 0$, so $r > 1$. If both are negative, $r < 0$.
3. Thus $r < 0$ or $r > 1$; the statement follows since r was arbitrary. ■

The case $x^2 \leq x$ needs no proof: the implication is vacuously true. A graph alone is not a proof.

PAST PAPER **T2 Q11 · A factor is at most $\sqrt{n}$**
For $n, a, b \in \mathbb{Z}^+$, prove $n = ab \to (a \leq \sqrt{n} \vee b \leq \sqrt{n})$. **Contraposition:** suppose $a > \sqrt{n}$ and $b > \sqrt{n}$. Then $ab > \sqrt{n} \cdot \sqrt{n} = n$ *(Appendix A, Theorem T27)*, hence $n \neq ab$. ■
Contradiction: suppose $n = ab$ and both factors exceed $\sqrt{n}$. The same inequality gives $ab > n$, contradicting $ab = n$. ■

TRAP **T2 Q4 · Closure claims**
$\mathbb{Z}$ is **not** closed under division ($3/2 = 1.5 \notin \mathbb{Z}$). Division is not a binary operation on **all** of $\mathbb{Q} \times \mathbb{Q}$, since $r/0$ is undefined; division of a rational by a **nonzero** rational does give a rational. In proving $\mathbb{Q}$ closed under $+$, the line "$bd \neq 0$ *(zero product property)*" is required, and is commonly omitted.

12 · Model proofs

PROOF **Sum of two even integers is even (Lecture 4, Ex. 4)**
1. Let m and n be arbitrary even integers.
   1.1. Then $m = 2r$ and $n = 2s$ for some integers r and s *(by definition of even)*.
   1.2. $m + n = 2r + 2s = 2(r + s)$ *(by basic algebra)*.
   1.3. $r + s$ is an integer *(by closure of integers under $+$)*, so $m + n$ is even *(by definition of even)*.
2. Therefore the sum of any two even integers is even. ■

Lecture 3.4.3: "by definition" and "by closure" apply universal modus ponens: instantiate the general rule at the chosen integers, then use its hypothesis.

WORKED EXAMPLE **Odd products and contraposition · Lecture 1, Ex. 1**
Prove that the product of two odd integers is odd; deduce that an even product has an even factor.
1. Let $a = 2r + 1$ and $b = 2s + 1$, where $r, s \in \mathbb{Z}$ *(definition of odd)*.
2. $ab = 2(2rs + r + s) + 1$, with $2rs + r + s \in \mathbb{Z}$ *(algebra; closure)*. Thus ab is odd.
3. By contraposition and the even/odd dichotomy, ab even implies a even or b even. ■

In particular, two consecutive odd integers have odd product. Taking $s = r + 1$ gives $ab = 2(2r^2 + 4r + 1) + 1$.

PROOF **Theorem 4.2.2 · The sum of any two rational numbers is rational**
1. Let r and s be arbitrary rational numbers.
   1.1. Then $r = a/b$ and $s = c/d$ for some integers a, b, c, d with $b \neq 0$, $d \neq 0$ *(by definition of rational)*.
   1.2. $r + s = (ad + bc)/(bd)$ *(by basic algebra)*.
   1.3. $ad + bc$ and bd are integers *(by closure under $+$ and $\times$)*, and $bd \neq 0$ *(zero product property)*, so $r + s$ is rational *(by definition of rational)*.
2. Therefore the sum of any two rational numbers is rational. ■

Corollary 4.2.3: if $r \in \mathbb{Q}$, then $2r = r + r \in \mathbb{Q}$ by this theorem.

PROOF Theorem 4.3.1 · For all $a, b \in \mathbb{Z}^+$, if $a \mid b$ then $a \le b$

1. Let a, b be positive integers with $a \mid b$.
   1.1. Then $b = ak$ for some integer k *(by definition of divisibility)*.
   1.2. Since a and b are positive, k is positive, i.e. $k \ge 1$.
   1.3. Therefore $a \le ak = b$. ■

PROOF Theorem 4.3.2 · The only divisors of 1 are 1 and −1 (division into cases)

1. Let m be any integer with $m \mid 1$. Then $1 = mk$ for some integer k *(by definition of divisibility)*.
   1.1. Since $mk > 0$, m and k are both positive or both negative.
   1.2. **Case 1:** both positive. Then $m \le 1$ *(by Theorem 4.3.1)*, so $m = 1$.
   1.3. **Case 2:** both negative. Then $1 = (-m)(-k)$, so $-m$ is a positive divisor of 1; as in Case 1, $-m = 1$, i.e. $m = -1$.
2. Therefore the only divisors of 1 are 1 and −1. ■

PROOF Theorem 4.3.3 · Transitivity of divisibility

1. Suppose a, b, c are integers with $a \mid b$ and $b \mid c$.
   1.1. Then $b = ar$ and $c = bs$ for some integers r, s *(by definition of divisibility)*.
   1.2. $c = bs = (ar)s = a(rs)$ *(by substitution and associativity)*.
   1.3. rs is an integer *(by closure under $\times$)*, so $a \mid c$ *(by definition of divisibility)*. ■

PROOF Every integer n has $n^2 \bmod 4 \in \{0, 1\}$ (division into cases, via parity)

1. Let n be any integer. n is even or odd *(Assumption 1)*.
   1.1. **Case 1:** $n = 2k$ for some integer k. Then $n^2 = 4k^2 = 4(k^2) + 0$, and $k^2 \in \mathbb{Z}$ *(closure)*.
   1.2. **Case 2:** $n = 2k + 1$. Then $n^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1$, and $k^2 + k \in \mathbb{Z}$ *(closure)*.
2. In either case $n^2 = 4q + r$ with $r \in \{0, 1\}$, so $n^2 \bmod 4 \in \{0, 1\}$ *(by the quotient–remainder theorem, r is unique)*. ■

PROOF Theorem 4.6.1 · There is no greatest integer (contradiction)

1. Suppose not; that is, there is a greatest integer g, so $g \ge n$ for every integer n.
   1.1. Let $G = g + 1$. Then G is an integer *(by closure under $+$)* and $G > g$.
   1.2. This contradicts that g is the greatest integer.
2. Hence the supposition is false, and there is no greatest integer. ■

PROOF Theorem 4.7.1 · $\sqrt{2}$ is irrational (contradiction)

1. Suppose not; that is, $\sqrt{2}$ is rational.
   1.1. Then $\sqrt{2} = a/b$ for some integers a, b with $b \neq 0$ *(by definition of rational)*.
   1.2. Reduce a/b to lowest terms m/n *(Assumption 2)* then $m^2 = 2n^2$ *(by basic algebra)*.
   1.3. So m^2 is even *(by definition of even, as $n^2 \in \mathbb{Z}$)*, hence m is even *(by Proposition 4.6.4)*.
   1.4. Let $m = 2k$. Then $4k^2 = 2n^2$, so $n^2 = 2k^2$ is even, hence n is even *(by Proposition 4.6.4)*.
   1.5. So 2 divides both m and n, contradicting that m/n is in lowest terms.
2. Therefore $\sqrt{2}$ is irrational. ■

PROOF rational + irrational is irrational (contradiction)

1. Suppose not: some rational r and irrational s have $r + s$ rational.
   1.1. Then $r = a/b$ and $r + s = c/d$ with $a, b, c, d \in \mathbb{Z}$, $b, d \neq 0$ *(by definition of rational)*.
   1.2. So $s = c/d - a/b = (bc - ad)/(bd)$, with $bc - ad, bd \in \mathbb{Z}$ and $bd \neq 0$ *(closure; zero product property)*.
   1.3. Hence s is rational *(by definition of rational)*, contradicting that s is irrational. ■

PROOF The set of primes is infinite (contradiction)

1. Suppose not: the primes are exactly $p_1, p_2, ..., p_n$. Let $N = p_1 p_2 \cdots p_n + 1$.
   1.1. $N > 1$, so some prime p divides N *(existence of a prime divisor, proved below)* $p = p_i$ for some i *(the list is complete)*.
   1.2. p_i divides $p_1 \cdots p_n = N - 1$ as well. But a prime dividing a cannot divide $a + 1$ *(consecutive-integer divisibility result, proved below)*. Contradiction.
2. Hence there are infinitely many primes. ■

PROOF There exist irrational p, q with p^q rational (Lecture 1, Ex. 7; non-constructive proof by cases)

1. $\sqrt{2}$ is irrational *(Theorem 4.7.1)*. Consider $\sqrt{2}^{\sqrt{2}}$: it is rational or irrational.
   1.1. **Case 1:** rational. Take $p = q = \sqrt{2}$.
   1.2. **Case 2:** irrational. Take $p = \sqrt{2}^{\sqrt{2}}$, $q = \sqrt{2}$: then $p^q = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2$, rational.
2. In either case the required p and q exist. ■

PROOF **Any two consecutive integers have opposite parity**
1. Let m and $m+1$ be two arbitrary consecutive integers. m is even or odd *(Assumption 1)*.
   1.1. **Case 1:** $m = 2k$ for some $k \in \mathbb{Z}$. Then $m+1 = 2k+1$ is odd *(by definition of odd)*.
   1.2. **Case 2:** $m = 2k+1$. Then $m+1 = 2(k+1)$ with $k+1 \in \mathbb{Z}$ *(closure)*, so $m+1$ is even.
2. In both cases m and $m+1$ have opposite parity. ■

PROOF **The square of any odd integer has the form $8m+1$ (division into cases via QR)**
1. Let n be a arbitrary odd integer. By the QR theorem with $d = 4$, $n = 4q$, $4q+1$, $4q+2$ or $4q+3$ for some $q \in \mathbb{Z}$; as n is odd, $n = 4q+1$ or $n = 4q+3$.
   1.1. **Case 1:** $n^2 = 16q^2 + 8q + 1 = 8(2q^2+q) + 1$.
   1.2. **Case 2:** $n^2 = 16q^2 + 24q + 9 = 8(2q^2+3q+1) + 1$.
2. In both cases $n^2 = 8m+1$ for an integer m *(closure)*. ■

PROOF **Theorem 4.4.6 · Triangle inequality $|x+y| \le |x| + |y|$ for all $x, y \in \mathbb{R}$**
1. By Lemma 4.4.4, $-|x| \le x \le |x|$ and $-|y| \le y \le |y|$; adding, $-(|x|+|y|) \le x+y \le |x|+|y|$.
   1.1. **Case 1:** $x+y \ge 0$. Then $|x+y| = x+y \le |x|+|y|$ *(definition of absolute value)*.
   1.2. **Case 2:** $x+y < 0$. Then $|x+y| = -(x+y) \le |x|+|y|$ *(from the left inequality)*.
2. So $|x+y| \le |x|+|y|$ in all cases. ■

PROOF **For any integer a and prime p: if $p \mid a$ then $p \nmid (a+1)$ (contradiction)**
1. Suppose not: $p \mid a$ and $p \mid (a+1)$ for some integer a and prime p.
   1.1. Then $a = pr$ and $a+1 = ps$ for some $r, s \in \mathbb{Z}$ *(definition of divisibility)*, so $1 = ps - pr = p(s-r)$.
   1.2. So $p \mid 1$ *(as $s - r \in \mathbb{Z}$)*, hence $p = 1$ or $p = -1$ *(Theorem 4.3.2)*. This is a contradiction, since a prime is greater than 1. ■

PROOF **If $a \mid b$ and $a \mid c$ then $a \mid (mb+nc)$ for all integers m, n**
1. Suppose $a \mid b$ and $a \mid c$: $b = ar$ and $c = as$ for some $r, s \in \mathbb{Z}$ *(definition of divisibility)*.
2. Then $mb + nc = a(mr+ns)$ *(basic algebra)*, and $mr + ns \in \mathbb{Z}$ *(closure)*, so $a \mid (mb+nc)$. ■

PROOF **Lecture 1 · If x and y are "colorful" (n colorful $\leftrightarrow \exists k \in \mathbb{Z}\,(n = 3k)$), so is $x+2y$**
Work backwards first: the goal "$x+2y$ is colorful" means we need an integer c with $x + 2y = 3c$.
1. Let x, y be colorful integers. Then $x = 3a$ and $y = 3b$ for some $a, b \in \mathbb{Z}$ *(by definition of colorful)*.
2. $x + 2y = 3a + 6b = 3(a+2b)$ *(basic algebra)*. Let $c = a + 2b$, an integer *(closure)*.
3. So $x + 2y = 3c$, and $x+2y$ is colorful *(by definition of colorful)*. ■

WORKED EXAMPLE **Disproof by counterexample**
- "$\forall a, b, c \in \mathbb{Z}\,(a \mid bc \to a \mid b \vee a \mid c)$" is false: $4 \mid 2 \cdot 2$ but $4 \nmid 2$.
- "$\forall a, b \in \mathbb{Z}\,(a \mid b \wedge b \mid a \to a = b)$" is false: take $a = 1$, $b = -1$. (True on $\mathbb{Z}^+$: Theorem 4.3.1.)
- "The product of two irrational numbers is irrational" is false: $\sqrt{2} \cdot \sqrt{8} = 4$ (Lecture 1).
- "$\forall a, b \in \mathbb{R}\,(a^2 = b^2 \to a = b)$" is false: take $a = 1$, $b = -1$ (Lecture 4).
- "There is no integer that is both even and odd" is **true**: if $n = 2a = 2b+1$ then $1 = 2(a-b)$, so $1/2 \in \mathbb{Z}$, a contradiction.

PROOF **Theorem 4.2.1 · Every integer is a rational number**
1. Let a be a arbitrary integer. Then $a = a/1$, a quotient of integers with non-zero denominator.
2. Hence a is rational *(by definition of rational)*. ■

PROOF **Proof by exhaustion (Lecture 4, Ex. 3) · every even n with $4 \le n \le 26$ is a sum of two primes**
$4 = 2+2$, $6 = 3+3$, $8 = 3+5$, $10 = 5+5$, $12 = 5+7$, $14 = 3+11$, $16 = 5+11$, $18 = 7+11$, $20 = 7+13$, $22 = 5+17$, $24 = 5+19$, $26 = 7+19$. Every case is checked. ■

PROOF **For every integer n, $3 \mid (n^3 - n)$ (division into cases, QR with $d = 3$)**
1. Let $n \in \mathbb{Z}$. By the QR theorem, $n = 3q$, $3q+1$ or $3q+2$ for some $q \in \mathbb{Z}$. Note $n^3 - n = (n-1)n(n+1)$ *(basic algebra)*.
   1.1. **Case $n = 3q$:** the factor $n = 3q$, so $n^3 - n = 3 \cdot q(n-1)(n+1)$.
   1.2. **Case $n = 3q+1$:** the factor $n - 1 = 3q$, so $n^3 - n = 3 \cdot qn(n+1)$.
   1.3. **Case $n = 3q+2$:** the factor $n + 1 = 3(q+1)$, so $n^3 - n = 3 \cdot (q+1)(n-1)n$.
2. In each case the cofactor is an integer *(closure)*, so $3 \mid (n^3 - n)$ *(definition of divisibility)*. ■

PROOF **Every prime $p > 3$ has the form $6k+1$ or $6k+5$**
1. Let $p > 3$ be prime. By the QR theorem, $p = 6k + r$ with $r \in \{0, 1, 2, 3, 4, 5\}$.

1.1. If $r \in \{0, 2, 4\}$, then $p = 2(3k + r/2)$ is even and > 2, so not prime.
1.2. If $r = 3$, then $p = 3(2k + 1)$ with $p > 3$, so not prime.
2. Hence $r = 1$ or $r = 5$. ■

PROOF **Every integer $n > 1$ is divisible by a prime (via well-ordering)**
1. Let $n > 1$ be an integer, and let $D = \{d \in \mathbb{Z} : d > 1 \wedge d \mid n\}$. $D \neq \varnothing$, since $n \in D$.
2. D has a smallest element p *($(\mathbb{N}, \leq)$ is well-ordered)*.
3. p is prime: suppose instead $p = rs$ with $1 < r < p$. Then $r \mid p$ and $p \mid n$, so $r \mid n$ *(Theorem 4.3.3)*, so $r \in D$ with $r < p$, contradicting the minimality of p.
4. Hence the prime p divides n. ■

PROOF **Lemma 4.4.4 · For every real r: $-|r| \leq r \leq |r|$ (division into cases)**
1. **Case $r \geq 0$:** $|r| = r$ *(definition of absolute value)*, and $-|r| = -r \leq 0 \leq r$. So $-|r| \leq r = |r|$.
2. **Case $r < 0$:** $|r| = -r > 0$, so $-|r| = r$ and $r < 0 < |r|$. So $-|r| = r \leq |r|$. ■

PAST PAPER **Using unique factorization · AY25 Q6(iii) in full**
Unique factorization: every integer $n > 1$ is a product of primes, unique up to order.
Claim: if $n = m^2$ is a perfect square and a prime p divides n, then $p^2 \mid n$.
If $m = 0$, then $n = 0$ and $p^2 \mid 0$. Otherwise write $|m| = p_1^{e_1} \cdots p_k^{e_k}$ *(unique factorization)* then $n = p_1^{2e_1} \cdots p_k^{2e_k}$. A prime dividing n is some p_i (uniqueness of the factorization), and its exponent $2e_i \geq 2$, so $p_i^2 \mid n$. ■
In a perfect square every prime appears to an even power.

WORKED EXAMPLE **Division into cases**
Prove: for every integer n, $n^2 + n$ is even.
1. Let n be a arbitrary integer. Then n is even or odd *(Assumption 1)*.
1.1. **Case 1:** $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, and $2k^2 + k \in \mathbb{Z}$ *(closure)*.
1.2. **Case 2:** $n = 2k + 1$. Then $n^2 + n = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1)$, and $(2k + 1)(k + 1) \in \mathbb{Z}$ *(closure)*.
2. In both cases $n^2 + n$ is even *(by definition of even)*. ■

WORKED EXAMPLE **Contradiction**
Prove: there is no smallest positive rational number.
1. Suppose not: let r be the smallest positive rational number.
1.1. $r = a/b$ for some integers a, b with $b \neq 0$ *(definition of rational)*. Then $r/2 = a/(2b)$, where $a, 2b \in \mathbb{Z}$ and $2b \neq 0$, so $r/2$ is rational.
1.2. $r/2 > 0$ and $r/2 < r$ *(as $r > 0$)*, contradicting the choice of r as the smallest.
2. Hence there is no smallest positive rational number. ■

Part IV · Set Theory

13 · Describing sets

DEFINITION **Set · Membership · Cardinality**
A **set** is an unordered collection of objects, its **members** or **elements**; order and repetition do not matter: $\{9, 8, 7\} = \{7, 8, 7, 9, 9\}$. $x \in S$ means x is an element of S; $x \notin S$ means it is not. The **cardinality** $|S|$ is the size of S, the number of elements of S.

DEFINITION **Set-roster, set-builder and replacement notation**
Set-roster: list all the elements between braces, e.g. $\{1, 2, 3\}$, $\{1, 2, ..., 100\}$, $\{1, 2, 3, ...\}$ (the ellipsis "..." reads "and so forth").
Set-builder: $\{x \in U : P(x)\}$ is the set of all $x \in U$ for which $P(x)$ is true. z is a member iff $z \in U$ **and** $P(z)$.
Replacement: $\{t(x) : x \in A\}$ is the set of all objects $t(x)$ as x ranges over A. z is a member iff $t(x) = z$ for **some** $x \in A$.
E.g. $\{x + 1 : x \in \mathbb{Z}_{\geq 0}\} = \mathbb{Z}^+$; $\{x \in \mathbb{Z} : -2 < x < 5\} = \{-1, 0, 1, 2, 3, 4\}$.

DEFINITION Interval notation (Lecture 5)
For real numbers $a \le b$: $(a,b) = \{x \in \mathbb{R} : a < x < b\}$, $[a,b] = \{x \in \mathbb{R} : a \le x \le b\}$, $(a,b] = \{x \in \mathbb{R} : a < x \le b\}$, $[a,b) = \{x \in \mathbb{R} : a \le x < b\}$.
Unbounded: $(a,\infty) = \{x \in \mathbb{R} : x > a\}$, $[a,\infty) = \{x \in \mathbb{R} : x \ge a\}$, $(-\infty,b) = \{x \in \mathbb{R} : x < b\}$, $(-\infty,b] = \{x \in \mathbb{R} : x \le b\}$.
The interval (a,b) and the ordered pair (a,b) look the same; context tells them apart.

Symbol	Set
$\mathbb{N}$	natural numbers $\{0,1,2,...\}$, **including 0**
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}$	integers, rationals, reals
$\mathbb{Z}^+, \mathbb{Z}^-, \mathbb{Z}_{\ge 0}$	positive, negative, non-negative integers (0 is neither positive nor negative)
$(a,b), [a,b]$	open / closed real intervals; also $(a,b], [a,\infty)$ …

PAST PAPER AY25 Q17(b) · Name these subsets of $\mathbb{N}$
$A = \{0,1,3,7,15,31,...\}$ in replacement notation: $A = \{2^n - 1 : n \in \mathbb{N}\}$.
$B = \{3,9,15,21,27,...\}$ in set-builder notation: $B = \{n \in \mathbb{N} : 3 \mid n \wedge n \text{ is odd}\}$.
(Many correct answers; check the first few terms.)

14 · Subsets, equality and the empty set

DEFINITION Subset, superset · Proper subset · Set equality · Empty set, singleton
$A \subseteq B \Leftrightarrow \forall x(x \in A \to x \in B)$: every element of A is in B ("A is contained in B"). $A \not\subseteq B \Leftrightarrow \exists x(x \in A \wedge x \notin B)$.
If $A \subseteq B$ we may write $B \supseteq A$: "B contains / includes A", "B is a **superset** of A".
$A \subsetneq B$ (proper, or strict, inclusion) iff $A \subseteq B$ and $A \neq B$. $A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A \Leftrightarrow \forall x(x \in A \leftrightarrow x \in B)$.
The **empty set** $\varnothing = \{\}$ is the set with no elements. A set with exactly one element is a **singleton**.

THEOREM Theorem 6.2.4 · The empty set
$\varnothing \subseteq A$ for every set A (and $\varnothing$ is unique). **Proof:** $\forall x(x \in \varnothing \to x \in A)$ is vacuously true, as $x \in \varnothing$ is always false. **Uniqueness:** the empty set is unique: two empty sets are subsets of one another by Theorem 6.2.4, hence equal.

Claim		Why
$\varnothing \in \{1,2,3\}$	F	$\varnothing$ is not listed
$\varnothing \subseteq \{1,2,3\}$	T	Thm 6.2.4
$\{2\} \in \{1,2,3\}$	F	elements are numbers
$\{2\} \subseteq \{1,2,3\}$	T	2 is an element
$\{2\} \in \{\{1\},\{2\},\{3\}\}$	T	$\{2\}$ is listed
$\{2\} \subseteq \{\{1\},\{2\},\{3\}\}$	F	2 is not listed
$\{9\} = \{\{9\}\}$	F	different elements

The $\varnothing$ ladder	$\lvert\cdot\rvert$
$\varnothing = \{\}$	0
$\{\varnothing\} = \mathcal{P}(\varnothing)$	1
$\{\varnothing, \{\varnothing\}\} = \mathcal{P}(\mathcal{P}(\varnothing))$	2
$\mathcal{P}(\mathcal{P}(\mathcal{P}(\varnothing)))$	4
$\mathcal{P}^4(\varnothing)$	16

$\varnothing \neq \{\varnothing\}$: the first is empty, the second has one element. $\varnothing \in \{\varnothing\}$ and $\varnothing \subseteq \{\varnothing\}$ are both true.

TRAP $\subseteq$ is transitive but $\in$ is not · AY23 Q8
Which hold for all sets? (i) $A \in B \wedge B \subseteq C \Rightarrow A \in C$ is **true**, since A is an element of B and hence of C.
(ii) $A \in B \wedge B \subseteq C \Rightarrow A \subseteq C$ is false: take $A = \{1\}$, $B = C = \{\{1\}\}$, but $1 \notin C$.
(iii), (iv) $A \subseteq B \wedge B \in C \Rightarrow A \in C$ or $A \subseteq C$ are false: take $A = \{x\}$, $B = \{x,y\}$, $C = \{\{x,y\}\}$. Only (i): **A**.

15 · Ordered pairs and Cartesian products

DEFINITION Ordered pair, n-tuple · Cartesian product
$(a,b) = (c,d) \Leftrightarrow a = c \wedge b = d$; likewise for n-tuples componentwise. $(1,2) \neq (2,1)$ though $\{1,2\} = \{2,1\}$.
$A \times B = \{(a,b) : a \in A \wedge b \in B\}$; $A_1 \times \cdots \times A_n = \{(a_1,...,a_n) : a_i \in A_i\}$; $A^n = A \times \cdots \times A$.

Facts. $|A \times B| = |A| \cdot |B|$. $A \times \varnothing = \varnothing \times A = \varnothing$. $A \times B \neq B \times A$ in general. $(A \times B) \times C \neq A \times B \times C$: elements $((a,b),c)$ versus (a,b,c). $A \times (B \cup C) = (A \times B) \cup (A \times C)$, and likewise for $\cap$ and $\setminus$.

PAST PAPER **AY24 Q6 · Which hold for all sets A, B, C?**
(i) $\mathcal{P}(A) \subseteq \mathcal{P}(B) \to A \subseteq B$ is **true** (proof in §17). (ii) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ is **true** (proof in §17). (iii) $A \times B \subseteq B \times C \to A \subseteq C$ is **false**: take $B = \varnothing$; then $A \times \varnothing = \varnothing$ is a subset of every set, whatever A is. Answer **A**.

TRAP **T3 Q3(c) · $A \times \mathcal{P}(B) \neq \mathcal{P}(A \times B)$**
For every A, B, $\varnothing \in \mathcal{P}(A \times B)$ *($\varnothing \subseteq A \times B$)*, but $\varnothing \notin A \times \mathcal{P}(B)$ *(every member of a Cartesian product is an ordered pair; $\varnothing$ is not one)*. This named witness proves the sets differ. A cardinality "proof" ($n \cdot 2^k \neq 2^{nk}$ for $|A| = n$, $|B| = k$) is wrong: equality holds when $n = 1$, or $n = 2$ and $k = 1$.

16 · Operations on sets

DEFINITION **Universal set · Union, intersection, difference, complement**
A **universal set** (universe of discourse) U is the set containing every object under discussion, e.g. $\mathbb{R}$ when all sets considered are sets of reals.
For $A, B \subseteq U$: $A \cup B = \{x \in U : x \in A \vee x \in B\}$, $A \cap B = \{x \in U : x \in A \wedge x \in B\}$, $B \setminus A = \{x \in U : x \in B \wedge x \notin A\}$ (the **relative complement** of A in B), $\overline{A} = \{x \in U : x \notin A\} = U \setminus A$.
Procedural versions (the steps in element proofs): $a \in X \cup Y \Leftrightarrow a \in X \vee a \in Y$; $a \in X \cap Y \Leftrightarrow a \in X \wedge a \in Y$; $a \in X \setminus Y \Leftrightarrow a \in X \wedge a \notin Y$; $a \in \overline{X} \Leftrightarrow a \notin X$; $(a, b) \in X \times Y \Leftrightarrow a \in X \wedge b \in Y$.

DEFINITION **Symmetric difference (Tutorial 3) · Disjoint**
$A \oplus B = (A \setminus B) \cup (B \setminus A)$, the set of elements in exactly one of A and B. A and B are **disjoint** iff $A \cap B = \varnothing$; $A_1, A_2, \ldots$ are **mutually disjoint** iff $A_i \cap A_j = \varnothing$ whenever $i \neq j$. $\bigcup_{i=0}^{n} A_i$, $\bigcap_{i=0}^{n} A_i$ extend $\cup, \cap$.

THEOREM **Theorem 6.2.1 · Subset relations**
$A \cap B \subseteq A$, $A \cap B \subseteq B$ (inclusion of intersection); $A \subseteq A \cup B$, $B \subseteq A \cup B$ (inclusion in union); $A \subseteq B \wedge B \subseteq C \Rightarrow A \subseteq C$ (transitivity). The **set identities** (Theorem 6.2.2) are the right column of the table in §2.

PAST PAPER **AY23 Q7 · For which can no non-empty A, B exist?**
(i) $A \setminus B = A \cap B$: these two sets are disjoint, so both must be $\varnothing$; then $A = (A \setminus B) \cup (A \cap B) = \varnothing$. **Impossible.**
(ii) $A \setminus B = A \cup B$: then $B \subseteq A \cup B = A \setminus B$, so every element of B is not in B; $B = \varnothing$. **Impossible.**
(iii) $A \cup B = A \cap B$: take $A = B$. (iv) $A \oplus B = A \cup B$: take $A \cap B = \varnothing$. Answer **B** (only (i), (ii)).

17 · Proving and disproving set statements

METHOD **Three methods**
Element method. To show $X \subseteq Y$: "Let $z \in X$ … so $z \in Y$". To show $X = Y$: prove both inclusions, or one chain of $\Leftrightarrow$ steps each justified by a definition or a logic law.
Algebraic method. Rewrite using the set identities (§2), one named law per line.
Disproof. Give one concrete counterexample with every set written out. A Venn diagram helps in finding one.

PROOF $\{x \in \mathbb{Z} : x^2 = 1\} = \{1, -1\}$ **(two inclusions, Lecture 5)**
1. ($\subseteq$) Take any $z \in \{x \in \mathbb{Z} : x^2 = 1\}$.
   1.1. Then $z \in \mathbb{Z}$ and $z^2 = 1$, so $(z - 1)(z + 1) = 0$ *(by basic algebra)*.
   1.2. So $z = 1$ or $z = -1$, i.e. $z \in \{1, -1\}$.
2. ($\supseteq$) Take any $z \in \{1, -1\}$. Then $z \in \mathbb{Z}$ and $z^2 = 1$, so $z \in \{x \in \mathbb{Z} : x^2 = 1\}$.
3. Therefore the sets are equal *(by definition of set equality, from 1 and 2)*. ■

PROOF **De Morgan: $\overline{A \cup B} = \overline{A} \cap \overline{B}$ (chain of $\Leftrightarrow$)**
1. Let $z \in U$. Then
   1.1. $z \in \overline{A \cup B} \Leftrightarrow \sim(z \in A \cup B)$ *(by definition of complement)*
   1.2. $\Leftrightarrow \sim(z \in A \vee z \in B)$ *(by definition of $\cup$)*
   1.3. $\Leftrightarrow z \notin A \wedge z \notin B$ *(by De Morgan's law for logic)*
   1.4. $\Leftrightarrow z \in \overline{A} \wedge z \in \overline{B} \Leftrightarrow z \in \overline{A} \cap \overline{B}$ *(by definitions of complement, $\cap$)*. ■

PROOF $(A \cap B) \cup (A \setminus B) = A$ **(algebraic, Lecture 5)**

$(A \cap B) \cup (A \setminus B) = (A \cap B) \cup \left(A \cap \overline{B}\right)$ *(set difference law)*
$= A \cap \left(B \cup \overline{B}\right)$ *(distributive law)*
$= A \cap U = A$ *(complement law; identity law)*

■

PROOF $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ **(algebraic)**

$A \setminus (B \cap C)$
$= A \cap \overline{B \cap C}$ *(set difference)*
$= A \cap \left(\overline{B} \cup \overline{C}\right)$ *(De Morgan)*
$= \left(A \cap \overline{B}\right) \cup \left(A \cap \overline{C}\right)$ *(distributive)*
$= (A \setminus B) \cup (A \setminus C)$ *(set difference ×2)*.

■

PROOF **For all sets A, B: $A \subseteq B \Leftrightarrow \mathcal{P}(A) \subseteq \mathcal{P}(B)$**

1. ($\Rightarrow$) Suppose $A \subseteq B$. Let $X \in \mathcal{P}(A)$; then $X \subseteq A$ *(definition of power set)*, so $X \subseteq B$ *(transitivity of $\subseteq$)*, so $X \in \mathcal{P}(B)$.
2. ($\Leftarrow$) Suppose $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. $A \subseteq A$, so $A \in \mathcal{P}(A)$, hence $A \in \mathcal{P}(B)$, i.e. $A \subseteq B$ *(definition of power set)*.

■

WORKED EXAMPLE **T3 Q4 · Set equality by double inclusion**

Let $A = \{2n + 1 : n \in \mathbb{Z}\}$ and $B = \{2n - 5 : n \in \mathbb{Z}\}$. Prove $A = B$.

1. If $a \in A$, write $a = 2n + 1 = 2(n + 3) - 5$ for $n \in \mathbb{Z}$. Since $n + 3 \in \mathbb{Z}$ *(closure)*, $a \in B$ *(definition of B)*.
2. If $b \in B$, write $b = 2n - 5 = 2(n - 3) + 1$ for $n \in \mathbb{Z}$. Since $n - 3 \in \mathbb{Z}$, $b \in A$.
3. Thus $A = B$ *(double inclusion)*.

■

A substitution $k = n + 3$ must range over **all** integers: given k, use $n = k - 3$. Closure alone does not show this.

WORKED EXAMPLE **T3 Q5 · Element method as a set-builder chain**

Prove that for all sets A, B, C: $A \cap (B \setminus C) = (A \cap B) \setminus C$.

$A \cap (B \setminus C) = \{x : x \in A \wedge x \in (B \setminus C)\}$ *(definition of $\cap$)*
$= \{x : x \in A \wedge (x \in B \wedge x \notin C)\}$ *(definition of $\setminus$)*
$= \{x : (x \in A \wedge x \in B) \wedge x \notin C\}$ *(associativity of $\wedge$)*
$= \{x : x \in A \cap B \wedge x \notin C\}$ *(definition of $\cap$)*
$= (A \cap B) \setminus C$ *(definition of $\setminus$)*.

■

WORKED EXAMPLE **T3 Q6 · Symmetric difference by set identities only**

(a) $A = \{1, 4, 9, 16\}$, $B = \{2, 4, 6, ..., 16\}$: find $A \oplus B$. (b) Using Theorem 6.2.2, prove $A \oplus B = (A \cup B) \setminus (A \cap B)$.

(a) $A \setminus B = \{1, 9\}$, $B \setminus A = \{2, 6, 8, 10, 12, 14\}$, so $A \oplus B = \{1, 2, 6, 8, 9, 10, 12, 14\}$.

(b) $A \oplus B = (A \setminus B) \cup (B \setminus A)$ *(definition of $\oplus$)*
1. $= \left(A \cap \overline{B}\right) \cup \left(B \cap \overline{A}\right)$ *(set difference law)*
2. $= \left(\left(A \cap \overline{B}\right) \cup B\right) \cap \left(\left(A \cap \overline{B}\right) \cup \overline{A}\right)$ *(distributive law)*
3. $= \left((A \cup B) \cap \left(\overline{B} \cup B\right)\right) \cap \left(\left(A \cup \overline{A}\right) \cap \left(\overline{B} \cup \overline{A}\right)\right)$ *(distributive, commutative laws)*
4. $= \left((A \cup B) \cap U\right) \cap \left(U \cap \left(\overline{A} \cup \overline{B}\right)\right)$ *(commutative, complement laws)*
5. $= (A \cup B) \cap \left(\overline{A} \cup \overline{B}\right)$ *(commutative, identity laws)*
6. $= (A \cup B) \cap \overline{A \cap B}$ *(De Morgan's law)*
7. $= (A \cup B) \setminus (A \cap B)$ *(set difference law)*.

■

WORKED EXAMPLE **Set identity**

Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

$A \setminus (B \cup C)$
$= A \cap \overline{B \cup C}$ *(set difference law)*
$= A \cap \left(\overline{B} \cap \overline{C}\right)$ *(De Morgan's law)*
$= (A \cap A) \cap \left(\overline{B} \cap \overline{C}\right)$ *(idempotent law)*
$= \left(A \cap \overline{B}\right) \cap \left(A \cap \overline{C}\right)$ *(associative, commutative laws)*
$= (A \setminus B) \cap (A \setminus C)$ *(set difference law)*.

■

WORKED EXAMPLE **Prove or disprove**

(a) $A \cup (B \setminus C) = (A \cup B) \setminus C$; (b) $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$; (c) $A \subseteq C \wedge B \subseteq C \rightarrow A \cup B \subseteq C$.

(a) **False:** $A = \{1\}$, $B = \varnothing$, $C = \{1\}$: left side $\{1\}$, right side $\varnothing$.

(b) **False:** $A = \{1\}$, $B = \{2\}$: $\{1, 2\} \in \mathcal{P}(A \cup B)$ but $\{1, 2\} \notin \mathcal{P}(A) \cup \mathcal{P}(B)$. (Only $\supseteq$ holds.)

(c) **True:** let $x \in A \cup B$; then $x \in A$ or $x \in B$ *(definition of $\cup$)* in either case $x \in C$ *(as $A \subseteq C$, $B \subseteq C$)*.

■

18 · Partitions and power sets

DEFINITION **Partition (Lecture 6)**
$\mathcal{C}$ is a **partition** of A iff (1) every element of $\mathcal{C}$ is a **non-empty** subset of A, and (2) every element of A lies in **exactly one** element of $\mathcal{C}$: $\forall x \in A\, \exists! S \in \mathcal{C}\, (x \in S)$. The elements of $\mathcal{C}$ are its **components**. (Equivalently: non-empty, mutually disjoint, union is A.)

DEFINITION **Power set · Theorem 6.3.1**
$\mathcal{P}(A)$ is the set of all subsets of A. If A is finite and $|A| = n$, then $|\mathcal{P}(A)| = 2^n$. Always $\varnothing \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$.
$\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$, but only $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.

PAST PAPER **AY24 Q7 and Q8 · Partitions, power sets**
Q7. (i) $\{\{2,3\},\{4\},\{3,2\},\{1\}\} = \{\{2,3\},\{4\},\{1\}\}$ **is** a partition of $\{1,2,3,4\}$, because repeated elements count once. (ii) $\{\mathbb{Z}\}$ **is** a partition of $\mathbb{Z}$ (one component). (iii) $\mathbb{Z}$ is **not**: its elements are numbers, not subsets. **C.**
Q8. (i) $|\mathcal{P}(S) \setminus \mathcal{P}(\varnothing)| = 2^{|S|} - 1$ is true, as $\mathcal{P}(\varnothing) = \{\varnothing\}$. (ii) $|S \cap \mathcal{P}(S)| = 0$ is false: $S = \{a, \{a\}\}$ gives $S \cap \mathcal{P}(S) = \{\{a\}\}$. (iii) $|S| \cdot |\mathcal{P}(S)| = |S \times \mathcal{P}(S)|$ is true. **D.**

PAST PAPER **AY25 Q7, Q8 · Counting**
Q7. A, B, C pairwise share exactly one element; $D = \mathcal{P}(A) \cap \mathcal{P}(B) \cap \mathcal{P}(C) = \mathcal{P}(A \cap B \cap C)$. The triple intersection lies inside each one-element pairwise intersection, so it has 0 or 1 elements: $|D| \in \{1,2\}$. **B.**
Q8. $|A| = 2|B|$, $|B| = |C| + 2$, and $\{B, C, D\}$ partitions A, so $|A| = |B| + |C| + |D|$: $2|B| = |B| + (|B| - 2) + |D|$, giving $|D| = 2$. **C.**

PAST PAPER **AY24 Q16 (5 marks) · $A = \{\varnothing, x\}$**
$\mathcal{P}(A) = \{\varnothing, \{\varnothing\}, \{x\}, \{\varnothing, x\}\}$ and $\mathcal{P}(\varnothing) = \{\varnothing\}$, $\mathcal{P}(\mathcal{P}(\varnothing)) = \{\varnothing, \{\varnothing\}\}$.
- (a) $\mathcal{P}(A) \setminus \mathcal{P}(\mathcal{P}(\varnothing)) = \{\{x\}, \{\varnothing, x\}\}$.
- (b) $(\mathcal{P}(A) \setminus \mathcal{P}(\varnothing)) \setminus \{A\} = \{\{\varnothing\}, \{x\}\}$, so its power set is $\{\varnothing, \{\{\varnothing\}\}, \{\{x\}\}, \{\{\varnothing\}, \{x\}\}\}$.
- (c) $\mathcal{P}(\varnothing) \times \mathcal{P}(A) = \{(\varnothing, S) : S \in \mathcal{P}(A)\}$ and $\mathcal{P}(A) \times \mathcal{P}(\varnothing) = \{(S, \varnothing) : S \in \mathcal{P}(A)\}$ share only $(\varnothing, \varnothing)$, so the answer is $\{(\varnothing, \{\varnothing\}), (\varnothing, \{x\}), (\varnothing, \{\varnothing, x\})\}$.

PAST PAPER **AY23 Q6 and Q17 · $A_1 = \{\varnothing\}$, $A_2 = \{a\}$, $A_3 = \{a, \{a\}\}$**
Q6. Target $\varnothing \cup \mathcal{P}(\varnothing) = \{\varnothing\}$. $\varnothing \times \{\varnothing\} = \varnothing$ ✗; $\mathcal{P}(\varnothing \cup \mathcal{P}(\varnothing)) = \{\varnothing, \{\varnothing\}\}$ ✗; $\mathcal{P}(\varnothing) \setminus \varnothing = \{\varnothing\}$ ✓; $(\varnothing \cap \{\varnothing\}) \cup (\varnothing \cup \{\varnothing\}) = \{\varnothing\}$ ✓. **D.**
Q17. (a) $(A_1 \cap A_2) \setminus A_3 = \varnothing$. (b) $(A_1 \cup A_2) \setminus A_3 = \{\varnothing\}$. (c) $(A_1 \times A_2) \cup A_3 = \{(\varnothing, a), a, \{a\}\}$. (d) $(A_1 \cup A_2) \times A_3 = \{(\varnothing, a), (\varnothing, \{a\}), (a, a), (a, \{a\})\}$.

PROOF **Distributive law $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, by the element method with cases**
1. ($\subseteq$) Let $x \in A \cup (B \cap C)$. Then $x \in A$ or $x \in B \cap C$ *(definition of $\cup$)*.
   1.1. **Case $x \in A$:** then $x \in A \cup B$ and $x \in A \cup C$ *(definition of $\cup$)*, so $x \in (A \cup B) \cap (A \cup C)$.
   1.2. **Case $x \in B \cap C$:** then $x \in B$ and $x \in C$, so again $x \in A \cup B$ and $x \in A \cup C$.
2. ($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$, so $x \in A \cup B$ and $x \in A \cup C$.
   2.1. **Case $x \in A$:** then $x \in A \cup (B \cap C)$.
   2.2. **Case $x \notin A$:** then $x \in B$ and $x \in C$ *(as $x \in A \cup B$, $x \in A \cup C$)*, so $x \in B \cap C \subseteq A \cup (B \cap C)$.
3. Hence the two sets are equal *(definition of set equality)*. ■

PROOF **For all sets A, B: $A \subseteq B \leftrightarrow A \cup B = B$**
1. ($\rightarrow$) Suppose $A \subseteq B$. $B \subseteq A \cup B$ *(inclusion in union)*. Conversely let $x \in A \cup B$: if $x \in A$ then $x \in B$ *(as $A \subseteq B$)* if $x \in B$, done. So $A \cup B = B$.
2. ($\leftarrow$) Suppose $A \cup B = B$. Let $x \in A$; then $x \in A \cup B$ *(inclusion in union)* $= B$. So $A \subseteq B$. ■

PROOF **For every set A, $A \times \varnothing = \varnothing$ (proof by contradiction)**
1. Suppose not: some $z \in A \times \varnothing$. Then $z = (a, b)$ with $a \in A$ and $b \in \varnothing$ *(definition of $\times$)*.
2. But $\varnothing$ has no elements, a contradiction. So $A \times \varnothing$ has no elements: $A \times \varnothing = \varnothing$. ■

PROOF **$A \times (B \cup C) = (A \times B) \cup (A \times C)$ (chain of $\leftrightarrow$)**
$(x, y) \in A \times (B \cup C)$
$\leftrightarrow x \in A \wedge (y \in B \vee y \in C)$ *(definitions of $\times$, $\cup$)*
$\leftrightarrow (x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)$ *(distributive law of logic)*
$\leftrightarrow (x, y) \in (A \times B) \cup (A \times C)$ *(definitions of $\times$, $\cup$)*. ■

PROOF $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$ **(algebraic)**
$(A \cup B) \setminus C$
$= (A \cup B) \cap \overline{C}$ *(set difference law)*
$= \left(A \cap \overline{C}\right) \cup \left(B \cap \overline{C}\right)$ *(distributive, commutative)*
$= (A \setminus C) \cup (B \setminus C)$ *(set difference law).* ■

WORKED EXAMPLE **Disproving a set identity · "$(A \setminus B) \cup (B \setminus C) = A \setminus C$ for all sets"**
False. Counterexample: $A = \varnothing$, $B = \{1\}$, $C = \varnothing$. Then $(A \setminus B) \cup (B \setminus C) = \varnothing \cup \{1\} = \{1\}$, while $A \setminus C = \varnothing$. Write out every set and compute both sides.

DRILL **Nested sets and power sets (answers after the arrow)**
- $|\{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}| \to$ **3**
- $\mathcal{P}(\{\varnothing\}) \to \{\varnothing, \{\varnothing\}\}$
- $\{\varnothing\} \times \{\varnothing\} \to \{(\varnothing, \varnothing)\}$
- $\varnothing \times \{\varnothing\} \to \varnothing$
- $\mathcal{P}(\{a, b\}) \setminus \{\varnothing\} \to \{\{a\}, \{b\}, \{a, b\}\}$
- $\{1, \{1\}\} \cap \mathcal{P}(\{1\}) \to \{\{1\}\}$
- $|\mathcal{P}(A \times B)|$ with $|A| = 2$, $|B| = 3 \to 2^6 = 64$
- $\{x \in \mathbb{Z} : x^2 < 5\} \to \{-2, -1, 0, 1, 2\}$
- $\{2n + 1 : n \in \{0, 1, 2\}\} \to \{1, 3, 5\}$
- $\{\{1, 2\}, \{3\}\}$ a partition of $\{1, 2, 3\}$? → **yes**; $\{\{1, 2\}, \{2, 3\}\}$ → **no** (2 twice); $\{\{1, 2\}, \{3\}, \varnothing\}$ → **no** ($\varnothing$ component)

PROOF **If $A \subseteq B$ then $A \cap C \subseteq B \cap C$ and $A \cup C \subseteq B \cup C$**
1. Let $x \in A \cap C$: $x \in A$ and $x \in C$ *(definition of $\cap$)* $x \in B$ *(as $A \subseteq B$)* so $x \in B \cap C$.
2. Let $x \in A \cup C$: if $x \in A$ then $x \in B \subseteq B \cup C$; if $x \in C$ then $x \in B \cup C$ *(inclusion in union).* ■

Reading only (Lecture 5.3). Unrestricted comprehension breaks: Russell's $R = \{x : x \notin x\}$ gives $R \in R \leftrightarrow R \notin R$; a set of "everything" $\mathcal{D}$ contradicts $|\mathcal{P}(\mathcal{D})| > |\mathcal{D}|$ (Cantor). Zermelo (1908), later Fraenkel and Skolem, fixed this with the ZF axioms. Cantor, Russell, Zermelo and Fraenkel contributed to modern set theory; Gauss did not (AY25 Q1).

WORKED EXAMPLE **Power set of a set that contains $\varnothing$**
Let $A = \{\varnothing, \{\varnothing\}\}$. Find $\mathcal{P}(A)$ and $A \cap \mathcal{P}(A)$.
$\mathcal{P}(A) = \{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\}$, with four subsets since $|A| = 2$. For $A \cap \mathcal{P}(A)$ test each element of A: $\varnothing \in \mathcal{P}(A)$ (as $\varnothing \subseteq A$), and $\{\varnothing\} \in \mathcal{P}(A)$ (as $\{\varnothing\} \subseteq A$, since $\varnothing \in A$). So $A \cap \mathcal{P}(A) = \{\varnothing, \{\varnothing\}\} = A$, i.e. $A \subseteq \mathcal{P}(A)$.

Part V · Relations

19 · Relations and inverses

DEFINITION **Relation · Domain, co-domain, range**
A (binary) **relation from A to B** is a subset $R \subseteq A \times B$; xRy ("x is R-related to y") means $(x, y) \in R$, and $x \not R y$ means $(x, y) \notin R$. A **relation on A** is a relation from A to A, i.e. a subset of $A \times A = A^2$. Its **directed graph** draws each element of A once, with an arrow $x \to y$ whenever xRy and a loop at x when xRx.
$Dom(R) = \{a \in A : aRb \text{ for some } b \in B\}$, $coDom(R) = B$, $Range(R) = \{b \in B : aRb \text{ for some } a \in A\}$.
An **n-ary relation** on $A_1 \times \cdots \times A_n$ is a subset of it (binary, ternary, quaternary for $n = 2, 3, 4$). Relational databases are built on n-ary relations.

DEFINITION **Inverse relation**
$R^{-1} = \{(y, x) \in B \times A : (x, y) \in R\}$, i.e. $\forall x \in A, y \in B\,((y, x) \in R^{-1} \leftrightarrow (x, y) \in R)$. Reverse every arrow.

PAST PAPER **AY25 Q9, Q10 · Relations are sets of pairs**
Q9. $A = \{1, 2, 3\}$, $B = \{2, 5\}$. $S_2 = \mathcal{P}(A \times B)$ is **exactly** the set S_3 of relations from A to B; $S_1 = \mathcal{P}(A) \times \mathcal{P}(B)$ holds **pairs of sets**, a different kind of object. So $S_4 \subseteq S_3 \subseteq S_2$. Answer **A**.
Q10. The relation "$p \to q$" on $\{T, F\}$ is $R = \{(T, T), (F, T), (F, F)\}$. $R^{-1} = \{(T, T), (T, F), (F, F)\}$ is "$q \to p$", the **converse**. The converse is equivalent to the inverse, so the answer key accepted **B or D**.

20 · Composition of relations

DEFINITION **Composition (Lecture 6.1.4)**
For $R \subseteq A \times B$ and $S \subseteq B \times C$, the **composition** $S \circ R$ is the relation from A to C with
$$\forall x \in A, \forall z \in C\,(x(S \circ R)z \leftrightarrow \exists y \in B\,(xRy \wedge ySz)).$$
x and z are related iff there is a path $x \overset{R}{\to} y \overset{S}{\to} z$. $\boldsymbol{S \circ R}$ **applies** $\boldsymbol{R}$ **first.**

METHOD **Computing $S \circ R$ by hand**
For each pair $(x, y) \in R$, list every $(y, z) \in S$ that **starts where it ends**, and record (x, z). For $R \circ R \circ R$, compute $R_2 = R \circ R$ first, then $R \circ R_2$. Stop and re-check pairs that could join through two different middles.

PAST PAPER **AY25 Q17(a) (4 marks) · $S = \{(1,3),(3,6),(6,9),(9,12)\}$, $R = \{(6,6),(9,9)\}$**
$R \circ S$ (apply S then R): $(3,6)(6,6) \to (3,6)$; $(6,9)(9,9) \to (6,9)$. So $R \circ S = \{(3,6),(6,9)\}$.
$S \circ R$ (apply R then S): $(6,6)(6,9) \to (6,9)$; $(9,9)(9,12) \to (9,12)$. So $S \circ R = \{(6,9),(9,12)\}$.
(i) $(R \circ S) \setminus (S \circ R) = \{\mathbf{(3,6)}\}$.
(ii) $(R^{-1} \circ S^{-1}) \circ (S \circ R) = (S \circ R)^{-1} \circ (S \circ R)$; with $(S \circ R)^{-1} = \{(9,6),(12,9)\}$ this is $\{\mathbf{(6,6)},\mathbf{(9,9)}\}$.

PAST PAPER **AY24 Q11, Q17(e), Q12 · Powers and formula relations**
Q11. $R = \{(a,a),(a,b),(b,c),(c,a),(c,b)\}$ on $\{a,b,c\}$. $R \circ R$ is all of $A \times A$ except (b,c); then bRc and $c(R \circ R)c$ put (b,c) into $R \circ R \circ R$, which is $\boldsymbol{A \times A}$. Answer B.
Q17(e). $xR_1y \leftrightarrow y = 2x$, $xR_2y \leftrightarrow x = y^2$ on $\mathbb{Z}^+$. $x(R_1 \circ R_2)z \leftrightarrow \exists y(x = y^2 \wedge z = 2y)$, so $\boldsymbol{x = z^2/4}$ (with z even).
Q12. $|R \circ R|$ has no fixed relation to $|R|$: on $\{a,b\}$, $R = \{(a,a),(a,b),(b,a)\}$ gives $|R \circ R| = 4 > 3$; $\{(a,a),(a,b)\}$ gives $2 = 2$ but $\neq 2^2$; $\{(a,a)\}$ gives $1 = 1^2$. None of (i)–(iii) always holds. Answer **E**.

METHOD **Compositions and powers on the digraph (T4 Q4)**
$(x,z) \in S \circ R$ iff one R-arrow followed by one S-arrow leads from x to z. Hence (x,y) lies in $R \circ \cdots \circ R$ (k copies) iff the digraph of R has a walk of **exactly $\boldsymbol{k}$ arrows** from x to y; the same arrow or loop may be used more than once. For each starting vertex, trace where exactly k arrows can end. For larger k square first:
$R \circ R \circ R \circ R = (R \circ R) \circ (R \circ R)$ *(associativity)*.

PAST PAPER **AY23 Q11 · $|R \circ R \circ R \circ R|$ from a digraph on $\{1,2,3,4\}$**
Arrows: $1 \to 2$, $1 \to 3$, $3 \to 2$, $3 \to 4$, $4 \to 3$, so $R = \{(1,2),(1,3),(3,2),(3,4),(4,3)\}$; vertex 2 is a dead end.
Two arrows: $1 \to 3 \to 2$, $1 \to 3 \to 4$, $3 \to 4 \to 3$, $4 \to 3 \to 2$, $4 \to 3 \to 4$: $R \circ R$
$= \{(1,2),(1,4),(3,3),(4,2),(4,4)\}$.
Four arrows $= (R \circ R) \circ (R \circ R)$: from 1 only via 4, reaching 2 and 4; from 3 via 3, reaching 3; from 4 via 4, reaching 2 and 4. So $R \circ R \circ R \circ R = \{(1,2),(1,4),(3,3),(4,2),(4,4)\}$ and the answer is **5** (option B).

WORKED EXAMPLE **T4 Q4 · $R \circ R$ and $R \circ R \circ R$ read off the digraph**
$A = \{a,b,c\}$ with arrows $a \to b$, $b \to a$, $b \to c$, $c \to b$ and a loop at c: $R = \{(a,b),(b,a),(b,c),(c,b),(c,c)\}$.
(a) **Exactly two arrows:** from a reach a, c; from b reach b, c; from c reach a, b, c. $R \circ R$
$= \{(a,a),(a,c),(b,b),(b,c),(c,a),(c,b),(c,c)\}$. **Not transitive:** (a,c) and (c,b) are in it but (a,b) is not.
(b) **Exactly three arrows:** from a reach b, c; from b and from c reach everything. $R \circ R \circ R$
$= \{(a,b),(a,c),(b,a),(b,b),(b,c),(c,a),(c,b),(c,c)\}$. **Not transitive:** (a,c) and (c,a) are in it but (a,a) is not.
(c) $(R \circ R) \cup R = A \times A$, which is **transitive**.

PROOF $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$ **(Lecture 6, slide 18)**
Let $R \subseteq A \times B$ and $S \subseteq B \times C$. For all $z \in C$, $x \in A$: $(z,x) \in (S \circ R)^{-1} \leftrightarrow (x,z) \in S \circ R$ *(definition of inverse)*
$\leftrightarrow \exists y \in B\,(xRy \wedge ySz)$ *(definition of $\circ$)*
$\leftrightarrow \exists y \in B\,(zS^{-1}y \wedge yR^{-1}x)$ *(definition of inverse; commutative law)*
$\leftrightarrow (z,x) \in R^{-1} \circ S^{-1}$ *(definition of $\circ$)*. ■

PROOF **Proposition (Lecture 6, slide 18):** $(T \circ S) \circ R = T \circ (S \circ R)$
Let $R \subseteq A \times B$, $S \subseteq B \times C$, $T \subseteq C \times D$, $x \in A$ and $w \in D$.
$x((T \circ S) \circ R)w$
$\leftrightarrow \exists y(xRy \wedge \exists z(ySz \wedge zTw))$
$\leftrightarrow \exists z \exists y(xRy \wedge ySz \wedge zTw)$ *(z is not free in xRy; same-kind quantifiers commute; associative law)*
$\leftrightarrow \exists z(\exists y(xRy \wedge ySz) \wedge zTw)$ *(y is not free in zTw)*
$\leftrightarrow \exists z(x(S \circ R)z \wedge zTw) \leftrightarrow x(T \circ (S \circ R))w$. ■

PAST PAPER **T4 Q6 (AY23/24 exam) · Seven copies of an equivalence relation**
$R = \{(1,1),(1,2),(2,1),(2,2),(3,3)\}$ on $\{1,2,3\}$; find $R \circ R \circ \cdots \circ R$ (7 copies). R is an equivalence relation, so $R \circ R = R$ (AY23 Q12, §23). By associativity, $R^7 = ((R \circ R) \circ (R \circ R)) \circ ((R \circ R) \circ R) = (R \circ R) \circ (R \circ R)$ $= R \circ R = R$.

WORKED EXAMPLE **T4 Q9 · Inverse and composition by the element method**
$S = \{(m,n) \in \mathbb{Z}^2 : m^2 + n^2 \text{ is even}\}$. Determine (a) S^{-1}, (b) $S \circ S$, (c) $S \circ S^{-1}$.
(a) $S^{-1} = \{(x,y) \in \mathbb{Z}^2 : (y,x) \in S\}$ *(definition of inverse)*
$= \{(x,y) \in \mathbb{Z}^2 : y^2 + x^2 \text{ is even}\}$ *(definition of S)*
$= \{(x,y) \in \mathbb{Z}^2 : x^2 + y^2 \text{ is even}\}$ *(commutativity of $+$)*
$= S$.
(b) $S \circ S = S$:
1. ($\subseteq$) Let $(x,z) \in S \circ S$. Then $(x,y) \in S$ and $(y,z) \in S$ for some $y \in \mathbb{Z}$ *(definition of composition)*.
   1.1. So $x^2 + y^2$ and $y^2 + z^2$ are even *(definition of S)*, hence $x^2 + 2y^2 + z^2$ is even *(sum of two even integers)*.
   1.2. As $2y^2$ is even *(definition of even, closure)*, $x^2 + z^2$ is even *(difference of two even integers)*, so $(x,z) \in S$ *(definition of S)*.
2. ($\supseteq$) Let $(x,z) \in S$. Note $(x,x) \in S$, as $x^2 + x^2$ is even. Since $(x,x) \in S$ and $(x,z) \in S$, $(x,z) \in S \circ S$ *(definition of composition)*.
3. Therefore $S \circ S = S$. ■

(c) $S \circ S^{-1} = S \circ S = S$ by (a) and (b).

WORKED EXAMPLE **Lecture 6 · Composition as "going through" a middle set**
"takes" $\subseteq$ Students $\times$ Modules and "held in" $\subseteq$ Modules $\times$ Venues. Ann takes CS1010, CS1231, MA1101; Bryan takes CS1010, IS1103; Candy takes CS1231, IS1103; Danny takes nothing. CS1010 is held in LT15; CS1231 in ICube and SR1; IS1103 in SR1; CS2100 in LT15; MA1101 nowhere.
"held in $\circ$ takes" = "goes to": Ann $\to$ {LT15, ICube, SR1}; Bryan $\to$ {LT15, SR1}; Candy $\to$ {ICube, SR1}; Danny $\to$ nothing. CS2100 contributes no pair, since nobody takes it.

21 · Properties of relations on a set A

DEFINITION **Reflexive, symmetric, transitive (Lecture 6.2) · Antisymmetric (6.4) · Asymmetric (Tutorial 5)**

	R has the property iff	R fails it iff
Reflexive	$\forall x \in A\,(xRx)$	$\exists x \in A\,(x \not R x)$
Symmetric	$\forall x,y \in A\,(xRy \to yRx)$	$\exists x,y\,(xRy \wedge y \not R x)$
Transitive	$\forall x,y,z \in A\,(xRy \wedge yRz \to xRz)$	$\exists x,y,z\,(xRy \wedge yRz \wedge x \not R z)$
Antisymmetric	$\forall x,y \in A\,(xRy \wedge yRx \to x = y)$	$\exists x,y\,(xRy \wedge yRx \wedge x \neq y)$
Asymmetric	$\forall x,y \in A\,(xRy \to y \not R x)$	$\exists x,y\,(xRy \wedge yRx)$

THEOREM **Useful facts**
- R symmetric $\Leftrightarrow R = R^{-1}$ (Tutorial 4).
- R transitive $\Leftrightarrow R \circ R \subseteq R$.
- R reflexive $\Rightarrow R \subseteq R \circ R$.
- Asymmetric $\Rightarrow$ antisymmetric (and irreflexive).
- **Antisymmetric is not "not symmetric":** any $R \subseteq \{(x,x) : x \in A\}$ is both.
- "$=$" is an equivalence relation **and** a partial order (AY23 Q10, AY24 Q9).
- The empty relation on $A \neq \varnothing$ is symmetric, transitive, antisymmetric and asymmetric, but not reflexive.
- If R, S **both** have the property: reflexivity survives $\cup, \cap, \circ$; symmetry survives $\cup, \cap$ but not $\circ$; transitivity and antisymmetry survive only $\cap$.
- Write "0 is related to itself", not "0 is reflexive".

THEOREM **More facts for MCQs (T5 D1–D4, T4 Q3, T4 Q6)**

- **Alternative definition of antisymmetry (T5 D4):** $\forall x, y \in A\,(x \neq y \to (xRy \to y\not{R}x))$. It is the usual definition rewritten: $(xRy \wedge yRx) \to x = y \equiv x \neq y \to \sim(xRy \wedge yRx)$ *(contrapositive)* $\equiv x \neq y \to (x\not{R}y \vee y\not{R}x)$ *(De Morgan's law)* $\equiv x \neq y \to (xRy \to y\not{R}x)$ *(implication law)*. Distinct elements are never related in both directions.
- **Symmetric and antisymmetric** $\Rightarrow R \subseteq \{(x, x) : x \in A\}$: if xRy then yRx *(symmetry)*, so $x = y$ *(antisymmetry)*. If R is also reflexive, R is exactly the identity relation "=" (T5 D2): the only relation on A that is both an equivalence relation and a partial order.
- **Symmetric and asymmetric** (T5 D3): only $R = \varnothing$, since one pair xRy would give both yRx and $y\not{R}x$.
- **Empty relation (T5 D1):** on $A \neq \varnothing$ it is symmetric, transitive, antisymmetric and asymmetric (each vacuously) but not reflexive. On $A = \varnothing$ it is the only relation ($A \times A = \varnothing$) and is reflexive too, so it is an equivalence relation and a partial order. This is why questions say "on a non-empty set" (AY23 Q10).
- **Size of an equivalence relation:** if the classes are $S_1, ..., S_k$, then $R = (S_1 \times S_1) \cup \cdots \cup (S_k \times S_k)$ and $|R| = |S_1|^2 + \cdots + |S_k|^2$. T4 Q6: classes $\{1, 2\}$ and $\{3\}$ give $|R| = 2^2 + 1^2 = 5$. On $\{1, 2, 3\}$ the five equivalence relations (AY25 Q4) have 3, 5, 5, 5 and 9 pairs.
- **T4 Q3 on $\mathbb{Q}$:** $xy \geq 0$ is reflexive and symmetric but not transitive $(1, 0, -1)$; $xy > 0$ is symmetric and transitive but **not reflexive** $(0 \cdot 0 = 0)$. Neither is an equivalence relation on $\mathbb{Q}$; removing 0 repairs both (T4 Q10, §23).

METHOD **Proof skeletons for the properties**

Reflexive: "Let $x \in A$. … so xRx." **Symmetric:** "Let $x, y \in A$ with xRy. … so yRx."
Transitive: "Let $x, y, z \in A$ with xRy and yRz. … so xRz." **Antisymmetric:** "Let xRy and yRx. … so $x = y$."
Disproof: one explicit counterexample, named elements, each membership checked.

PAST PAPER **AY23 Q9 and Q10 · Counterexamples on small sets**

Q9 (R reflexive, S symmetric). On $\{a, b, c\}$ take $R = \{(a, a), (b, b), (c, c), (a, c)\}$, $S = \{(a, b), (b, a)\}$: $R \circ S = \{(a, b), (b, a), (b, c)\}$ is not symmetric; $R \cup S$ is not symmetric; $(R \circ S)^{-1}$ is not transitive. Only "$R \cup S$ is reflexive" survives: **A**.
Q10. Asymmetric $\Rightarrow$ antisymmetric: true. "Not reflexive $\Rightarrow$ not asymmetric": false ($R = \{(a, b)\}$). "Not reflexive $\Rightarrow$ asymmetric": false ($R = \{(a, a)\}$ on $\{a, b\}$). "No relation is both an equivalence relation and a partial order": false (=). Only (i): **A**.

PAST PAPER **AY24 Q10, AY25 Q11, AY25 Q14**

AY24 Q10. $pRq \leftrightarrow (p \to q)$ on $\{T, F\}$: $R = \{(T, T), (F, F), (F, T)\}$ is reflexive, antisymmetric, transitive, not symmetric: **D**.
AY25 Q11. $ARB \leftrightarrow |A \cap B| = 1$ on $\mathcal{P}(\mathbb{Z})$: symmetric; not reflexive ($|\varnothing \cap \varnothing| = 0$); not transitive. **Symmetric but not an equivalence relation.** Answer B.
AY25 Q14. Reflexive R on $\{1, 2, 3, 4\}$ with $(xRy \wedge xRz) \to y = z$: reflexivity forces xRx, so each x is related to **nothing else**. Only the identity relation qualifies, so $|\mathcal{F}| = 1$. Answer B.

PROOF **R is symmetric $\leftrightarrow R = R^{-1}$ (Tutorial 4)**

1. ($\to$) Suppose R symmetric. $(x, y) \in R \to (y, x) \in R$ *(symmetry)* $\to (x, y) \in R^{-1}$ *(definition of inverse)*, so $R \subseteq R^{-1}$. Conversely $(x, y) \in R^{-1} \to (y, x) \in R \to (x, y) \in R$ *(symmetry)*, so $R^{-1} \subseteq R$.
2. ($\leftarrow$) Suppose $R = R^{-1}$ and xRy. Then $(y, x) \in R^{-1} = R$, i.e. yRx. ■

PROOF **R is transitive $\leftrightarrow R \circ R \subseteq R$ (AY23 Q12 model answer, extended)**

1. ($\to$) Let $(x, z) \in R \circ R$. There is some y with xRy and yRz *(definition of composition)* so xRz *(transitivity)*, i.e. $(x, z) \in R$.
2. ($\leftarrow$) Let xRy and yRz. Then $(x, z) \in R \circ R$ *(definition of composition)* $\subseteq R$, so xRz. ■

Similarly, **reflexive $\Rightarrow R \subseteq R \circ R$**: given $(x, z) \in R$, use xRx and xRz with middle $y = x$.

WORKED EXAMPLE **T5 Q7 · Asymmetric vs antisymmetric**

On $A = \{a, b, c\}$ find a relation that is (a) asymmetric and antisymmetric; (b) antisymmetric but not asymmetric; (c) asymmetric but not antisymmetric; (d) neither.
(a) $\{(a, b), (a, c)\}$. (b) $\{(a, a), (b, c)\}$. (c) **No solution:** every asymmetric relation is antisymmetric (below). (d) $\{(a, b), (b, a), (a, c)\}$.

1. Let R be an asymmetric relation on A: $\forall x, y \in A\,(xRy \to y\not{R}x)$ *(definition of asymmetry)*.
   1.1. $\equiv \forall x, y \in A\,(x\not{R}y \vee y\not{R}x)$ *(implication law)*.
   1.2. $\Rightarrow \forall x, y \in A\,((x\not{R}y \vee y\not{R}x) \vee x = y)$ *(generalization)*.

1.3. $\equiv \forall x, y \in A\,(\sim(xRy \wedge yRx) \vee x = y)$ *(De Morgan's law)*.
1.4. $\equiv \forall x, y \in A\,(xRy \wedge yRx \to x = y)$ *(implication law)*.
2. The final formula is the definition of antisymmetry, so R is antisymmetric: asymmetry makes it vacuously true. ■

22 · Closures

DEFINITION **Transitive closure (Lecture 6.2.2); reflexive and symmetric closures**
The **transitive closure** of R on A is the relation R^t such that (1) R^t is transitive, (2) $R \subseteq R^t$, (3) $R^t \subseteq S$ for every transitive $S \supseteq R$. It is the **smallest** transitive relation containing R. Likewise the **reflexive closure** $R^r = R \cup \{(x, x) : x \in A\}$ and the **symmetric closure** $R^s = R \cup R^{-1}$.

METHOD **Computing R^t**
$R^t = R \cup R^2 \cup R^3 \cup \cdots$ where $R^2 = R \circ R$; on a finite set stop once a power adds nothing new (at most $|A|$ powers). On the digraph: draw $x \to z$ whenever some path leads from x to z. A cycle back to the starting point creates a loop.

PAST PAPER **AY24 Q17(a), (b) (5 marks)**
(a) $R = \{(a, b), (a, c)\}$ on $\{a, b, c\}$: $|R^r| = 2 + 3 = \mathbf{5}$; $|R^s| = |\{(a, b), (a, c), (b, a), (c, a)\}| = \mathbf{4}$; $|R^t| = \mathbf{2}$, since no pair ends where another begins, so R is already transitive.
(b) $S = \{(a, b), (b, a), (b, d), (d, c), (b, b), (d, d)\}$; $T = S^t \setminus S$ consists of the new pairs obtained from paths: $a \to b \to a$ gives (a, a); $a \to b \to d$ gives (a, d); $a \to b \to d \to c$ gives (a, c); $b \to d \to c$ gives (b, c). $\boldsymbol{T = \{(a, a), (a, c), (a, d), (b, c)\}}$.

WORKED EXAMPLE **T5 Q6 · Reflexive closure (the closure-proof template)**
Let R be a relation on a non-empty set A, and define S on A by $xSy \leftrightarrow ((x = y) \vee (xRy))$. Show: (a) S is reflexive; (b) $R \subseteq S$; (c) if S' is a reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$. What is S called?
(a) Let $x \in A$. $x = x$, so xSx *(definition of S)*. Therefore S is reflexive.
(b) Suppose $(x, y) \in R$. Then xSy *(definition of S)*, so $(x, y) \in S$. Therefore $R \subseteq S$ *(definition of $\subseteq$)*.
(c) Let $(x, y) \in S$: $x = y$ or xRy *(definition of S)*.
1. **Case 1 ($x = y$):** $xS'y$ *(S' is reflexive)*, so $(x, y) \in S'$.
2. **Case 2 (xRy):** $(x, y) \in R \subseteq S'$, so $(x, y) \in S'$.

In all cases $(x, y) \in S'$. Therefore $S \subseteq S'$. ■
S is the **reflexive closure** of R: the smallest reflexive relation on A containing R. Parts (a)–(c) mirror the three conditions defining a closure.

WORKED EXAMPLE **Composition and closure**
$A = \{1, 2, 3, 4\}$, $R = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$. Find $R \circ R$, $R^{-1} \circ R$ and $|R^t|$.
$R \circ R = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$ (two steps round the 4-cycle). $R^{-1} \circ R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$: follow an arrow of R, then return along the same arrow. From any vertex a path reaches every vertex, itself included, so $R^t = A \times A$ and $|R^t| = 16$.

WORKED EXAMPLE **All the closures**
$A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 3)\}$. Find R^r, R^s, R^t and the smallest equivalence relation containing R.
$R^r = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\}$ (5 pairs); $R^s = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$ (4); $R^t = \{(1, 2), (2, 3), (1, 3)\}$ (3).
The smallest equivalence relation containing R is $A \times A$ (9 pairs): ignoring directions, 1–2–3 is one connected piece, so there is a single class.

THEOREM **Closure facts**
The transitive closure of a **symmetric** relation is symmetric, and of a **reflexive** relation is reflexive (reverse or extend the paths). Hence the **smallest equivalence relation containing R** is $(R \cup R^{-1} \cup \{(x, x) : x \in A\})^t$. Its classes are the connected components of the digraph of R when arrow directions are ignored.

23 · Equivalence relations

DEFINITION **Relation induced by a partition · Equivalence relation**
Given a partition $\mathcal{C}$ of A, the **induced relation** is $xRy \leftrightarrow x, y$ lie in the same component; it is reflexive, symmetric and transitive (Theorem 8.3.1).
R on A is an **equivalence relation** iff it is **reflexive, symmetric and transitive**. Notation: $\sim$.

DEFINITION **Equivalence class · Quotient $A/\sim$**
For an equivalence relation $\sim$ on A and $a \in A$, the **equivalence class** of a (the **class of a**) is $[a]_\sim = \{x \in A : a \sim x\}$; procedurally, $\forall x \in A\,(x \in [a]_\sim \leftrightarrow a \sim x)$.
$A/\sim = \{[x]_\sim : x \in A\}$, the set of all equivalence classes, read "the quotient of A by $\sim$". Its **elements** are the equivalence classes.

DEFINITION **Congruence modulo n (Lecture 6.3.4)**
Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$. a is **congruent to b modulo n**, written $a \equiv b \pmod n$, iff $a - b = nk$ for some $k \in \mathbb{Z}$; in other words, $n \mid (a-b)$.
E.g. $7 \equiv 1 \pmod 2$; $-3 \equiv 12 \pmod 5$; $-4 \not\equiv 5 \pmod 7$. The classes are $[x] = \{x + nk : k \in \mathbb{Z}\}$, and $\mathbb{Z}/\sim_n = \{[0], [1], ..., [n-1]\}$.

PROOF **Lemma Rel.1 (Lecture 6)**
Let $\sim$ be an equivalence relation on A. For $x, y \in A$, these are equivalent: (i) $x \sim y$; (ii) $[x] = [y]$; (iii) $[x] \cap [y] \neq \varnothing$.
1. ((i) $\Rightarrow$ (ii)) Suppose $x \sim y$; then $y \sim x$ *(symmetry)*.
   1.1. For every $z \in [x]$: $x \sim z$ *(definition of $[x]$)*, so $y \sim z$ *(transitivity, as $y \sim x$)*, so $z \in [y]$. Thus $[x] \subseteq [y]$.
   1.2. Symmetrically $[y] \subseteq [x]$, so $[x] = [y]$.
2. ((ii) $\Rightarrow$ (iii)) Suppose $[x] = [y]$. $x \sim x$ *(reflexivity)*, so $x \in [x] = [x] \cap [y]$ *(idempotent law)* hence $[x] \cap [y] \neq \varnothing$.
3. ((iii) $\Rightarrow$ (i)) Take $z \in [x] \cap [y]$. Then $x \sim z$ and $y \sim z$ *(definitions of $[x], [y]$)* so $z \sim y$ *(symmetry)* and $x \sim y$ *(transitivity)*. ■

PROOF **Proposition (Lecture 6, slide 54) · Congruence mod n is an equivalence relation on $\mathbb{Z}$ ($a \equiv b \pmod n \leftrightarrow n \mid (a-b)$)**
1. (Reflexive) For all $a \in \mathbb{Z}$: $a - a = 0 = n \cdot 0$, so $a \equiv a \pmod n$ *(definition of congruence)*.
2. (Symmetric) Let $a \equiv b \pmod n$: $a - b = nk$ for some $k \in \mathbb{Z}$. Then $b - a = n(-k)$ with $-k \in \mathbb{Z}$ *(closure)*, so $b \equiv a \pmod n$.
3. (Transitive) Let $a \equiv b$ and $b \equiv c \pmod n$: $a - b = nk$, $b - c = nl$. Then $a - c = n(k+l)$ with $k + l \in \mathbb{Z}$, so $a \equiv c \pmod n$. ■

PAST PAPER **AY23 Q19 (6 marks) · $aSb \leftrightarrow \exists k \in \mathbb{Z}\,(a - b = k/2)$ on $\mathbb{Q}$**
(a) Prove S transitive (official model answer):
1. Let $a, b, c \in \mathbb{Q}$ such that aSb and bSc.
2. There exist $m, n \in \mathbb{Z}$ such that $a - b = m/2$ and $b - c = n/2$ *(by definition of S)*.
3. Now $a - c = (a-b) + (b-c) = m/2 + n/2 = (m+n)/2$ *(by basic algebra)*.
4. $m + n \in \mathbb{Z}$ *(by closure of integers under addition)*.
5. Therefore aSc *(by definition of S)*. ■

(b) Two distinct classes: $[0] = \{k/2 : k \in \mathbb{Z}\}$ and $[0.1]$. **(c)** $\mathbb{Q}/S = \{[x] : x \in \mathbb{Q} \wedge 0 \leq x < 1/2\}$. Each class has exactly one representative in $[0, 1/2)$, so there are no duplicates.

PAST PAPER **AY25 Q17(c), (d), Q15 · AY24 Q9 · AY23 Q12**
AY25 Q17(c). "Same biological father **and** mother": **yes**, an equivalence relation. "Share **a** biological parent": **no**, because it is not transitive (half-siblings).
AY25 Q17(d). $x \sim y \leftrightarrow \exists d \in \mathbb{Z}^+(d \mid x \wedge d \mid y)$: $d = 1$ always works, so everything is related: $\mathbb{Z}^+/\sim = \{\mathbb{Z}^+\} = \{[1]\}$.
AY25 Q15. "$\forall x \in A\,([x] \subseteq A/\sim)$" is **false**: classes are **elements** of $A/\sim$, not subsets.
"$\exists x \in A/\sim \forall y \in x\,(y \in A)$" and "$|A| = \sum_{x \in A/\sim} |x|$" are true: **E**.
AY24 Q9. Divisibility on 10 primes is the identity relation: 10 classes **and** 10 minimal elements. Answer **C**.
AY23 Q12. For an equivalence relation R, all four hold: $R = R^{-1}$ (symmetry), so $R^{-1} \circ R = R \circ R^{-1} = R \circ R$; $R \circ R \subseteq R$ (transitivity: $xRy \wedge yRz \Rightarrow xRz$); $R \subseteq R \circ R$ (reflexivity: take the middle $y = x$). Hence $R \circ R^{-1} = R$: **D**.

PROOF **Theorem Rel.2 · Theorem 8.3.4 · If $\sim$ is an equivalence relation on A, then $A/\sim$ is a partition of A (Lecture 6)**

1. $A/\sim$ is a set *(by definition)*.
2. Every element is a non-empty subset of A: $[x] = \{y \in A : x \sim y\} \subseteq A$, and $x \in [x]$ *(reflexivity)*.
3. Every $x \in A$ is in exactly one class:
   3.1. $x \in [x]$, so x is in at least one class.
   3.2. If $x \in [y]$ and $x \in [z]$, then $[y] \cap [z] \neq \varnothing$, so $[y] = [z]$ *(Lemma Rel.1, (iii) ⇒ (ii))*. ■

PROOF **Theorem 8.3.1 · The relation induced by a partition $\mathcal{C}$ of A is an equivalence relation**

1. (Reflexive) Let $x \in A$. x lies in some component S *(definition of partition)*, so $x, x \in S$ and xRx.
2. (Symmetric) If xRy, then x, y lie in one component S, so $y, x \in S$ and yRx.
3. (Transitive) Let xRy and yRz: $x, y \in S_1$ and $y, z \in S_2$ for components S_1, S_2. Then $y \in S_1 \cap S_2$, so $S_1 = S_2$ *(each element lies in exactly one component)*, hence $x, z \in S_1$ and xRz. ■

Together with Theorem Rel.2, this gives a one-to-one correspondence between partitions and equivalence relations on A. Counts: §55.

PROOF **On $A = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, $(a,b) \sim (c,d) \leftrightarrow ad = bc$ is an equivalence relation**

1. (Reflexive) $ab = ba$ *(commutativity)*, so $(a,b) \sim (a,b)$.
2. (Symmetric) If $ad = bc$ then $cb = da$ *(commutativity)*, so $(c,d) \sim (a,b)$.
3. (Transitive) Let $ad = bc$ and $cf = de$, with $b, d, f \neq 0$.
   3.1. Multiply: $adf = bcf = bde$, so $d(af - be) = 0$ *(basic algebra)*.
   3.2. $d \neq 0$, so $af = be$ *(zero product property)*, i.e. $(a,b) \sim (e,f)$. ■

Each class is one rational number: $[(1,2)] = \{(1,2), (2,4), (-3,-6), ...\}$; $A/\sim$ "is" $\mathbb{Q}$.

METHOD **Writing $A/\sim$ without duplicates**

Pick **one representative per class** from a region that meets every class exactly once, then write $\{[x] : x \in$ that region$\}$.

- Congruence mod 3: $\mathbb{Z}/\equiv_3 = \{[0], [1], [2]\}$, with $[r] = \{3k + r : k \in \mathbb{Z}\}$.
- AY23 Q19: $\mathbb{Q}/S = \{[x] : x \in \mathbb{Q} \wedge 0 \leq x < 1/2\}$.
- Lecture Ex. 15: nonempty $A \subseteq \{1,2,3\}$, related iff they have the same least element: $[\{1\}]$ $= \{\{1\}, \{1,2\}, \{1,3\}, \{1,2,3\}\}$, $[\{2\}] = \{\{2\}, \{2,3\}\}$, $[\{3\}] = \{\{3\}\}$, three classes.

PAST PAPER **T4 Q8 (AY20/21 exam) · Classes of ordered pairs**

On $\mathbb{Z}^+ \times \mathbb{Z}^+$, $(a,b) \sim (c,d) \leftrightarrow ab = cd$. Write the set-builder line first: $[(1,1)]$
$= \{(x,y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : (1,1) \sim (x,y)\}$ *(definition of class)*
$= \{(x,y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 1\}$ *(definition of $\sim$)*
$= \{(1,1)\}$. Likewise $[(4,3)] = \{(x,y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 12\} = \{(1,12), (2,6), (3,4), (4,3), (6,2), (12,1)\}$.

WORKED EXAMPLE **T4 Q10 · Equivalence relation using Appendix A**

On $\mathbb{Z} \setminus \{0\}$ let $a \sim b \leftrightarrow ab > 0$. (a) Prove $\sim$ is an equivalence relation. (b) Determine all distinct equivalence classes.

1. (Reflexivity) Let $a \in \mathbb{Z} \setminus \{0\}$. Since $a \neq 0$, $a^2 > 0$ *(T21)*. Thus $a \sim a$ *(definition of $\sim$)*.
2. (Symmetry) Let $a, b \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$. Then $ab > 0$ *(definition of $\sim$)*, so $ba > 0$ *(commutative law of $\times$)*, so $b \sim a$.
3. (Transitivity) Let $a, b, c \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$ and $b \sim c$. Then $ab > 0$ and $bc > 0$ *(definition of $\sim$)*.
   3.1. Multiplying, $ab^2c > 0$ *(Ord1)*, so $(ac)b^2 > 0$ *(associative, commutative laws of $\times$)*.
   3.2. So ac and b^2 are both positive or both negative *(T25)*.
   3.3. Since $b^2 > 0$ *(T21, as $b \neq 0$)*, $ac > 0$. Thus $a \sim c$.
4. Therefore $\sim$ is an equivalence relation. ■

(b) T25 rules out pairs of opposite signs; the product of two positive or two negative integers is positive. The classes are $[1] = \mathbb{Z}^+$ and $[-1] = \mathbb{Z}^-$, so $(\mathbb{Z} \setminus \{0\})/\sim = \{\mathbb{Z}^+, \mathbb{Z}^-\}$.

DEFINITION **Representative · Well-defined operation on $A/\sim$**

Any $b \in [a]$ is a **representative** of $[a]$, since then $[b] = [a]$: mod 5, $[2] = [7] = [12]$.

Let $\sim$ be an equivalence relation on A and $*$ a binary operation on A (so $a * b \in A$ for all $a, b \in A$). The induced rule $[a] * [b] = [a * b]$ is **well defined** iff the answer depends only on the classes, not on the representatives chosen: $a \sim a' \wedge b \sim b' \Rightarrow a * b \sim a' * b'$.

PROOF **$+$ and $\cdot$ are well defined on the classes of congruence mod n**

1. Fix $n \in \mathbb{Z}^+$ and let $a, a', b, b' \in \mathbb{Z}$ with $a \equiv a' \pmod n$ and $b \equiv b' \pmod n$: $a - a' = nk$ and $b - b' = nm$ for some $k, m \in \mathbb{Z}$ *(definition of congruence)*.
2. (Addition) $(a+b) - (a'+b') = n(k+m)$ with $k+m \in \mathbb{Z}$ *(closure)*, so $a + b \equiv a' + b' \pmod n$.
3. (Multiplication) $ab - a'b' = (a-a')b + a'(b-b') = n(kb + a'm)$ with $kb + a'm \in \mathbb{Z}$ *(closure)*, so $ab \equiv a'b' \pmod n$. ■

E.g. mod 5: $[2] \cdot [4] = [8] = [3]$ and $[7] \cdot [9] = [63] = [3]$.

TRAP **Check reflexive, symmetric, transitive first · then check well-definedness**

- $a \sim b \leftrightarrow ab \geq 0$ is reflexive and symmetric, but **not** transitive on either $\mathbb{Z}$ or $\mathbb{R}$ ($1 \sim 0$ and $0 \sim -1$, but $1 \nsim -1$), so it does not give a partition into equivalence classes. On $\mathbb{Z} \setminus \{0\}$ its classes are $\mathbb{Z}^+$ and $\mathbb{Z}^-$ (§23, T4 Q10). On $\mathbb{R} \setminus \{0\}$ it means "same sign": $(\mathbb{R} \setminus \{0\})/\sim = \{[1], [-1]\} = \{(0, \infty), (-\infty, 0)\}$.
- $a \sim b \leftrightarrow a - b = 2k$ for some $k \in \mathbb{Z}$, on $\mathbb{R}$: $[a] = \{a + 2k : k \in \mathbb{Z}\}$ and $\mathbb{R}/\sim = \{[r] : 0 \leq r < 2\}$ (infinitely many classes). $+$ is well defined, but $\cdot$ is **not**: $[0.5] = [2.5]$, yet $[0.5] \cdot [0.5] = [0.25] \neq [1.25] = [2.5] \cdot [0.5]$, since $1.25 - 0.25 = 1$ is not of the form $2k$.

WORKED EXAMPLE **Equivalence relation and quotient**

On $\mathbb{Z}$ let $xRy \leftrightarrow 3 \mid (x^2 - y^2)$. Prove R is an equivalence relation and find $\mathbb{Z}/R$.

1. (Reflexive) $x^2 - x^2 = 0 = 3 \cdot 0$, so xRx *(definition of divisibility)*.
2. (Symmetric) If $x^2 - y^2 = 3k$ then $y^2 - x^2 = 3(-k)$ with $-k \in \mathbb{Z}$, so yRx.
3. (Transitive) If $x^2 - y^2 = 3k$ and $y^2 - z^2 = 3l$, then $x^2 - z^2 = 3(k+l)$, so xRz.
4. (Classes) By the QR theorem $x = 3q + r$ with $r \in \{0, 1, 2\}$; $x^2 = 9q^2 + 6qr + r^2$, so $x^2 \bmod 3 = 0$ if $r = 0$ and 1 if $r = 1, 2$. Hence $\mathbb{Z}/R = \{[0], [1]\} = \{\{3k : k \in \mathbb{Z}\}, \{n \in \mathbb{Z} : 3 \nmid n\}\}$. ■

WORKED EXAMPLE **Classes of a relation on pairs**

On $\mathbb{Z} \times \mathbb{Z}$ let $(a, b) \sim (c, d) \leftrightarrow a + d = b + c$. Show $\sim$ is an equivalence relation and describe $(\mathbb{Z} \times \mathbb{Z})/\sim$.

1. Rewrite: $(a, b) \sim (c, d) \leftrightarrow a - b = c - d$ *(basic algebra)*. Then reflexivity, symmetry and transitivity are those of $=$ on $\mathbb{Z}$ (e.g. $a - b = c - d$ and $c - d = e - f$ give $a - b = e - f$).
2. So $[(a, b)] = \{(c, d) : c - d = a - b\}$, a diagonal line of lattice points, and each class contains exactly one point $(k, 0)$ with $k = a - b$.
3. $(\mathbb{Z} \times \mathbb{Z})/\sim = \{[(k, 0)] : k \in \mathbb{Z}\}$, with one class for each integer and no duplicates. ■

24 · Partial orders

DEFINITION **Partial order (Lecture 6.4.2) · Poset**

R on A is a **partial order** iff it is **reflexive, antisymmetric and transitive**; then (A, R) is a **partially ordered set** (poset). A general partial order is written $\preccurlyeq$ ("curly less than or equal to"). Examples: $\leq$ on $\mathbb{R}$; $\subseteq$ on any set of sets; $\mid$ on $\mathbb{Z}^+$. Non-examples: $<$ (not reflexive); congruence mod n (not antisymmetric); $\mid$ on $\mathbb{Z}$ ($2 \mid -2$ and $-2 \mid 2$, but $2 \neq -2$).

TRAP **T5 Q1 · Disprove with one named counterexample**

On strings over $\{s, u\}$, $aRb \leftrightarrow \text{len}(a) \leq \text{len}(b)$ is **not** a partial order: sRu and uRs but $s \neq u$, so R is not antisymmetric. One counterexample suffices; there is no need to check reflexivity or transitivity.

PROOF **$\mid$ is a partial order on any set $A \subseteq \mathbb{Z}^+$ (Lecture 6, Ex. 20)**

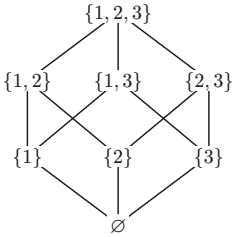
1. (Reflexive) Let $a \in A$. $a = 1 \cdot a$, so $a \mid a$ *(by definition of divisibility)*.
2. (Antisymmetric) Let $a, b \in A$ with $a \mid b$ and $b \mid a$. Then $a \leq b$ and $b \leq a$ *(by Theorem 4.3.1, as $a, b \in \mathbb{Z}^+$)*, so $a = b$.
3. (Transitive) Let $a \mid b$ and $b \mid c$. Then $a \mid c$ *(by Theorem 4.3.3)*. ■

DEFINITION **Hasse diagram (Lecture 6.4.3)**

For a partial order $\preccurlyeq$ on a finite A and distinct x, y: **if $x \preccurlyeq y$ and no $m \in A$ has $x \preccurlyeq m \preccurlyeq y$ (with $m \neq x, y$), draw x below y joined by a line; otherwise no line.**

From the digraph: place arrows pointing up; delete (1) loops, (2) arrows implied by transitivity, (3) arrowheads.

Reading it: $x \preccurlyeq y$ iff you can climb from x to y along lines.



DEFINITION **Comparable · Compatible (AY24 paper) · Chain, antichain (AY24 paper) · Total order**

a, b are **comparable** iff $a \preccurlyeq b$ or $b \preccurlyeq a$; otherwise **noncomparable**. a, b are **compatible** iff $\exists c \in A\,(a \preccurlyeq c \wedge b \preccurlyeq c)$.

A **chain** is a subset C in which every two elements are comparable; a **maximal chain** M has no $t \notin M$ with $M \cup \{t\}$ a chain; the **length** of a chain is one less than its number of elements. An **antichain** is a subset in which no two distinct elements are comparable.

$\preccurlyeq$ is a **total (linear) order** iff it is a partial order and $\forall x, y \in A\,(x \preccurlyeq y \vee y \preccurlyeq x)$; its Hasse diagram is a single chain. $\mid$ on $\mathbb{Z}^+$ is partial, not total ($3 \nmid 5$, $5 \nmid 3$); $\leq$ on $\mathbb{Q}$ is total.

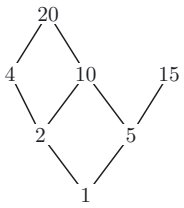
WORKED EXAMPLE **T5 Q8 · Comparable versus compatible on $(\{1, 2, 4, 5, 10, 15, 20\}, \mid)$**

Comparable (one divides the other), 14 pairs: $\{1,2\}$, $\{1,4\}$, $\{1,5\}$, $\{1,10\}$, $\{1,15\}$, $\{1,20\}$, $\{2,4\}$, $\{2,10\}$, $\{2,20\}$, $\{4,20\}$, $\{5,10\}$, $\{5,15\}$, $\{5,20\}$, $\{10,20\}$.

Compatible (a common upper bound in A), 17 pairs: those 14 and $\{2,5\}$ (both divide 10), $\{4,5\}$, $\{4,10\}$ (both divide 20).

Not compatible: $\{2,15\}$, $\{4,15\}$, $\{10,15\}$, $\{15,20\}$; nothing in A lies above 15 except 15 itself.

Comparable ⇒ compatible: if $a \preccurlyeq b$ take $c = b$; then $a \preccurlyeq b$ and $b \preccurlyeq b$ *(reflexivity)*. **The converse fails:** $\{2,5\}$ is compatible but not comparable.



DEFINITION **Maximal, minimal, largest, smallest (Lecture 6.4.5)**

For a poset $(A, \preccurlyeq)$ and $c \in A$:

- c **maximal** $\Leftrightarrow \forall x \in A\,(c \preccurlyeq x \to c = x)$: nothing lies strictly above c. Equivalently, every $x \in A$ has $x \preccurlyeq c$ or is not comparable with c.
- c **minimal** $\Leftrightarrow \forall x \in A\,(x \preccurlyeq c \to c = x)$: nothing lies strictly below c. Equivalently, every $x \in A$ has $c \preccurlyeq x$ or is not comparable with c.
- c **largest** $\Leftrightarrow \forall x \in A\,(x \preccurlyeq c)$: every element lies below c.
- c **smallest** $\Leftrightarrow \forall x \in A\,(c \preccurlyeq x)$: every element lies above c.

Largest = greatest = maximum; smallest = least = minimum. On a Hasse diagram: maximal = no line going up; minimal = no line going down.

DEFINITION **Well-ordered set (Lecture 6.4.7)**

Let $\preccurlyeq$ be a total order on a set A. A is **well-ordered** iff every non-empty subset of A contains a smallest element. Symbolically,

$$\forall S \in \mathcal{P}(A)\,(S \neq \varnothing \to \exists x \in S\,\forall y \in S\,(x \preccurlyeq y)).$$

$(\mathbb{N}, \leq)$ is well-ordered. $(\mathbb{Z}, \leq)$ is not: $\mathbb{Z}$ itself (or $\mathbb{Z}^-$) is a non-empty subset with no smallest element.

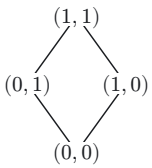
THEOREM **Extremal-element facts**

- **Proposition (Lecture 6, slide 83):** smallest ⇒ minimal; largest ⇒ maximal. If c is smallest and $x \preccurlyeq c$, then also $c \preccurlyeq x$, so $x = c$ by antisymmetry.
- A largest (smallest) element, if it exists, is **unique** (two largest elements precede each other, hence are equal by antisymmetry), and is then the **only** maximal (minimal) element.
- Conversely, in a **finite** poset a unique maximal element is largest. This fails for infinite posets.
- A finite non-empty poset has at least one maximal and one minimal element.
- Distinct maximal elements are noncomparable (so are distinct minimal ones).
- An element can be both maximal and minimal: an isolated point.
- Infinite posets may have none: $(\mathbb{Z}, \leq)$.
- Every total order on a finite set is well-ordered: a non-empty subset has a minimal element, and in a total order a minimal element is smallest.

PAST PAPER **AY23 Q13, Q14 · Facts about partial orders**
(i) "No element is both smallest and largest" is false: $(\{a\}, \{(a,a)\})$. (ii) "An element can be maximal and minimal but neither largest nor smallest" is true: $(\{a,b\}, \{(a,a),(b,b)\})$. (iii) "Distinct maximal elements are comparable" is false, as it would contradict maximality. (iv) "There may be no maximal or minimal element" is true: $(\mathbb{Z}, \leq)$. **C**.
Q14. On $\{1,2,3\}$: $P = \{(1,1),(1,2),(2,2)\}$ is not reflexive ($(3,3)$ missing); $Q = \{(1,1),(1,2),(1,3),(2,2),(3,2),(3,3)\}$ **is** a partial order; $R = \{(1,1),(1,2),(2,2),(2,3),(3,3),(3,1)\}$ is not transitive ($1R2$, $2R3$, not $1R3$), and $R^t = A \times A$ is not antisymmetric. Only Q: **B**.

WORKED EXAMPLE **Partial order and extremal elements**
On $\mathbb{Z}^+$ let $a \preccurlyeq b \leftrightarrow b = a \cdot 2^k$ for some $k \in \mathbb{N}$. Prove $\preccurlyeq$ is a partial order; find its minimal and maximal elements.
1. (Reflexive) $a = a \cdot 2^0$ and $0 \in \mathbb{N}$.
2. (Antisymmetric) If $b = a2^j$ and $a = b2^k$, then $a = a2^{j+k}$, so $2^{j+k} = 1$, $j = k = 0$, and $a = b$.
3. (Transitive) If $b = a2^j$ and $c = b2^k$ then $c = a2^{j+k}$ with $j + k \in \mathbb{N}$ *(closure)*.
4. (Minimal) $b \preccurlyeq a$ with $b \neq a$ means $a = b2^k$, $k \geq 1$, which is possible iff a is even. So the minimal elements are exactly the **odd** numbers.
5. (Maximal) None: $a \preccurlyeq 2a$ and $2a \neq a$. Hence no largest or smallest element either. ■

WORKED EXAMPLE **T5 Q5 · Componentwise order on $\{0,1\}^2$**
Let $B = \{0,1\}$ and $(a,b)R(c,d) \leftrightarrow (a \leq c) \wedge (b \leq d)$ on $B \times B$. Prove R is a partial order.
1. (Reflexive) Let $(a,b) \in B \times B$. Then $a \leq a$ and $b \leq b$, so $(a,b)R(a,b)$ *(definition of R)*.
2. (Antisymmetric) Let $(a,b)R(c,d)$ and $(c,d)R(a,b)$. Then $a \leq c$, $b \leq d$, $c \leq a$ and $d \leq b$ *(definition of R)*, so $a = c$ and $b = d$ *(antisymmetry of $\leq$)*, so $(a,b) = (c,d)$ *(equality of ordered pairs)*.
3. (Transitive) Let $(a,b)R(c,d)$ and $(c,d)R(e,f)$. Then $a \leq c$, $b \leq d$, $c \leq e$ and $d \leq f$ *(definition of R)*, so $a \leq e$ and $b \leq f$ *(transitivity of $\leq$)*, so $(a,b)R(e,f)$ *(definition of R)*.
4. Therefore R is a partial order on $B \times B$. ■



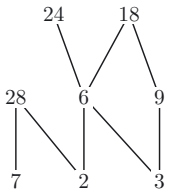
Maximal and largest: $(1,1)$. **Minimal and smallest:** $(0,0)$. **Not well-ordered:** it is not even a total order, since $(0,1)$ and $(1,0)$ are noncomparable.

25 · Linearization and Kahn's algorithm

DEFINITION **Linearization (Lecture 6.4.7)**
A **linearization** of a partial order $\preccurlyeq$ on A is a **total order** $\preccurlyeq^*$ on A with $\forall x, y \in A\,(x \preccurlyeq y \rightarrow x \preccurlyeq^* y)$.
It lists the elements one at a time without breaking any precedence.

METHOD **Kahn's algorithm (1962) · Testing a proposed line-up**
Kahn: set $A_0 := A$, $i := 0$; while $A_i \neq \varnothing$: pick a **minimal** element c_i of A_i, set $A_{i+1} := A_i \setminus \{c_i\}$, $i := i + 1$. Output $c_0 \preccurlyeq^* c_1 \preccurlyeq^* \cdots$. Different choices of minimal element give different linearizations.
Test: a sequence is a linearization iff **for every line of the Hasse diagram, the lower element comes earlier**. A single reversed pair means it is not a linearization.

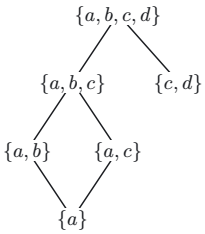
METHOD **Is there a linearization with $a \preccurlyeq^* b$? (T5 Q2) · Listing them all (T5 Q4)**
For distinct a, b, some linearization has $a \preccurlyeq^* b$ **iff not $b \preccurlyeq a$**. If $b \preccurlyeq a$, every linearization has $b \preccurlyeq^* a$; if $a \preccurlyeq b$, every one has $a \preccurlyeq^* b$; noncomparable elements can appear in either order. For a chain $a_1 \preccurlyeq^* a_2 \preccurlyeq^* \cdots \preccurlyeq^* a_k$: possible iff no later element lies strictly below an earlier one.
T5 Q2 on $(\{2,3,5,7,21,30,84,99\}, |)$: (i) $21 \preccurlyeq^* 7$ is impossible, as $7 \mid 21$. (ii) $3 \preccurlyeq^* 2$ is possible (noncomparable). (iii) $21 \preccurlyeq^* 5 \preccurlyeq^* 84$ is possible: 5 is noncomparable with 21 and with 84, and $21 \mid 84$ agrees. (iv) $99 \preccurlyeq^* 84 \preccurlyeq^* 30$ is possible: pairwise noncomparable. True: **(ii), (iii), (iv)**.
All linearizations (T5 Q4): run Kahn's algorithm through every choice of minimal element. For the order of AY23 Q15 (11, 13 below 14, 15 below 16 below 12) there are $2 \cdot 2 = \mathbf{4}$: $11,13,14,15,16,12$; $11,13,15,14,16,12$; $13,11,14,15,16,12$; $13,11,15,14,16,12$.
Total order (L6 s87): its Hasse diagram is one chain, so its only linearization is itself.



PAST PAPER **AY24 Q13–Q15 · $(\{2,3,6,7,9,18,24,28\}, |)$**
Covers: $2 \mid 6$, $2 \mid 28$, $3 \mid 6$, $3 \mid 9$, $7 \mid 28$, $6 \mid 18$, $6 \mid 24$, $9 \mid 18$.
Q13 Longest maximal chains have length 2:
$\{2,6,18\}, \{2,6,24\}, \{3,6,18\}, \{3,6,24\}, \{3,9,18\}$. There are **5**; answer B.
Q14 $\{2,3,28\}$ is not an antichain ($2 \mid 28$); $\{7,9,24\}$ is; $\{6\}$ is, vacuously. **D**.
Q15 $2,3,6,7,9,18,24,28$ ✓; $7,3,2,28,9,6,18,24$ ✓; $7,2,3,6,18,9,24,28$ ✗, since $9 \mid 18$ but 18 comes first. **B**.

PAST PAPER **AY24 Q17(c), (d) · $(E = \{2,4,6,8,10,12\}, |)$**
Covers: $2 \mid 4$, $2 \mid 6$, $2 \mid 10$, $4 \mid 8$, $4 \mid 12$, $6 \mid 12$. **Minimal:** 2. **Maximal:** 8, 10, 12. **Smallest:** 2. **Largest:** none.
Two maximal chains of different lengths: $\{2,4,8\}$ (length 2) and $\{2,10\}$ (length 1); $\{2,4,12\}$, $\{2,6,12\}$ also work for the first.

PAST PAPER **AY25 Q17(e) · $(\{\{a\},\{a,b\},\{a,c\},\{a,b,c\},\{a,b,c,d\},\{c,d\}\}, \subseteq)$**
Covers: $\{a\} \subset \{a,b\}, \{a,c\} \subset \{a,b,c\} \subset \{a,b,c,d\}$, and $\{c,d\} \subset \{a,b,c,d\}$. **Smallest element:** no, since $\{a\}$ and $\{c,d\}$ are both minimal and noncomparable. **Maximal:** yes, $\{a,b,c,d\}$ (which is also largest).



PAST PAPER **AY23 Q15, Q18 · $x \preccurlyeq z \leftrightarrow F_x = F_z \vee |F_x| < |F_z|$ on $\{11, ..., 16\}$, F_x = positive divisors**
Divisor counts: $11 \to 2$, $13 \to 2$, $14 \to 4$, $15 \to 4$, $16 \to 5$, $12 \to 6$. The order compares counts only, so 11, 13 are incomparable, as are 14, 15. **Minimal:** 11, 13. **Smallest:** none. **Maximal:** 12. **Largest:** 12.
Q15: a linearization must list counts in non-decreasing order: $13,11,15,14,16,12$ ✓ and $11,13,14,15,16,12$ ✓; the other two put 12 too early. **B**.

PROOF **AY24 Q17(f), (g) · $(a,b)W(c,d) \leftrightarrow a \le c \wedge b \ge d$ on $\mathbb{Z} \times \mathbb{Z}$**
(f) "Every two elements are comparable" is false. Counterexample: $(1,3)$ and $(2,4)$, since neither $1 \le 2 \wedge 3 \ge 4$ nor $2 \le 1 \wedge 4 \ge 3$ holds.
(g) "Every two elements are compatible" is true.
1. Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be arbitrary elements of $\mathbb{Z} \times \mathbb{Z}$.
2. Let $c = (\max(x_1, y_1), \min(x_2, y_2)) \in \mathbb{Z} \times \mathbb{Z}$.
3. Then $x_1 \le c_1$, $y_1 \le c_1$ *(definition of max)* and $x_2 \ge c_2$, $y_2 \ge c_2$ *(definition of min)*.
4. So xWc and yWc *(definition of W)*: x and y are compatible. ■

Relation	Refl.	Sym.	Trans.	Antisym.	Verdict
$=$ on any set	✓	✓	✓	✓	equivalence **and** partial order
$\le$ on $\mathbb{R}$	✓	✗	✓	✓	total order
$<$ on $\mathbb{R}$	✗	✗	✓	✓	neither (asymmetric)
$\mid$ on $\mathbb{Z}^+$	✓	✗	✓	✓	partial order, not total
$\mid$ on $\mathbb{Z}$	✓	✗	✓	✗	neither ($2 \mid -2 \mid 2$)
$\equiv \pmod n$ on $\mathbb{Z}$	✓	✓	✓	✗	equivalence, n classes
$\subseteq$ on $\mathcal{P}(A)$	✓	✗	✓	✓	partial order
$\lvert X\rvert = \lvert Y\rvert$ on $\mathcal{P}(A)$	✓	✓	✓	✗	equivalence, classes by size
$xy \ge 0$ on $\mathbb{Z}$ or $\mathbb{Q}$ (T4 Q3)	✓	✓	✗	✗	neither: $-1, 0, 1$
$xy > 0$ on $\mathbb{Q}$ (T4 Q3)	✗	✓	✓	✗	neither: $0 \not< 0$
$\lvert x-y\rvert \le 1$ on $\mathbb{Z}$	✓	✓	✗	✗	neither: $0, 1, 2$
$\lvert x-y\rvert \le 2$ on $\mathbb{Z}$ (T4 Q3)	✓	✓	✗	✗	neither: $-2, 0, 2$
$\varnothing$ on $A \ne \varnothing$	✗	✓	✓	✓	neither
$A \times A$	✓	✓	✓	✗ if $\lvert A\rvert \ge 2$	equivalence, one class

WORKED EXAMPLE **Lecture 6, Ex. 26 · Kahn's algorithm on $\{d \in \mathbb{Z}^+ : d \mid 30\}$**

Step	Minimal elements of A_i	Choose
A_0	1	$c_0 = 1$
A_1	2, 3, 5	$c_1 = 3$
A_2	2, 5	$c_2 = 2$
A_3	5, 6	$c_3 = 6$
A_4	5	$c_4 = 5$

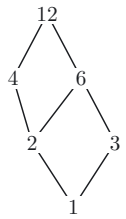
A_5	10, 15	$c_5 = 15$
A_6	10	$c_6 = 10$
A_7	30	$c_7 = 30$

Result: $1 \preccurlyeq^* 3 \preccurlyeq^* 2 \preccurlyeq^* 6 \preccurlyeq^* 5 \preccurlyeq^* 15 \preccurlyeq^* 10 \preccurlyeq^* 30$. Any other choice at steps 1–3 or 5 gives another valid linearization.

WORKED EXAMPLE **Hasse diagram and Kahn**

For $(\{1,2,3,4,6,12\}, |)$: draw the Hasse diagram; give the minimal, maximal, smallest, largest elements; produce a linearization with Kahn's algorithm.

Covers: $1 \mid 2$, $1 \mid 3$, $2 \mid 4$, $2 \mid 6$, $3 \mid 6$, $4 \mid 12$, $6 \mid 12$ (diagram right). Minimal and smallest: **1**. Maximal and largest: **12**. Kahn: A_0 has the single minimal element 1; then 2 and 3 are minimal and we take 2; then 3 and 4, and we take 3; then 4 and 6, and we take 4; then 6; then 12. Linearization: $1 \preccurlyeq^* 2 \preccurlyeq^* 3 \preccurlyeq^* 4 \preccurlyeq^* 6 \preccurlyeq^* 12$ (also valid: $1, 3, 2, 6, 4, 12$).



Part VI · Functions

26 · Functions, images and preimages

DEFINITION **Function (L7 s7)**

A **function** f from a set X to a set Y, written $f : X \to Y$, is a relation $f \subseteq X \times Y$ satisfying

(F1) $\forall x \in X\, \exists y \in Y\, ((x,y) \in f)$; every input has an output.

(F2) $\forall x \in X\, \forall y_1, y_2 \in Y\, (((x,y_1) \in f \wedge (x,y_2) \in f) \to y_1 = y_2)$; the output is unique.

Equivalently, $\forall x \in X\, \exists! y \in Y\, ((x,y) \in f)$. On an arrow diagram: every element of X has exactly one arrow leaving it.

DEFINITION **Image, preimage (L7 s11) · Setwise image and preimage (L7 s12) · Domain, co-domain, range (L7 s16)**

Write $f(x) = y$ iff $(x,y) \in f$; f **maps** x to y ($f : x \mapsto y$); x is the **argument**, $f(x)$ the **image** of x, and x is **a preimage** of y.

For $A \subseteq X$ and $B \subseteq Y$:

$$f(A) = \{f(x) : x \in A\} \quad \text{(setwise image)}, \qquad f^{-1}(B) = \{x \in X : f(x) \in B\} \quad \text{(setwise preimage)}.$$

X is the **domain**, Y the **co-domain**, and the **range** is $f(X) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$. Always range $\subseteq$ co-domain.

TRAP **Two meanings of f^{-1} (L7 s12, s15)**

- $f^{-1}(B)$ with B a **subset** of the co-domain is the setwise preimage. It exists for every function and is a set.
- $f^{-1}(y)$ with y an **element** of the co-domain is the inverse function (§28). It exists only when f is a bijection.
- For the preimage of a single element write $f^{-1}(\{b\})$, never $f^{-1}(b)$, unless f is known to be bijective.

WORKED EXAMPLE **L7 Ex. 1, 4, 6 · T6 D1, Q1, Q2**

- **Not a function.** "$f(x)$ is the real y with $x^2 + y^2 = 1$" on $\mathbb{R}$: $x = 2$ has no y (F1 fails); $x = 0$ has $y = \pm 1$ (F2 fails).
- **T6 D1.** $f(z) = 1$ if $2 \mid z$, 2 if $3 \mid z$: not a function ($f(6)$ has two values, $f(1)$ has none). $f : \mathbb{R} \to \mathbb{Z}$, $f(x) = 2x$: not a function ($f(0.3) = 0.6 \notin \mathbb{Z}$). $f : \mathbb{Z} \to \mathbb{R}$, $f(x) = 2x$: a function.
- **T6 Q1** on $\mathbb{N}$. $xR_1y \leftrightarrow x^2 = y^2$: a function (F1: take $y = x$; F2: $y_1^2 = y_2^2$ with $y_1, y_2 \geq 0$ gives $y_1 = y_2$). $xR_2y \leftrightarrow y \mid x$: not a function ($6R_2 2$ and $6R_2 3$). $xR_3y \leftrightarrow y = x + 1$: a function.
- **Ex. 4.** $g : \mathbb{Z} \to \mathbb{Z}$, $g(x) = x^2$: $g(\{-1,0,1\}) = \{0,1\}$ and $g^{-1}(\{0,1,2\}) = \{-1,0,1\}$.
- **Ex. 6.** $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 2x + 1$: domain $\mathbb{Z}$, co-domain $\mathbb{Z}$, range = the odd integers.
- **T6 Q2.** $\text{len} : \{s,u\}^* \to \mathbb{Z}_{\geq 0}$: $\text{len}(\{\varepsilon, ss, uu, ssss\}) = \{0,2,4\}$; $\text{len}^{-1}(\{3\})$ is the set of all 8 strings of length 3; the inverse function len^{-1} does not exist, since $\text{len}(s) = \text{len}(u)$.

PROOF **T6 Q9 · $X \subseteq f^{-1}(f(X))$ and $f(f^{-1}(Y)) \subseteq Y$; the reverse inclusions can fail**

Let $f : A \to B$, $X \subseteq A$, $Y \subseteq B$.

1. ($X \subseteq f^{-1}(f(X))$) Let $x \in X$. Then $f(x) \in f(X)$ *(definition of $f(X)$)*, so $x \in f^{-1}(f(X))$ *(definition of setwise preimage)*.

2. $(f(f^{-1}(Y)) \subseteq Y)$ Let $y \in f(f^{-1}(Y))$. Then $y = f(x)$ for some $x \in f^{-1}(Y)$ *(definition of setwise image)*, and $f(x) \in Y$ *(definition of setwise preimage)*. So $y \in Y$. ■

Counterexamples for the reverse. $f : \{-1,1\} \to \{0\}$, $X = \{1\}$: $f^{-1}(f(X)) = \{-1,1\} \not\subseteq X$.
$f : \{0\} \to \{-1,1\}$ with $f(0) = 1$, $Y = \{-1\}$: $f(f^{-1}(Y)) = \varnothing$, so $Y \not\subseteq f(f^{-1}(Y))$.
Equality holds in the first when f is injective and in the second when f is surjective.

27 · Sequences, strings and function equality

DEFINITION Sequence (L7 s18) · Fibonacci sequence (L7 s19) · String (L7 s20)

A **sequence** $a_0, a_1, a_2, \ldots$ is represented by a function a with domain $\mathbb{Z}_{\geq 0}$ and $a(n) = a_n$; any function with domain $\mathbb{Z}_{\geq m}$ represents a sequence.

Fibonacci: $F_0 = 0$, $F_1 = 1$, $F_{n+2} = F_{n+1} + F_n$: $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$

A **string** (word) over a set A is an expression $a_0 a_1 \cdots a_{l-1}$ with $l \in \mathbb{Z}_{\geq 0}$ and each $a_i \in A$; l is its **length**; the **empty string** ε has length 0; A^* is the set of all strings over A. Two sequences (strings of the same length) are equal iff they agree at every index.

DEFINITION Floor, ceiling (T6 D3 · AY23/24 final)

For a real number x: the **floor** $\lfloor x \rfloor$ is the unique integer n with $n \leq x < n+1$, i.e. $\lfloor x \rfloor = \max\{m \in \mathbb{Z} : m \leq x\}$; the **ceiling** $\lceil x \rceil$ is the unique integer n with $n - 1 < x \leq n$, i.e. $\lceil x \rceil = \min\{m \in \mathbb{Z} : m \geq x\}$.

$\lfloor 2.7 \rfloor = 2$, $\lceil 2.7 \rceil = 3$, $\lfloor -2.7 \rfloor = -3$, $\lceil -2.7 \rceil = -2$, $\lfloor 5 \rfloor = \lceil 5 \rceil = 5$. Read off the definitions:

- $\lfloor x \rfloor \leq x \leq \lceil x \rceil$; $\quad x \in \mathbb{Z} \Leftrightarrow \lfloor x \rfloor = x \Leftrightarrow \lceil x \rceil = x$; $\quad x \notin \mathbb{Z} \Rightarrow \lceil x \rceil = \lfloor x \rfloor + 1$.
- For $n \in \mathbb{Z}$: $\lfloor n/2 \rfloor = n/2$ if n is even and $(n-1)/2$ if n is odd; $\lceil n/2 \rceil = n/2$ if n is even and $(n+1)/2$ if n is odd.

THEOREM Theorem 7.1.1 · Function equality (L7 s22)

$f : A \to B$ and $g : C \to D$ are equal iff (i) $A = C$ and $B = D$, and (ii) $f(x) = g(x)$ for all $x \in A$.

WORKED EXAMPLE L7 Ex. 11–13 · T6 D3

- On $X = \{0,1,2\}$, $f(x) = (x^2 + x + 1) \bmod 3$ and $g(x) = (x+2)^2 \bmod 3$ both give $1, 0, 1$, so $f = g$.
- $f, g : \{0,2\} \to \mathbb{Z}$ with $f(x) = 2x$, $g(x) = x^2$: equal (same domain, co-domain, values $0, 4$).
- $f : \mathbb{Z} \to \mathbb{Z}$ and $g : \mathbb{Z} \to \mathbb{Q}$ with $f(x) = x^3 = g(x)$: **not** equal, since the co-domains differ.
- **T6 D3.** $f, g : \mathbb{Q} \to \mathbb{Q}$, $f(x) = \lfloor x \rfloor + 1$, $g(x) = \lceil x \rceil$: both have range $\mathbb{Z}$, but $f(0) = 1 \neq 0 = g(0)$, so $f \neq g$ (they agree exactly off the integers).

28 · Injections, surjections, bijections, inverses

DEFINITION Injection (L7 s25) · Surjection (L7 s27) · Bijection (L7 s29)

Let $f : X \to Y$.

- f is **injective** (one-to-one) iff $\forall x_1, x_2 \in X\,(f(x_1) = f(x_2) \Rightarrow x_1 = x_2)$. Contrapositive: $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$.
- f is **surjective** (onto) iff $\forall y \in Y\, \exists x \in X\,(y = f(x))$: range = co-domain.
- f is **bijective** (a one-to-one correspondence) iff it is injective and surjective: $\forall y \in Y\, \exists! x \in X\,(y = f(x))$.

Arrows **into** each element of the co-domain: at most one (injective), at least one (surjective), exactly one (bijective).

METHOD Negations (to disprove)

- **Not injective:** $\exists x_1, x_2 \in X\,(f(x_1) = f(x_2) \wedge x_1 \neq x_2)$. Name two inputs.
- **Not surjective:** $\exists y \in Y\, \forall x \in X\,(y \neq f(x))$. Name one y and show no x reaches it.

METHOD Proof templates

- **Injective:** "Let $x_1, x_2 \in X$ with $f(x_1) = f(x_2)$" ... "so $x_1 = x_2$."
- **Surjective:** "Take any $y \in Y$. Let $x = \ldots$" (solve $y = f(x)$ on scratch paper). Check $x \in X$ and $f(x) = y$.
- **Bijective:** both, or exhibit an inverse (Theorem 7.2.3).

PROOF **L7 Ex. 14, 16 · $f : \mathbb{Q} \to \mathbb{Q}$, $f(x) = 3x + 1$ is a bijection** **Ex. 15, 17 · $g : \mathbb{Z} \to \mathbb{Z}$, $g(x) = x^2$ is neither**

1. (Injective) Let $x_1, x_2 \in \mathbb{Q}$ with $f(x_1) = f(x_2)$. Then $3x_1 + 1 = 3x_2 + 1$, so $x_1 = x_2$ *(basic algebra)*.
2. (Surjective) Take any $y \in \mathbb{Q}$. Let $x = (y-1)/3$. Then $x \in \mathbb{Q}$ *(closure of $\mathbb{Q}$)* and $f(x) = 3 \cdot \frac{y-1}{3} + 1 = y$. ■

g is not injective: $g(1) = 1 = g(-1)$ but $1 \neq -1$. g is not surjective: $g(x) = x^2 \geq 0 > -1$ for all $x \in \mathbb{Z}$, so -1 has no preimage.

The domain matters: $x \mapsto 3x + 1$ on $\mathbb{Z}$ is injective but not surjective ($y = 0$ would need $x = -1/3 \notin \mathbb{Z}$).

WORKED EXAMPLE **T6 Q4 · Classify four functions**

(a) $f : \mathbb{Q} \to \mathbb{Q}$, $x \mapsto 12x + 31$: bijective, with $f^{-1}(y) = (y - 31)/12$, because $y = 12x + 31 \Leftrightarrow x = (y - 31)/12$ for all $x, y \in \mathbb{Q}$.

(b) $g : \text{Bool}^2 \to \text{Bool}$, $(p, q) \mapsto p \wedge \sim q$: not injective ($g(F, T) = F = g(F, F)$); surjective ($g(T, F) = T$).

(c) $h : \text{Bool}^2 \to \text{Bool}^2$, $(p, q) \mapsto (p \wedge q, p \vee q)$: not injective ($h(T, F) = (F, T) = h(F, T)$); not surjective: if $p \wedge q = T$ then $p \vee q = T$, so (T, F) is not in the range.

(d) $k : \mathbb{Z} \to \mathbb{Z}$, $k(x) = x$ if x is even, $2x - 1$ if x is odd.

1. k preserves parity: x even gives $k(x) = x$ even; x odd gives $k(x) = 2(x-1) + 1$ odd. So x is even iff $k(x)$ is even.
2. (Injective) Let $k(x_1) = k(x_2)$. If this value is even, both x_i are even and $x_1 = k(x_1) = k(x_2) = x_2$. If it is odd, both are odd and $2x_1 - 1 = 2x_2 - 1$, so $x_1 = x_2$.
3. (Not surjective) Suppose $k(x) = 3$. Then x is odd *(line 1)*, so $2x - 1 = 3$ and $x = 2$, which is even: contradiction. So 3 has no preimage. ■

DEFINITION **Inverse function (L7 s31)**

Let $f : X \to Y$. A function $g : Y \to X$ is an **inverse** of f iff

$$\forall x \in X \, \forall y \in Y \, (y = f(x) \Leftrightarrow x = g(y)).$$

The inverse, when it exists, is unique and is written f^{-1}. Example: $f(x) = 3x + 1$ on $\mathbb{Q}$ has $f^{-1}(y) = (y - 1)/3$.

PROOF **Proposition · Uniqueness of inverses (L7 s34)** **Theorem 7.2.3 · f is bijective iff f has an inverse; then f^{-1} is a bijection (L7 s35–36)**

Uniqueness. If $g_1, g_2 : Y \to X$ are inverses of f, then for all $x \in X$, $y \in Y$: $x = g_1(y) \Leftrightarrow y = f(x) \Leftrightarrow x = g_2(y)$. So $g_1 = g_2$ *(Theorem 7.1.1)*.

Theorem 7.2.3.

1. ("if") Suppose f has an inverse $g : Y \to X$.
   1.1. (Injective) Let $f(x_1) = f(x_2) = y$. Then $x_1 = g(y)$ and $x_2 = g(y)$ *(g is an inverse)*, so $x_1 = x_2$.
   1.2. (Surjective) Let $y \in Y$ and $x = g(y)$. Then $y = f(x)$ *(g is an inverse)*.
2. ("only if") Suppose f is bijective. Then $\forall y \in Y \, \exists! x \in X \, (y = f(x))$ *(definition of bijection)*. Define $g(y)$ to be that unique x. Then g is well defined and $y = f(x) \Leftrightarrow x = g(y)$, so g is an inverse of f.
3. (f^{-1} is a bijection) f is an inverse of f^{-1}, since the defining condition is symmetric in f and g. So f^{-1} is bijective by line 1. ■

THEOREM **T6 Q5 (AY22/23 S2 exam) · Left and right inverses**

$g : B \to A$ is a **left inverse** of $f : A \to B$ iff $g(f(a)) = a$ for all $a \in A$; $h : B \to A$ is a **right inverse** iff $f(h(b)) = b$ for all $b \in B$.

f is injective iff it has a left inverse; f is surjective iff it has a right inverse (for $A \neq \varnothing$). Neither kind of inverse implies the other. In T6 Q4: left inverses for f, k; right inverses for f, g.

29 · Composition of functions

DEFINITION **Composition (L7 s39) · Identity function (L7 s41)**

For $f : X \to Y$ and $g : Y \to Z$, the **composition** $g \circ f : X \to Z$ is defined by $(g \circ f)(x) = g(f(x))$ for all $x \in X$. **f is applied first.**

The **identity function** on X is $\text{id}_X : X \to X$ with $\text{id}_X(x) = x$.

THEOREM **Theorems on composition (L7 s44–52)**

- **7.3.1.** For $f : X \to Y$: $f \circ \mathrm{id}_X = f$ and $\mathrm{id}_Y \circ f = f$.
- **7.3.2.** If $f : X \to Y$ is a bijection: $f^{-1} \circ f = \mathrm{id}_X$ and $f \circ f^{-1} = \mathrm{id}_Y$.
- **Associativity.** For $f : A \to B$, $g : B \to C$, $h : C \to D$: $(h \circ g) \circ f = h \circ (g \circ f)$.
- **Not commutative.** $f(x) = 3x$, $g(x) = x + 1$ on $\mathbb{Z}$: $(g \circ f)(0) = 1 \neq 3 = (f \circ g)(0)$.
- **7.3.3.** f, g injective $\Rightarrow g \circ f$ injective. **7.3.4.** f, g surjective $\Rightarrow g \circ f$ surjective. Hence bijections compose to a bijection.

PROOF **Associativity (L7 s47) · Theorem 7.3.3 (L7 s50) · Theorem 7.3.4 (L7 s52)**

Associativity. Both sides have domain A and co-domain D, and for every $x \in A$: $((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x))) = h((g \circ f)(x)) = (h \circ (g \circ f))(x)$. Equal by Theorem 7.1.1.

7.3.3.

1. Let $f : X \to Y$ and $g : Y \to Z$ be injective, and let $x_1, x_2 \in X$ with $(g \circ f)(x_1) = (g \circ f)(x_2)$.
2. Then $g(f(x_1)) = g(f(x_2))$ *(definition of composition)*.
3. So $f(x_1) = f(x_2)$ *(g is injective)*, and then $x_1 = x_2$ *(f is injective)*. ■

7.3.4.

1. Let $f : X \to Y$ and $g : Y \to Z$ be surjective, and let $z \in Z$.
2. There is $y \in Y$ with $g(y) = z$ *(g is surjective)*, and $x \in X$ with $f(x) = y$ *(f is surjective)*.
3. Then $(g \circ f)(x) = g(f(x)) = g(y) = z$. ■

TRAP **T6 D4 · A wrong surjectivity proof**

"$\forall y \exists x\, f(x) = y$, and $\forall z \exists y\, g(y) = z$, so $(g \circ f)(x) = g(y) = z$." The x, y, z in the last line were never fixed: the proof must **start** with an arbitrary $z \in Z$, **then** obtain y from g, **then** obtain x from f for that particular y.

PROOF **T6 Q3 · For bijections $f : A \to B$, $g : B \to C$: $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$**

1. f^{-1}, g^{-1} are bijections *(Theorem 7.2.3)* $g \circ f$ and $f^{-1} \circ g^{-1}$ are bijections *(Theorems 7.3.3, 7.3.4)* so $(g \circ f)^{-1}$ exists.
2. Both $(g \circ f)^{-1}$ and $f^{-1} \circ g^{-1}$ have domain C and co-domain A.
3. Let $z \in C$. Let $y = g^{-1}(z)$, so $z = g(y)$; let $x = f^{-1}(y)$, so $y = f(x)$ *(definition of inverse)*.
4. Then $(g \circ f)(x) = g(f(x)) = g(y) = z$, so $(g \circ f)^{-1}(z) = x$ *(definition of inverse)*.
5. Also $(f^{-1} \circ g^{-1})(z) = f^{-1}(g^{-1}(z)) = f^{-1}(y) = x$.
6. The two functions agree at every $z \in C$, so they are equal *(Theorem 7.1.1)*. ■

PROOF **T6 Q6 · If $g \circ f$ is injective, then f is injective · If $g \circ f$ is surjective, then g is surjective**

1. Let $f : B \to C$ and let g have domain C with $g \circ f$ injective. Let $x_1, x_2 \in B$ with $f(x_1) = f(x_2)$.
2. Then $(g \circ f)(x_1) = g(f(x_1)) = g(f(x_2)) = (g \circ f)(x_2)$, so $x_1 = x_2$ *($g \circ f$ is injective)*. ■

For the second: given z, surjectivity of $g \circ f$ gives x with $g(f(x)) = z$, and $y = f(x)$ is a preimage of z under g.

The other halves fail: $X = Z = \{0\}$, $Y = \{0, 1\}$, $f(0) = 0$, $g(0) = g(1) = 0$. Then $g \circ f = \mathrm{id}_{\{0\}}$ is bijective, yet g is not injective and f is not surjective.

PAST PAPER **Order of a bijection · T6 Q7, Q8 · T8 Q10(c) · AY25 final Q11, Q12, Q26(b)**

The **order** of a bijection $f : A \to A$ is the least $n \in \mathbb{Z}^+$ with $f \circ f \circ \cdots \circ f = \mathrm{id}_A$ (n copies). The **order of an input** x is the least $m \in \mathbb{Z}^+$ with $f^{(m)}(x) = x$.

T6 Q7. On $\{1, 2, 3\}$, g swaps 1, 2 and h swaps 2, 3: orders 2 and 2. $g \circ h$ sends $1 \mapsto 2 \mapsto 3 \mapsto 1$: order 3; likewise $h \circ g$.

T6 Q8. $g(x) = 3x \bmod 5$: $g^{(3)}(21) = g^{(2)}(3) = g(4) = 2$. Order of 3: $3 \mapsto 4 \mapsto 2 \mapsto 1 \mapsto 3$, so 4. Orders of $0, 1, 2, 3, 4$ are $1, 4, 4, 4, 4$: classes $\{0\}$ and $\{1, 2, 3, 4\}$.

T8 Q10(c). $f = \{(1, 2), (2, 4), (3, 1), (4, 3)\}$, $g = \{(1, 4), (2, 1), (3, 2), (4, 3)\}$: orders 4 and 4; $f^{-1} \circ g$ sends $1 \mapsto 2 \mapsto 3 \mapsto 1$, $4 \mapsto 4$: order 3.

AY25 Q11, Q12. $f(p, q) = (p, p \oplus q)$ and $g(p, q) = (p \oplus q, q)$ on "Bool"^{2}. $f \circ f = \mathrm{id}$, so $f = f^{-1}$; but $f^{-1} \circ f = \mathrm{id} \neq f$. $(g \circ f)(p, q) = (q, p \oplus q)$ has order 3.

AY25 Q26(b). $z : \mathbb{Z}_5 \to \mathbb{Z}_5$, $z(x) = 2x + 1$: $z^{-1}([3]) = [1]$; $z^{(2)}(x) = 4x + 3$; $[3] \mapsto [2] \mapsto [0] \mapsto [1] \mapsto [3]$, so the order of $[3]$ is 4.

30 · Addition and multiplication on $\mathbb{Z}_n$

DEFINITION **$\mathbb{Z}_n$ (L7 s54–55) · Addition and multiplication on $\mathbb{Z}_n$ (L7 s56)**

$\mathbb{Z}_n$ denotes the quotient $\mathbb{Z}/\underset{n}{\sim}$, where $\underset{n}{\sim}$ is congruence mod n: $\mathbb{Z}_n = \{[0], [1], \ldots, [n-1]\}$ with $[r] = \{nk + r : k \in \mathbb{Z}\}$.

For $[x], [y] \in \mathbb{Z}_n$: $[x] + [y] = [x + y]$ and $[x] \cdot [y] = [x \cdot y]$. In $\mathbb{Z}_3$: $[0] + [1] = [1]$, $[2] \cdot [2] = [4] = [1]$.

TRAP **$\mathbb{Z}_n$ has two meanings in the slides**
Lecture 7 uses $\mathbb{Z}_n = \mathbb{Z}/\underset{n}{\sim}$ (a set of n classes). Lecture 9, slide 9 writes $\mathbb{Z}_n = \{1, 2, ..., n\}$ when defining finite sets. Read the definition in the question.

PROOF **Propositions $\cdot$ + and $\cdot$ on $\mathbb{Z}_n$ are well defined (L7 s57–58)**
Claim: for all $[x_1], [y_1], [x_2], [y_2] \in \mathbb{Z}_n$, if $[x_1] = [x_2]$ and $[y_1] = [y_2]$ then $[x_1] + [y_1] = [x_2] + [y_2]$ and $[x_1] \cdot [y_1] = [x_2] \cdot [y_2]$.
1. Let $[x_1] = [x_2]$ and $[y_1] = [y_2]$. Then $x_1 \equiv x_2$ and $y_1 \equiv y_2 \pmod n$ *(Lemma Rel.1)*.
2. So $x_1 - x_2 = nk$ and $y_1 - y_2 = nl$ for some $k, l \in \mathbb{Z}$ *(definition of congruence)*.
3. $(x_1 + y_1) - (x_2 + y_2) = n(k + l)$, so $x_1 + y_1 \equiv x_2 + y_2 \pmod n$, hence $[x_1 + y_1] = [x_2 + y_2]$ *(Lemma Rel.1)*.
4. $x_1y_1 - x_2y_2 = (nk + x_2)(nl + y_2) - x_2y_2 = n(nkl + ky_2 + lx_2)$ with $nkl + ky_2 + lx_2 \in \mathbb{Z}$ *(closure)*, hence $[x_1y_1] = [x_2y_2]$. ■

The representative-level version of this proof is in §23.

METHOD **Is a rule on classes well defined?**
A rule $F([x]) = ...$ defines a function on $A/\sim$ only if the output does not depend on the representative: **show $x \sim x' \Rightarrow$ same output**. To disprove, name two representatives of one class with different outputs.
- $[x]_3 \mapsto [x]_2$ from $\mathbb{Z}_3$ to $\mathbb{Z}_2$: **not** well defined ($[0]_3 = [3]_3$ but $[0]_2 \neq [3]_2$). From $\mathbb{Z}_4$ to $\mathbb{Z}_2$ it is well defined, as $4 \mid (x - x') \Rightarrow 2 \mid (x - x')$.
- **T6 Q10.** On $\mathbb{Q}/\sim$ with $x \sim y \leftrightarrow x - y \in \mathbb{Z}$: + is well defined ($(x_1 + y_1) - (x_2 + y_2) = k + l \in \mathbb{Z}$); $\cdot$ is **not**: $[1/2] = [-1/2]$, but $[1/2] \cdot [1/2] = [1/4] \neq [-1/4] = [1/2] \cdot [-1/2]$, as $1/4 - (-1/4) = 1/2 \notin \mathbb{Z}$.
- **T8 Q10 (AY22/23 S2).** If $g : X \to Y$ satisfies $g(a) = g(b) \Leftrightarrow a \sim b$, then $f([x]) = g(x)$ is well defined ($[x] = [x'] \Rightarrow x \sim x' \Rightarrow g(x) = g(x')$) **and** injective ($f([x]) = f([x']) \Rightarrow g(x) = g(x') \Rightarrow x \sim x' \Rightarrow [x] = [x']$).

Part VII · Induction and Recursion

31 · Sequences, sums and products

DEFINITION **Sequence, term, explicit formula (L8 s5–6) · Summation (L8 s7) · Product (L8 s11)**
A **sequence** is an ordered list $a_m, a_{m+1}, ..., a_n$ (or infinite) of **terms**; k in a_k is the **index**. An **explicit formula** gives a_k in terms of k. For integers $m \leq n$:

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + \cdots + a_n, \qquad \prod_{k=m}^{n} a_k = a_m \cdot a_{m+1} \cdot \cdots \cdot a_n.$$

k is the **index**, m the **lower limit**, n the **upper limit**. **Recursive form:** $\sum_{k=m}^{m} a_k = a_m$ and $\sum_{k=m}^{n} a_k = \left(\sum_{k=m}^{n-1} a_k\right) + a_n$ for $n > m$; likewise for $\prod$. An **empty sum** is 0; an **empty product** is 1.

THEOREM **Theorem 5.1.1 (L8 s14) · Standard sequences (L8 s18–20)**
For real sequences a_k, b_k, real c and $n \geq m$:

$$\sum_{k=m}^{n} a_k + \sum_{k=m}^{n} b_k = \sum_{k=m}^{n} (a_k + b_k), \quad c \cdot \sum_{k=m}^{n} a_k = \sum_{k=m}^{n} c \cdot a_k, \quad \left(\prod_{k=m}^{n} a_k\right) \cdot \left(\prod_{k=m}^{n} b_k\right) = \prod_{k=m}^{n} (a_k \cdot b_k).$$

Arithmetic: $a_k = a_{k-1} + d$, so $a_n = a_0 + dn$ and $\sum_{k=0}^{n-1} a_k = \frac{n}{2}(2a_0 + (n-1)d)$.
Geometric: $a_k = ra_{k-1}$, so $a_n = a_0 r^n$ and $\sum_{k=0}^{n-1} a_k = a_0 \frac{1-r^n}{1-r}$ for $r \neq 1$.
Squares $1, 4, 9, 16, ...$; triangle numbers $1, 3, 6, 10, ...$ $(n(n+1)/2)$; lazy caterer $1, 2, 4, 7, 11, 16, ...$ $(n(n+1)/2 + 1)$.

WORKED EXAMPLE **L8 Ex. 4, 5, 7 · Telescoping, a product, change of variable**
- **Telescoping.** $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, so $\sum_{k=1}^{n} \frac{1}{k(k+1)} = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1}$.
- $\prod_{k=1}^{5} (k+2) = 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 = 2520$.
- **Change of variable.** The index is a dummy: $\sum_{k=1}^{3} k^2 = \sum_{i=1}^{3} i^2 = \sum_{k=3}^{5} (k-2)^2$. Put $j = k - 1$: $\sum_{k=1}^{n+1} \frac{k}{n+k} = \sum_{j=0}^{n} \frac{j+1}{n+j+1}$. Change the limits and every occurrence of the index together.

32 · Mathematical induction

DEFINITION Principle of Mathematical Induction, PMI (L8 s24) · Closed form (L8 s27)

Let $P(n)$ be a property defined for integers n and let a be a fixed integer. Suppose

1. $P(a)$ is true (**basis step**)
2. for all integers $k \geq a$, if $P(k)$ is true then $P(k+1)$ is true. (**inductive step**; $P(k)$ is the **inductive hypothesis**)

Then $P(n)$ is true for all integers $n \geq a$. PMI is taken as an axiom.

A sum is in **closed form** when it is written without an ellipsis or a summation symbol.

METHOD The format used in class (L8 s26) · Mistakes to avoid (T7 Q1, D2, D3)

1. Let $P(n) \equiv (...)$ for $n \geq a$. (set up the predicate)
2. Basis step: verify $P(a)$.
3. Assume $P(k)$ is true for an arbitrary $k \geq a$. (inductive hypothesis)
4. Inductive step: derive $P(k+1)$, marking where $P(k)$ is used.
5. Therefore $P(n)$ is true for all $n \geq a$ by MI.

- $P(k)$ is a **statement**, not a number: never write "$(k+1) + P(k)$".
- Do not start from $P(k+1)$ and reduce it to a true identity; start from one side of $P(k+1)$ and reach the other.
- Do not assume "$P(k)$ for all k"; that is the conclusion.
- Check the **basis value matches the claim** (T7 D2: the claim is for $n \in \mathbb{N}$, so the basis must be $n = 0$, and the step must be shown, not asserted).
- The inductive step must cover an **arbitrary** object of size $k+1$ (T7 D3: start from any group of $k+1$ people and remove one, rather than building up from a group of k).

TRAP Nine common mistakes in induction proofs (Week 7 deck)

1. **Assuming what is to be proved.** The hypothesis is $P(k)$, not $P(k+1)$; "by induction, $P(k+1)$ is true" is circular.
2. **Never using the hypothesis.** A derivation of $P(k+1)$ that does not use $P(k)$ is a direct proof, not a proof by induction.
3. **Proving the wrong implication.** The step is $P(k) \Rightarrow P(k+1)$; the converse $P(k+1) \Rightarrow P(k)$ builds no chain.
4. **Treating k as a particular number.** The step must work for an arbitrary k in the required range.
5. **Using n in place of k in the step:** writing $P(n) \Rightarrow P(n+1)$ while proving $P(n)$ for all $n \geq 1$.
6. **"Assume $P(k)$ is true for all $k \geq 1$."** That assumes the whole theorem. Write "let $k \geq 1$ be arbitrary and assume $P(k)$".
7. **An unjustified algebraic or inequality step** that forces $P(k+1)$: reaching $2^{k+1} < 2k^2$ and then claiming $2k^2 < (k+1)^2$ without checking when it holds. The structure looks perfect, yet there is a gap in the row of dominoes.
8. **Using more than the hypothesis provides.** In 1PI, assuming $P(k)$ does not give $P(k-1)$, $P(k-2)$; if earlier cases are needed, use 2PI.
9. **Too few basis cases for 2PI.** Proving $P(k) \Rightarrow P(k+2)$ with only $P(1)$ in the basis gives $P(1), P(3), P(5), ...$ and none of $P(2), P(4), P(6), ...$

PROOF Theorem 5.2.2 (5th ed. 5.2.1) · $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ for all $n \geq 1$ (L8 s26)

1. Let $P(n) \equiv (1 + 2 + \cdots + n = n(n+1)/2)$ for $n \in \mathbb{Z}^+$.
2. Basis step: $1 = 1(1+1)/2$, so $P(1)$ is true.
3. Assume $P(k)$ is true for an arbitrary $k \geq 1$: $1 + 2 + \cdots + k = k(k+1)/2$.
4. Inductive step: $1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1)$ *(inductive hypothesis)* $= \frac{(k+1)(k+2)}{2} = \frac{(k+1)((k+1)+1)}{2}$. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for all $n \in \mathbb{Z}^+$. ■

PROOF Theorem 5.2.3 (5th ed. 5.2.2) · $\sum_{i=0}^{n} r^i = \frac{r^{n+1}-1}{r-1}$ for real $r \neq 1$ and $n \geq 0$ (L8 s28)

1. Let $P(n) \equiv \left(\sum_{i=0}^{n} r^i = (r^{n+1} - 1)/(r-1)\right)$ for $n \geq 0$.
2. Basis step: $r^0 = 1 = (r^1 - 1)/(r-1)$, so $P(0)$ is true.
3. Assume $P(k)$ for an arbitrary $k \geq 0$.
4. Inductive step: $\sum_{i=0}^{k+1} r^i = \sum_{i=0}^{k} r^i + r^{k+1} = \frac{r^{k+1}-1}{r-1} + r^{k+1}$ *(inductive hypothesis)* $= \frac{r^{k+1}-1+r^{k+1}(r-1)}{r-1} = \frac{r^{(k+1)+1}-1}{r-1}$.
5. Therefore $P(n)$ is true for all $n \geq 0$. ■

PROOF Proposition 5.3.1 (5th ed. 5.3.2) · $3 \mid (2^{2n} - 1)$ for all $n \geq 0$ (L8 s29)

1. Let $P(n) \equiv (3 \mid (2^{2n} - 1))$ for $n \geq 0$.

2. Basis step: $2^0 - 1 = 0 = 3 \cdot 0$, so $P(0)$ is true.
3. Assume $P(k)$ for an arbitrary $k \geq 0$: $2^{2k} - 1 = 3r$ for some $r \in \mathbb{Z}$ *(definition of divisibility)*.
4. Inductive step: $2^{2(k+1)} - 1 = 2^{2k} \cdot 4 - 1 = 2^{2k} \cdot 3 + \left(2^{2k} - 1\right) = 2^{2k} \cdot 3 + 3r = 3\left(2^{2k} + r\right)$, and $2^{2k} + r \in \mathbb{Z}$ *(closure)*. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for all $n \geq 0$. ■

PROOF **Proposition 5.3.2 (5th ed. 5.3.3) · $2n + 1 < 2^n$ for all $n \geq 3$ (L8 s30)**

1. Let $P(n) \equiv (2n + 1 < 2^n)$ for $n \geq 3$.
2. Basis step: $2 \cdot 3 + 1 = 7 < 8 = 2^3$, so $P(3)$ is true.
3. Assume $P(k)$ for an arbitrary $k \geq 3$.
4. Inductive step: $2(k+1) + 1 = (2k + 1) + 2 < 2^k + 2$ *(inductive hypothesis)* $< 2^k + 2^k$ *($2 < 2^k$ for $k \geq 2$)* $= 2^{k+1}$.
5. Therefore $P(n)$ is true for all $n \geq 3$. ■

The basis cannot be lowered: $n = 1$ gives $3 < 2$ and $n = 2$ gives $5 < 4$, both false.

TRAP **L8 Ex. 12 · "All cows have the same colour"**

The step "cows 1 to k share a colour, cows 2 to $k+1$ share a colour, and cows 2 to k are common to both" fails for $k = 1$: the overlap $\{2, ..., k\}$ is **empty**, so nothing links cow 1 to cow 2. $P(1) \Rightarrow P(2)$ is false, and the chain breaks at its first link. The inductive step must work for **every** $k \geq a$.

WORKED EXAMPLE **L8 Ex. 13 · Tiling by L-trominoes**

$P(n)$: every $2^n \times 2^n$ board with one square removed can be tiled by L-trominoes. $P(1)$: the board is one L-tromino. Step: split a $2^{k+1} \times 2^{k+1}$ board into four $2^k \times 2^k$ quadrants; place one tromino at the centre covering one square of each quadrant not containing the removed square. Each quadrant now has one square missing and is tiled by $P(k)$.

PROOF **T7 Q2 · $1^2 + 2^2 + \cdots + n^2 = n(n+1)\frac{2n+1}{6}$** **T7 Q3 · Bernoulli: $1 + nx \leq (1+x)^n$ for real $x \geq -1$**

Q2. $P(1)$: $1 = (1 \cdot 2 \cdot 3)/6$. Step: $1^2 + \cdots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$ *(IH)* $= \frac{(k+1)(2k^2+7k+6)}{6}$ $= \frac{(k+1)(k+2)(2k+3)}{6}$, which is $P(k+1)$.

Q3. $P(1)$: $1 + x = (1+x)^1$. Step: $(1+x)^{k+1} = (1+x)^k(1+x) \geq (1+kx)(1+x)$ *(IH, and $1 + x \geq 0$)* $= 1 + (k+1)x + kx^2 \geq 1 + (k+1)x$ *($kx^2 \geq 0$)*. ■

PROOF **T7 Q4 · For odd a: $2^{n+2} \mid (a^{2^n} - 1)$ for all $n \in \mathbb{Z}^+$**

1. Let $P(n) \equiv (2^{n+2} \mid (a^{2^n} - 1))$.
2. Basis step: $a = 2p + 1$ for some $p \in \mathbb{Z}$, so $a^2 - 1 = (a-1)(a+1) = 4p(p+1) = 8m$ for some $m \in \mathbb{Z}$ *(a product of two consecutive integers is even)*. So $P(1)$ is true.
3. Assume $P(k)$: $a^{2^k} - 1 = 2^{k+2}m$ for some $m \in \mathbb{Z}$.
4. Inductive step: $a^{2^{k+1}} - 1 = \left(a^{2^k}\right)^2 - 1 = \left(a^{2^k} - 1\right)\left(a^{2^k} + 1\right) = 2^{k+2}m\left(2^{k+2}m + 2\right) = 2^{k+3}m\left(2^{k+1}m + 1\right)$. So $P(k+1)$ is true.
5. Therefore $P(n)$ is true for all $n \in \mathbb{Z}^+$ by MI. ■

PAST PAPER **AY22/23 Q21 · For the Fibonacci sequence $f_0, f_1, f_2, ... = 0, 1, 1, 2, 3, ...$ and all $n \in \mathbb{N}$: $\sum_{i=0}^{n} f_i^2 = f_n f_{n+1}$**

1. For each $n \in \mathbb{N}$ let $P(n)$ be the proposition $\sum_{i=0}^{n} f_i^2 = f_n f_{n+1}$.
2. (Basis step) $f_0^2 = 0^2 = 0 = 0 \cdot 1 = f_0 f_1$, so $P(0)$ is true.
3. (Induction step) Let $k \in \mathbb{N}$ such that $P(k)$ is true. Then $\sum_{i=0}^{k+1} f_i^2 = \sum_{i=0}^{k} f_i^2 + f_{k+1}^2 = f_k f_{k+1} + f_{k+1}^2$ *(induction hypothesis)* $= f_{k+1}(f_k + f_{k+1})$ *(basic algebra)* $= f_{k+1} f_{k+2}$ *(definition of Fibonacci)*. So $P(k+1)$ is true.
4. Therefore $\forall n \in \mathbb{N}\, P(n)$ is true by MI. ■

Check with $n = 4$: $0 + 1 + 1 + 4 + 9 = 15 = 3 \cdot 5 = f_4 f_5$. 1PI is enough: the step uses only $P(k)$ and the recurrence.

33 · Strong induction and well-ordering

DEFINITION 1PI and 2PI (L8 s37)
Let $P(n)$ be a property of integers $n \geq a$.

- **Weak (regular) induction, 1PI:** if $P(a)$ holds, and $P(k) \Rightarrow P(k+1)$ for every $k \geq a$, then $P(n)$ holds for all $n \geq a$.
- **Strong induction, 2PI:** if $P(a)$ holds, and $(P(a) \wedge P(a+1) \wedge \cdots \wedge P(k)) \Rightarrow P(k+1)$ for every $k \geq a$, then $P(n)$ holds for all $n \geq a$.
- **2PI, variant with several basis cases:** if $P(a), P(a+1), ..., P(b)$ hold, and $P(k) \Rightarrow P(k+b-a+1)$ for every $k \geq a$, then $P(n)$ holds for all $n \geq a$.

1PI, 2PI and the well-ordering principle are equivalent in power.

METHOD Choosing the form
Use 2PI when $P(k+1)$ depends on an earlier case that is not $P(k)$: a factor of $k+1$, the case $k-3$, or two or three previous terms of a recurrence. **Prove as many basis cases as the step reaches back**: a step using $P(k-3)$ needs four consecutive basis cases; a recurrence $a_{n+3} = a_{n+2} + a_{n+1} + a_n$ needs three.

PROOF L8 Ex. 14 · Every integer greater than 1 is divisible by a prime (2PI, L8 s39)

1. Let $P(n) \equiv (n$ is divisible by a prime) for $n > 1$.
2. Basis step: $2 \mid 2$ and 2 is prime, so $P(2)$ is true.
3. Inductive step: let $k \geq 2$ and assume $P(i)$ for all integers i with $2 \leq i \leq k$.
   - 3.1. Case 1 ($k+1$ is prime): $k+1$ is divisible by the prime $k+1$.
   - 3.2. Case 2 ($k+1$ is not prime): $k+1 = ab$ with $1 < a < k+1$ and $1 < b < k+1$. Then $2 \leq a \leq k$, so a prime p divides a *(inductive hypothesis)* and $a \mid (k+1)$. So $p \mid (k+1)$ *(Theorem 4.3.3)*.
4. Therefore every integer greater than 1 is divisible by a prime. ■

The well-ordering proof of the same result is in §12.

PROOF L8 Ex. 15, 16 · Every amount $n \geq 12$ is $4a+5b$ with $a, b \in \mathbb{N}$ (1PI and 2PI, L8 s40–41) T7 Q5 · $n \geq 8$ is $3x+5y$

By 1PI. $P(12)$: $12 = 4 \cdot 3$. Step: suppose $k = 4a+5b$. Case 1 ($a \geq 1$): replace one 4 by a 5: $k+1 = 4(a-1)+5(b+1)$. Case 2 ($a = 0$): then $k = 5b \geq 12$ forces $b \geq 3$; replace three 5s by four 4s: $k+1 = 4 \cdot 4 + 5(b-3)$.

By 2PI.

1. Let $P(n) \equiv (n = 4a+5b$ for some $a, b \in \mathbb{N})$ for $n \geq 12$.
2. Basis step: $12 = 4 \cdot 3 + 5 \cdot 0$; $13 = 4 \cdot 2 + 5 \cdot 1$; $14 = 4 \cdot 1 + 5 \cdot 2$; $15 = 4 \cdot 0 + 5 \cdot 3$.
3. Assume $P(i)$ for $12 \leq i \leq k$, for an arbitrary $k \geq 15$.
4. Inductive step: $12 \leq k-3 \leq k$, so $k-3 = 4a+5b$ for some $a, b \in \mathbb{N}$ *(inductive hypothesis)* then $k+1 = 4(a+1)+5b$.
5. Therefore $P(n)$ is true for all $n \geq 12$. ■

T7 Q5. Same pattern: basis $8 = 3+5$, $9 = 3 \cdot 3$, $10 = 5 \cdot 2$; step $k+3 = 3(x+1)+5y$. By 1PI instead: if $y > 0$, $k+1 = 3(x+2)+5(y-1)$; if $y = 0$, then $x \geq 3$ and $k+1 = 3(x-3)+5 \cdot 2$. (11 is not $4a+5b$ and 7 is not $3x+5y$, so the bounds are sharp.)

PROOF T7 Q6 · Every positive integer is a sum of distinct powers of 2 (2PI) T7 Q7 · $a_n < 3^n$

Q6. $P(1)$: $1 = 2^0$. Let $k \geq 1$ and assume $P(1), ..., P(k)$. Write $k+1 = 2m$ or $2m+1$ with $m \in \mathbb{Z}$; then $1 \leq m \leq k$, so $m = 2^{i_1} + \cdots + 2^{i_l}$ with $i_1 < \cdots < i_l$ *(inductive hypothesis)*. If $k+1 = 2m$, then $k+1 = 2^{i_1+1} + \cdots + 2^{i_l+1}$. If $k+1 = 2m+1$, then $k+1 = 2^0 + 2^{i_1+1} + \cdots + 2^{i_l+1}$, and $0 < i_1 + 1$. Exponents stay distinct.

Q7. $a_0 = 0, a_1 = 2, a_2 = 7$, $a_{n+3} = a_{n+2} + a_{n+1} + a_n$. Basis: $0 < 1$, $2 < 3$, $7 < 9$. Step: if $P(k), P(k+1), P(k+2)$ hold, then $a_{k+3} < 3^{k+2} + 3^{k+1} + 3^k < 3 \cdot 3^{k+2} = 3^{k+3}$. ■

PAST PAPER **T7 Q8 (AY23/24 S1 exam) · T8 Q9 (AY22/23 S2 exam) · T7 Q9 · Fibonacci and Hanoi**

T7 Q8. $P(a,b) \equiv F(a+b) = F(a+1)F(b) + F(a)F(b-1)$. Basis: $P(0,b)$: $F(b) = 1 \cdot F(b) + 0 \cdot F(b-1)$; $P(1,b)$: $F(b+1) = F(b) + F(b-1) = F(2)F(b) + F(1)F(b-1)$. Step: $F(k+1+b) = F(k+b) + F(k-1+b)$ *(definition of F)* $= (F(k+1)F(b) + F(k)F(b-1)) + (F(k)F(b) + F(k-1)F(b-1))$ *($P(k,b)$, $P(k-1,b)$)* $= (F(k+1) + F(k))F(b) + (F(k) + F(k-1))F(b-1)$ *(distributive law, twice)* $= F(k+2)F(b) + F(k+1)F(b-1)$.

T8 Q9. $\text{Even}(F_n) \Leftrightarrow \text{Even}(F_{n+3})$. Basis $P(0)$: $F_0 = 0$, $F_3 = 2$; $P(1)$: $F_1 = 1$, $F_4 = 3$. Step ($k \geq 1$): $\text{Even}(F_{k+1})$ $\Leftrightarrow (\text{Even}(F_k) \Leftrightarrow \text{Even}(F_{k-1}))$ *(Fact 1)* $\Leftrightarrow (\text{Even}(F_{k+3}) \Leftrightarrow \text{Even}(F_{k+2}))$ *($P(k)$, $P(k-1)$)* $\Leftrightarrow \text{Even}(F_{k+4})$ *(Fact 1)*.

T7 Q9, Tower of Hanoi. $T(1) = 1$. To move $k+1$ disks: move the top k to the spare peg ($2^k - 1$ moves, IH), move the largest (1 move), move the k back on top ($2^k - 1$ moves). $T(k+1) = 2(2^k - 1) + 1 = 2^{k+1} - 1$.

DEFINITION **Well-Ordering Principle for the integers (L8 s43)**

Every non-empty subset of $\mathbb{Z}_{\geq 0}$ has a smallest element. (Epp's version allows any set of integers bounded below; the course uses the statement above.)

PROOF **Well-Ordering Principle, proved by 2PI (L8 s44) · L8 Ex. 17**

1. Suppose not: let $S \subseteq \mathbb{Z}_{\geq 0}$ be non-empty with no smallest element.
2. For $n \in \mathbb{Z}_{\geq 0}$ let $P(n) \equiv (n \notin S)$.
3. Let $k \in \mathbb{Z}_{\geq 0}$ with $P(0), ..., P(k-1)$ true, i.e. $0, 1, ..., k-1 \notin S$. If $k \in S$, then k is the smallest element of S, contradicting line 1. So $k \notin S$ and $P(k)$ is true. (For $k = 0$ the hypothesis is empty, so this also gives the basis.)
4. Hence $P(n)$ is true for all $n \in \mathbb{Z}_{\geq 0}$ *(2PI)*, so $S = \varnothing$, contradicting line 1. ■

Ex. 17. (a) The positive reals have no least element ($x/2 < x$); no violation, as WOP concerns sets of integers. (b) $\{n \in \mathbb{Z}_{\geq 0} : n^2 < n\} = \varnothing$; no violation, as WOP concerns non-empty sets. (c) Non-negative integers of the form $46 - 7k$: $..., 18, 11, 4$; least element 4.

PROOF **WOP by infinite descent · PMI derived from WOP (Week 7 deck)**

WOP, alternative proof.

1. Suppose not: let $S \subseteq \mathbb{Z}_{\geq 0}$ be non-empty with no smallest element. Choose some $a \in S$.
2. a is not the smallest element, so there is $a_1 \in S$ with $a_1 < a$. But a_1 is not the smallest either, so there is $a_2 \in S$ with $a_2 < a_1$.
3. Continuing, we get an infinite descending sequence $a > a_1 > a_2 > \cdots$, which cannot occur in $\mathbb{Z}_{\geq 0}$ (only a non-negative integers lie below a). Hence S must have a smallest element. ■

Induction from WOP (smallest counterexample).

1. Suppose $P(0)$ is true and $P(k) \Rightarrow P(k+1)$ for all $k \geq 0$. Suppose some $P(k)$ is false.
2. Then the set of counterexamples $C = \{k \in \mathbb{Z}_{\geq 0} : P(k) \text{ is false}\}$ is non-empty, so C has a smallest element m *(WOP)*.
3. $m > 0$ *($P(0)$ is true)*, and $P(m-1)$ is true *(minimality of m)*.
4. Then $P(m)$ is true *($P(m-1) \Rightarrow P(m)$)*, contradicting $m \in C$. Hence $P(n)$ is true for all $n \geq 0$. ■

Dominoes: if some domino does not fall, there is a **first** one that does not; it is not domino 1 (basis), so the domino before it falls, and then it falls too.

Template for $\forall n \geq a\, P(n)$ by WOP: suppose the set of counterexamples is non-empty; take its least element m; show $m \neq a$; use the truth of P below m to get $P(m)$ or a smaller counterexample: contradiction. Mathematical induction and WOP are equivalent: each proves the other.

34 · Recursion and structural induction

DEFINITION **Recurrence relation, initial conditions (L8 s48)**

A **recurrence relation** for $a_0, a_1, a_2, ...$ is a formula relating each term a_k to certain predecessors $a_{k-1}, a_{k-2}, ..., a_{k-i}$. The **initial conditions** give the first terms $a_0, ..., a_{i-1}$.

Examples: $F_n = F_{n-1} + F_{n-2}$ with $F_0 = 0, F_1 = 1$; $\quad 0! = 1$, $n! = n \cdot (n-1)!$; $\quad a^0 = 1$, $a^n = a^{n-1} \cdot a$.

PROOF **L8 Ex. 19, 20 · For real $a_1, ..., a_n$, $b_1, ..., b_n$ and every $n \in \mathbb{Z}^+$: $\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$**

1. Let $P(n) \equiv \left(\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i\right)$ for $n \geq 1$.
2. Basis step: $\sum_{i=1}^1 (a_i + b_i) = a_1 + b_1 = \sum_{i=1}^1 a_i + \sum_{i=1}^1 b_i$, so $P(1)$ is true.
3. Assume $P(k)$ is true for an arbitrary $k \geq 1$.
4. Inductive step: $\sum_{i=1}^{k+1} (a_i + b_i) = \sum_{i=1}^k (a_i + b_i) + (a_{k+1} + b_{k+1})$ *(definition of $\sum$)* $= \sum_{i=1}^k a_i + \sum_{i=1}^k b_i + (a_{k+1} + b_{k+1})$ *(inductive hypothesis)* $= \left(\sum_{i=1}^k a_i + a_{k+1}\right) + \left(\sum_{i=1}^k b_i + b_{k+1}\right)$ *(associative and commutative laws)* $= \sum_{i=1}^{k+1} a_i + \sum_{i=1}^{k+1} b_i$ *(definition of $\sum$)*. So $P(k+1)$ is true.

5. Therefore $P(n)$ is true for every positive integer n. ■

The recursive form of $\sum$ (§31) is what lets a statement about a sum of any length be proved by induction.

DEFINITION **Recursively defined set (L8 s56, s58) · Structural induction (L8 s58)**

A recursive definition of a set S has three clauses:

- **(base clause)** certain elements, the **founders**, are in S;
- **(recursion clause)** S is closed under certain functions, the **constructors**: if f is a constructor and $x \in S$, then $f(x) \in S$;
- **(minimality clause)** membership of S can always be demonstrated by finitely many successive applications of the clauses above.

Structural induction over S. To prove $\forall x \in S\, P(x)$, it suffices to show **(basis step)** $P(c)$ for every founder c; and **(induction step)** $\forall x \in S\,(P(x) \Rightarrow P(f(x)))$ for every constructor f.

WORKED EXAMPLE **L8 Ex. 21–24 · Recursively defined sets**

- $\mathbb{Z}_{\geq 0}$: founder 0; constructor $x \mapsto x+1$. $2\mathbb{Z}$: founder 0; constructors $x \mapsto x-2$ and $x \mapsto x+2$.
- **Legal parentheses P:** base $() \in P$; if $E \in P$ then $(E) \in P$; if $E, F \in P$ then $EF \in P$. Derivation of $(())()$: $()$ by the base; $(())$ by rule (a) with $E = ()$; $(())()$ by rule (b) with $E = (())$, $F = ()$.
- **Ex. 24.** $1 \in S$; $x \in S \Rightarrow 2x \in S \wedge 3x \in S$. Then $S = \{2^a 3^b : a, b \in \mathbb{N}\}$: $9, 12 \in S$ and $10, 11, 13 \notin S$ (each has a prime factor other than 2 and 3).

METHOD **Membership questions on a recursive set**

To show $x \in S$: write a derivation, one clause per line.

To show $x \notin S$: find an **invariant** P true of every founder and preserved by every constructor, with $P(x)$ false; prove $\forall y \in S\, P(y)$ by structural induction.

PROOF **T7 Q10 · $2 \in S$; $x \in S \Rightarrow 3x \in S \wedge x^2 \in S$. Which of $0, 6, 15, 16, 36$ are in S?**

$6 = 3 \cdot 2 \in S$; $36 = 6^2 \in S$; $4 = 2^2 \in S$, so $16 = 4^2 \in S$.

$0 \notin S$: every $n \in S$ satisfies $n \geq 2$.

1. Let $P(n) \equiv (n \geq 2)$ for $n \in S$.
2. (Basis step) $2 \geq 2$, so $P(2)$ is true.
3. (Induction step) Let $x \in S$ with $x \geq 2$. Then $3x \geq 6 \geq 2$ and $x^2 \geq 4 \geq 2$, so $P(3x)$ and $P(x^2)$ are true.
4. Hence $\forall n \in S\, P(n)$ by structural induction over S. ■

$15 \notin S$: every element of S is even (2 is even; if x is even so are $3x$ and x^2), by the same argument.

PAST PAPER **AY20/21 S2 Q11 (Week 7 deck) · $T \subseteq \{\mathrm{s}, \mathrm{u}\}^*$ with $\mathrm{su} \in T$; if $\sigma, \tau \in T$ then $\mathrm{s}\,\sigma\,\mathrm{u} \in T$ and $\mathrm{s}\,\sigma\tau\,\mathrm{u} \in T$**

Which are in T? A. ssssusuuusuu B. ssussususuuu C. ssussuusssuu D. ussususussuu

$\mathbf{A} \in T$: su (base); $\mathrm{s} \cdot \mathrm{su} \cdot \mathrm{su} \cdot \mathrm{u} = \mathrm{ssusuu}$; $\mathrm{s} \cdot \mathrm{ssusuu} \cdot \mathrm{u} = \mathrm{sssusuuu}$; $\mathrm{s} \cdot \mathrm{sssusuuu} \cdot \mathrm{su} \cdot \mathrm{u} = \mathrm{ssssusuuusuu}$.

Invariants, each proved by structural induction over T (founder su; constructors $\sigma \mapsto \mathrm{s}\,\sigma\,\mathrm{u}$ and $(\sigma, \tau) \mapsto \mathrm{s}\,\sigma\tau\,\mathrm{u}$; a constructor with two arguments assumes $P(\sigma)$ **and** $P(\tau)$):

- (i) every element starts with s and ends with u; (ii) every element of length at least 3 starts with ss and ends with uu;
- (iii) no element contains usus; (iv) every element has as many s's as u's.

Proof of (iv).

1. Let $P(\sigma) \equiv$ (the number of s's in σ equals the number of u's) for $\sigma \in T$.
2. (Basis step) su has one s and one u, so $P(\mathrm{su})$ is true.
3. (Induction step) Let $\sigma, \tau \in T$ with $P(\sigma)$ and $P(\tau)$ true, having a and b s's respectively. Then $\mathrm{s}\,\sigma\,\mathrm{u}$ has $a+1$ s's and $a+1$ u's, and $\mathrm{s}\,\sigma\tau\,\mathrm{u}$ has $a+b+1$ of each. So $P(\mathrm{s}\,\sigma\,\mathrm{u})$ and $P(\mathrm{s}\,\sigma\tau\,\mathrm{u})$ are true.
4. Hence $\forall \sigma \in T\, P(\sigma)$ by structural induction over T. ■

(i) is immediate from the shape of the clauses, and (ii) follows from (i): in $\mathrm{s}\,\sigma\,\mathrm{u}$ and $\mathrm{s}\,\sigma\tau\,\mathrm{u}$ the inner strings start with s and end with u. For (iii): an occurrence of usus in a new string avoids the outer s and u, so it lies inside σ or τ (excluded by the hypothesis) or straddles the join in $\mathrm{s}\,\sigma\tau\,\mathrm{u}$, which needs σ to end in s, or $\mathrm{s}\,\sigma$ to end in usu, or $\tau\,\mathrm{u}$ to start with sus; (i) and (ii) exclude all three.

B contains usus (letters 6 to 9), against (iii). **C** has 7 s's and 5 u's, against (iv). **D** starts with u, against (i).

Only A.

PAST PAPER **AY22/23 Q14 · $1 \in S$ and $2 \in S$; if $x, y \in S$ then $x/y \in S$**
Claim: $S = \{2^n : n \in \mathbb{Z}\} = \{..., \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, ...\}$.
1. ($\subseteq$) Let $P(x) \equiv (x = 2^n$ for some $n \in \mathbb{Z})$ for $x \in S$. (Basis step) $1 = 2^0$ and $2 = 2^1$. (Induction step) Let $x, y \in S$ with $x = 2^a$ and $y = 2^b$, $a, b \in \mathbb{Z}$. Then $x/y = 2^{a-b}$ and $a - b \in \mathbb{Z}$, so $P(x/y)$ is true. Hence $\forall x \in S\, P(x)$ by structural induction over S.
2. ($\supseteq$) $2^{-1} = 1/2 \in S$. If $2^n \in S$ then $2^{n+1} = 2^n/2^{-1} \in S$, so $2^n \in S$ for all $n \geq 0$ by MI; and then $2^{-n} = 1/2^n \in S$. ■

A. $\forall x \in S\,(x \in \mathbb{Z} \Rightarrow x$ is even): false, $x = 1$. B. $\forall x \in S\,(x \in \mathbb{Q} \Rightarrow x < 1)$: false, $x = 2$.
C. $\forall x, y \in S\,(x + y \geq 1)$: false, $\frac{1}{2} + \frac{1}{4} = \frac{3}{4} < 1$. D. $|S| = |\mathbb{Z}|$: **true**, since $f : \mathbb{Z} \to S$, $f(n) = 2^n$ is a bijection.

PAST PAPER **T7 Q11 · AY25 final Q14–Q16 · Recursive sets of sets and strings**
T7 Q11. $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 3, 5, 7, 9\}$ are founders; constructors $\cap$, $\cup$, $\setminus$. $C = \{2, 4, 7, 9\}$ $= (A \setminus B) \cup (B \setminus A) \in S$. $D = \{2, 3, 4, 5\} \notin S$: the invariant "$1 \in X \Leftrightarrow 3 \in X$" holds for A and B and is preserved by $\cap$, $\cup$, $\setminus$ (check the four cases of $1, 3 \in X$ or not, $1, 3 \in Y$ or not), but fails for D.
AY25 Q14, Q15. $a \in S$; $x \in S \Rightarrow xa \in S$; $x \in S \Rightarrow bxb \in S$. Elements of length at most 4: a; aa; aaa, bab; $aaaa$, $baba$, $baab$: **7**. A string of length n arises from one of length $n-1$ (append a) or $n-2$ (wrap in b), in exactly one way, so $G_n = G_{n-1} + G_{n-2}$ with $G_1 = G_2 = 1$: $G_n = F_n$.
AY25 Q16. $S_0 = \{1, 2, 3\}$, $S_{n+1} = (S_n \cup \{x + 3 : x \in S_n\}) \setminus \{2, 3\}$. $S_1 = \{1, 4, 5, 6\}$, $S_2 = \{1, 4, 5, 6, 7, 8, 9\}$, and so on: the union of all S_n is $\mathbb{Z}^+$ (2 and 3 come from S_0).

Part VIII · Cardinality

35 · Pigeonhole principle and finite sets

THEOREM **Pigeonhole Principle · Dual Pigeonhole Principle (L9 s6)**
Let A and B be **finite** sets.
- **PHP.** If there is an injection $f : A \to B$, then $|A| \leq |B|$. Contrapositive: if $m > n$ pigeons are put into n pigeonholes, some pigeonhole holds at least two pigeons.
- **Dual PHP.** If there is a surjection $f : A \to B$, then $|A| \geq |B|$. Contrapositive: if $m < n$ pigeons are put into n pigeonholes, some pigeonhole is empty.

PROOF **PHP and Dual PHP (L9 s7–8)**
PHP.
1. Let $A = \{a_1, a_2, ..., a_m\}$ with $m = |A|$.
2. If $a_i \neq a_j$ then $f(a_i) \neq f(a_j)$ *(f is injective)*, so $f(a_1), ..., f(a_m)$ are m different elements of B.
3. Hence $|B| \geq m = |A|$. ■

Dual PHP.
1. Let $B = \{b_1, b_2, ..., b_n\}$ with $n = |B|$. For each b_i pick $a_i \in A$ with $f(a_i) = b_i$ *(f is surjective)*.
2. If $b_i \neq b_j$ then $f(a_i) \neq f(a_j)$, so $a_i \neq a_j$ *(f is a function)*.
3. So $a_1, ..., a_n$ are n different elements of A, and $|A| \geq n = |B|$. ■

DEFINITION **Finite, infinite (L9 s9) · Cardinality of a finite set (L9 s10)**
Let $\mathbb{Z}_n = \{1, 2, ..., n\}$ (in this lecture only). A set S is **finite** iff $S = \varnothing$ or there is a bijection from S to $\mathbb{Z}_n$ for some $n \in \mathbb{Z}^+$; otherwise S is **infinite**.
The **cardinality** of a finite set S is $|S| = 0$ if $S = \varnothing$, and $|S| = n$ if there is a bijection $f : S \to \mathbb{Z}_n$.

THEOREM **Finite sets (L9 s10–11)**
- **Equality of cardinality.** For finite A, B: $|A| = |B|$ iff there is a bijection $f : A \to B$. ($\Leftarrow$: PHP and Dual PHP. $\Rightarrow$: map $a_i \mapsto b_i$.)
- **Theorem Cardinality.1.** Any subset of a finite set is finite. Contrapositive: if $A \subseteq B$ and A is infinite, then B is infinite.

PAST PAPER **T8 Q8 (AY22/23 S1 exam) · $|A| \leq |R|$ for a relation R on a finite set A?**
(i) Reflexive: true. Define $f : A \to R$ by $f(x) = (x, x)$; this is a function into R because R is reflexive. If $f(x_1) = f(x_2)$ then $(x_1, x_1) = (x_2, x_2)$, so $x_1 = x_2$ *(equality of ordered pairs)*. f is an injection, so $|A| \leq |R|$ *(PHP)*.
(ii) Symmetric, (iii) transitive: false. The empty relation on a non-empty A is symmetric and transitive, with $|A| > 0 = |\varnothing|$.

36 · Same cardinality

DEFINITION **Same cardinality (Cantor, L9 s12)**
For **any** sets A and B: A has the **same cardinality** as B, written $|A| = |B|$, iff there is a bijection $f : A \to B$.
This defines "$|A| = |B|$" without defining $|A|$ for infinite sets.

PROOF **Theorem 7.4.1 · Same-cardinality is an equivalence relation (L9 s14–16)**
1. (Reflexive) $\mathrm{id}_A : A \to A$ is injective ($\mathrm{id}_A(x_1) = \mathrm{id}_A(x_2)$ gives $x_1 = x_2$) and surjective ($\mathrm{id}_A(x) = x$). So $|A| = |A|$.
2. (Symmetric) Let $|A| = |B|$, with a bijection $f : A \to B$ *(Cantor's definition)*. Then $f^{-1} : B \to A$ is a bijection *(Theorem 7.2.3)*. So $|B| = |A|$.
3. (Transitive) Let $|A| = |B|$ and $|B| = |C|$, with bijections $f : A \to B$ and $g : B \to C$. Then $g \circ f$ has the inverse $f^{-1} \circ g^{-1}$ *(T6 Q3, §29)*, so $g \circ f : A \to C$ is a bijection *(Theorem 7.2.3)*. So $|A| = |C|$. ■

PROOF $|2\mathbb{Z}| = |\mathbb{Z}|$ **(L9 s17–18) · L9 Ex. 1, 2**
Define $H : \mathbb{Z} \to 2\mathbb{Z}$ by $H(n) = 2n$.
1. (Injective) If $H(n_1) = H(n_2)$ then $2n_1 = 2n_2$, so $n_1 = n_2$.
2. (Surjective) Let $m \in 2\mathbb{Z}$. Then $m = 2k$ for some $k \in \mathbb{Z}$ *(definition of even)*, and $H(k) = 2k = m$.
3. So H is a bijection and $|\mathbb{Z}| = |2\mathbb{Z}|$ *(Cantor's definition)*. ■

Likewise $|\mathbb{N}| = |\mathbb{N} \setminus \{0\}|$ via $x \mapsto x + 1$, and $|\mathbb{N}| = |\mathbb{N} \setminus \{1, 3, 5, ...\}|$ via $x \mapsto 2x$.
An infinite set can have the same cardinality as a proper subset of itself. For a finite A and a proper subset B, always $|B| < |A|$.
Alternative definition of infinite (Week 8 deck). Some mathematicians use this as the definition: A is infinite iff there is a set B with $B \subseteq A \wedge B \neq A \wedge |B| = |A|$.
Week 8 deck checks: $|\mathbb{Z}^-| = |\mathbb{Z}^+|$, $|\mathbb{Z} \cup \{a, b, c\}| = |\mathbb{Z}|$, $|\mathbb{Z}| = |\mathbb{Z}^+|$, $|\mathbb{Z}| = |2\mathbb{Z}|$, $|\mathbb{Q}| = |\mathbb{Z}|$ are all true; $|\{a, b, c\}| = |\{a, b\}|$ is false.

37 · Countable sets

DEFINITION **$\aleph_0$ · Countably infinite · Countable, uncountable (L9 s23)**
$\aleph_0 = |\mathbb{Z}^+|$ (some authors use $\mathbb{N}$; either may be used). A set S is **countably infinite** iff $|S| = \aleph_0$, i.e. there is a bijection between $\mathbb{Z}^+$ (or $\mathbb{N}$) and S.
A set is **countable** iff it is finite or countably infinite; otherwise it is **uncountable**.

THEOREM **Proposition 9.1 · Lemma 9.2 · Countability via sequences (L9 s33–34)**
For an **infinite** set B:
- **Proposition 9.1.** B is countable iff there is a sequence $b_0, b_1, b_2, ... \in B$ in which every element of B appears **exactly once**.
- **Lemma 9.2.** B is countable iff there is a sequence $b_0, b_1, b_2, ...$ in which every element of B **appears**.

A bijection $f : \mathbb{Z}_{\geq 0} \to B$ is the sequence $f(0), f(1), ...$: surjective = every element appears at least once; injective = at most once. For Lemma 9.2 ("if"): delete the terms not in B and all but the first appearance of each element; what remains lists B exactly once.

WORKED EXAMPLE **Standard countable sets (L9 s24–30, s35) · T8 Q1, D1**
- $\mathbb{Z}$**:** list $0, 1, -1, 2, -2, ...$ Formula on $\mathbb{Z}^+$: $f(n) = n/2$ if n is even, $-(n-1)/2$ if n is odd. On $\mathbb{N}$ (T8 Q1): $g(n) = (-1)^n \left\lfloor \frac{n+1}{2} \right\rfloor$, giving $0, -1, 1, -2, 2, ...$
- $\mathbb{Z}^+ \times \mathbb{Z}^+$**:** count along the diagonals $(1,1), (1,2), (2,1), (1,3), (2,2), (3,1), ...$ The bijection is $f(x, y) = \frac{(x+y-2)(x+y-1)}{2} + x$.
- $\mathbb{Q}^+$**:** put p/q in a grid, walk the diagonals, and skip values already counted: $\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{3}{1}, \frac{1}{3}, ...$ Skipping gives injectivity; every grid point is reached, giving surjectivity.
- $\mathbb{Q}$**:** $x \mapsto -x$ is a bijection $\mathbb{Q}^+ \to \mathbb{Q}^-$, so $\mathbb{Q}^-$ is countably infinite. List $\mathbb{Q}^+$ as $q_1, q_2, q_3, ...$; then $0, q_1, -q_1, q_2, -q_2, ...$ contains every element of $\mathbb{Q} = \mathbb{Q}^- \cup \{0\} \cup \mathbb{Q}^+$ exactly once, so $\mathbb{Q}$ is countable (Proposition 9.1) and $|\mathbb{Q}| = |\mathbb{Z}|$. Equivalently: Lemma 9.4 for $\mathbb{Q}^- \cup \mathbb{Q}^+$, then T8 Q2 for $\{0\}$.
- **Perfect squares** $\{0, 1, 4, 9, ...\}$ (T8 D1): $n \mapsto n^2$ is a bijection from $\mathbb{N}$; also a subset of $\mathbb{N}$ (Theorem 7.4.3).

PROOF **T8 Q1 ·** $g : \mathbb{N} \to \mathbb{Z}$**,** $g(n) = (-1)^n \left\lfloor \frac{n+1}{2} \right\rfloor$ **is a bijection (so $\mathbb{Z}$ is countable with $\aleph_0 = |\mathbb{N}|$)**
For even n: $g(n) = \lfloor (n+1)/2 \rfloor = n/2 \geq 0$. For odd n: $g(n) = -(n+1)/2 < 0$.
1. (Injective) Let $a, b \in \mathbb{N}$ with $g(a) = g(b)$. Then $g(a), g(b)$ are both non-negative or both negative.
   1.1. Case 1 (both non-negative): a and b are even, so $a/2 = b/2$ and $a = b$.

1.2. Case 2 (both negative): a and b are odd, so $-(a+1)/2 = -(b+1)/2$ and $a = b$.
2. (Surjective) Let $m \in \mathbb{Z}$.
   2.1. Case 1 ($m \geq 0$): let $n = 2m$. Then $n \in \mathbb{N}$ and $g(n) = (-1)^{2m}\lfloor(2m+1)/2\rfloor = m$.
   2.2. Case 2 ($m < 0$): let $n = -2m - 1$. Then $n \in \mathbb{N}$ is odd and $g(n) = (-1)^{-2m-1}\lfloor(-2m)/2\rfloor = -(-m) = m$.
3. Therefore g is a bijection from $\mathbb{N}$ to $\mathbb{Z}$. ■

THEOREM **Closure theorems (L9 s31, s41–45)**

- **Cartesian product.** If A and B are countably infinite, so is $A \times B$; likewise $A_1 \times \cdots \times A_n$ for $n \geq 2$.
- **Unions.** The union of countably many countable sets is countable: if $A_1, A_2, \ldots$ are countable, so is $\bigcup_{i=1}^{\infty} A_i$.
- **Theorem 7.4.3.** Any subset of a countable set is countable. **Corollary 7.4.4.** Any set with an uncountable subset is uncountable.
- **Proposition 9.3.** Every infinite set has a countably infinite subset.
- **Lemma 9.4.** If A and B are countably infinite, then $A \cup B$ is countable.

PROOF **Lemma 9.4 (L9 s45) · Proposition 9.3 (L9 s44) · Theorem 7.4.3 (L9 s41–42)**

Lemma 9.4. Take sequences $a_0, a_1, \ldots$ and $b_0, b_1, \ldots$ in which every element of A, respectively B, appears *(Lemma 9.2)*. Then $a_0, b_0, a_1, b_1, \ldots$ is a sequence in which every element of $A \cup B$ appears, so $A \cup B$ is countable *(Lemma 9.2)*.

Proposition 9.3. Let B be infinite. Choose $b_0, b_1, b_2, \ldots$ from B one at a time, each different from all earlier choices; this is always possible, since otherwise $B = \{b_0, \ldots, b_{n-1}\}$ would be finite. Then $\{b_0, b_1, \ldots\} \subseteq B$ is countably infinite.

Theorem 7.4.3. Let A be countable and $B \subseteq A$. If B is finite it is countable. Otherwise A is countably infinite; take a bijection $f : \mathbb{Z}^+ \to A$ and let $M = f^{-1}(B)$. Define $g(1) = f(i_1)$ with $i_1 = \min M$ and $g(k) = f(i_k)$ with $i_k = \min\{m \in M : m > i_{k-1}\}$. Then $g : \mathbb{Z}^+ \to B$ lists the elements of B in order of their indices, each exactly once, so it is a bijection. ■

PROOF **T8 Q2 · B countably infinite, C finite $\Rightarrow B \cup C$ countable** **T8 Q5 · A countable union of countably infinite sets is countable**

Q2. Take a sequence $b_0, b_1, \ldots$ in which every element of B appears *(Lemma 9.2)* and write $C = \{c_0, \ldots, c_{n-1}\}$. Then $c_0, \ldots, c_{n-1}, b_0, b_1, \ldots$ contains every element of $B \cup C$, so $B \cup C$ is countable *(Lemma 9.2)*. (As a bijection: with $C \setminus B = \{c_0, \ldots, c_{k-1}\}$ and a bijection $f : \mathbb{N} \to B$, set $g(x) = c_x$ if $x < k$ and $f(x-k)$ otherwise.)

Q5.
1. $\mathbb{Z}^+ \times \mathbb{Z}^+$ is countable, so there is a bijection $f : \mathbb{Z}^+ \to \mathbb{Z}^+ \times \mathbb{Z}^+$.
2. For each $i \in \mathbb{Z}^+$ take a sequence $b_{i,1}, b_{i,2}, \ldots$ in which every element of S_i appears *(Lemma 9.2)*.
3. Define $c_k = b_{i,j}$ where $(i, j) = f(k)$.
4. Let $x \in \bigcup_{i \in \mathbb{Z}^+} S_i$. Then $x \in S_i$ for some i, so $x = b_{i,j}$ for some j; and $f(k) = (i, j)$ for some k *(f is surjective)*. So $x = c_k$.
5. Every element of the union appears in $c_1, c_2, \ldots$, so the union is countable *(Lemma 9.2)*. ■

PROOF **T8 Q4(a) · If $A_1, A_2, A_3, \ldots$ are countable, then $\bigcup_{i=1}^{n} A_i$ is countable for every $n \in \mathbb{Z}^+$**

1. Let $P(n) \equiv \left(\bigcup_{i=1}^{n} A_i \text{ is countable}\right)$ for $n \in \mathbb{Z}^+$.
2. Basis step: $\bigcup_{i=1}^{1} A_i = A_1$ is countable, so $P(1)$ is true.
3. Inductive step: suppose $P(k)$ is true for an arbitrary $k \geq 1$. Write $\bigcup_{i=1}^{k+1} A_i = \left(\bigcup_{i=1}^{k} A_i\right) \cup A_{k+1} = U \cup A_{k+1}$.
   3.1. Case 1 (U and A_{k+1} both finite): $U \cup A_{k+1}$ is finite, hence countable.
   3.2. Case 2 (exactly one is finite): the other is countably infinite, so $U \cup A_{k+1}$ is countable *(T8 Q2)*.
   3.3. Case 3 (neither is finite): both are countably infinite, so $U \cup A_{k+1}$ is countable *(Lemma 9.4)*.
4. Therefore $\bigcup_{i=1}^{n} A_i$ is countable for every $n \in \mathbb{Z}^+$ by MI. ■

(b) This does not prove that $\bigcup_{i=1}^{\infty} A_i$ is countable; see the trap below and T8 Q5.

TRAP **T8 Q3, Q4 · Induction reaches every finite n, never infinity**

- "$\bigcup_{i=1}^{n} A_i$ is finite (or countable) for every n" is proved by induction on n. It does **not** follow that $\bigcup_{i=1}^{\infty} A_i$ is finite: $A_i = \{i\}$ gives $\mathbb{Z}^+$. The infinite union of countable sets **is** countable, but that needs the argument of T8 Q5.
- In an inductive step the new set A_{k+1} must be **arbitrary**; choosing $A_{k+1} = \varnothing$ proves nothing.
- T8 D2: "uncountable and infinite" is redundant, since every uncountable set is infinite.

38 · Uncountable sets

PROOF **Theorem 7.4.2 (Cantor) · The interval $(0, 1) = \{x \in \mathbb{R} : 0 < x < 1\}$ is uncountable (L9 s38–40)**

1. Suppose $(0, 1)$ is countable. It is not finite, so it is countably infinite.
2. List its elements as a sequence $x_1, x_2, x_3, \ldots$ with decimal expansions $x_n = 0.a_{n1}a_{n2}a_{n3}\ldots$, each $a_{ij} \in \{0, 1, \ldots, 9\}$ (use 0.5000... and not 0.4999...).

3. Construct $d = 0.d_1d_2d_3...$ with $d_n = 1$ if $a_{nn} \neq 1$, and $d_n = 2$ if $a_{nn} = 1$.
4. For every $n \in \mathbb{Z}^+$, $d_n \neq a_{nn}$, so $d \neq x_n$.
5. But $d \in (0,1)$, so d should be in the list: contradiction. Therefore $(0,1)$ is uncountable. ■

This is **Cantor's diagonalization argument**: build an object that differs from the n-th listed object at position n.

TRAP **Week 8 deck FAQ · Why diagonalization does not make $\mathbb{Z}^+$ uncountable**

List the positive integers in decimal, padded with leading zeros (...000001, ...000002, ...), and change one digit of each along a diagonal. The resulting object typically has **infinitely many non-zero digits** extending to the left, like ...3141592653. An integer has a finite decimal representation $d_nd_{n-1}\cdots d_2d_1$, so this object is **not a positive integer**: it is missing from the list, but it was never required to be on it. A real in $(0,1)$ may have infinitely many digits $0.d_1d_2d_3...$, so there the diagonal object **is** in the set.

Same contrast: the set of all **finite** binary strings $\{\varepsilon, 0, 1, 00, 01, ...\}$ is countable; the set of all **infinite** binary strings $b_1b_2b_3...$ is uncountable.

THEOREM **Consequences (L9 s43, s46–47)**

- $\mathbb{R}$ is uncountable, since $(0,1) \subseteq \mathbb{R}$ *(Corollary 7.4.4)*.
- $|\mathbb{R}| = |(0,1)|$: bend $(0,1)$ into a circle resting on the number line at 0 and project each point from the top of the circle onto the line; this projection is a bijection. (One formula: $x \mapsto \tan(\pi(x - 1/2))$.)
- The irrationals $\mathbb{R} \setminus \mathbb{Q}$ are uncountable: otherwise $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be countable.

PROOF **T8 Q7 · If A is countably infinite, then $\mathcal{P}(A)$ is uncountable**

1. Suppose $\mathcal{P}(A)$ is countable. It is infinite, since $\{a\} \in \mathcal{P}(A)$ for every $a \in A$.
2. There is a sequence $a_0, a_1, a_2, ...$ in which every element of A appears exactly once, and a sequence $B_0, B_1, B_2, ...$ in which every element of $\mathcal{P}(A)$ appears exactly once *(Proposition 9.1)*.
3. Define $B = \{a_i : a_i \notin B_i\}$. Then $B \in \mathcal{P}(A)$.
4. Let $i \in \mathbb{N}$. If $a_i \notin B_i$ then $a_i \in B$; if $a_i \in B_i$ then $a_i \notin B$ (each a_i appears once only). Either way $B \neq B_i$.
5. So B is not in the sequence, contradicting line 2. Therefore $\mathcal{P}(A)$ is uncountable. ■

PROOF **T8 Q6 · For X infinite and Y finite there is a bijection $X \cup Y \to X$**

1. Take a countably infinite $X_0 \subseteq X$ *(Proposition 9.3)* and let $Y_0 = Y \setminus X$, a finite set.
2. $X_0 \cup Y_0$ is countably infinite *(T8 Q2)*, so there is a bijection $f : X_0 \cup Y_0 \to X_0$.
3. Define $g : X \cup Y \to X$ by $g(z) = f(z)$ if $z \in X_0 \cup Y_0$, and $g(z) = z$ otherwise. Then g is a bijection. ■

METHOD **Countable or uncountable? (T8 D3 · AY25 final Q13, Q17, Q18)**
Show countable: exhibit a sequence containing every element (Lemma 9.2), or build the set from countable sets by subset, finite product, or countable union.
Show uncountable: diagonalize, or find an uncountable subset, or biject with $\mathcal{P}(\mathbb{N})$ or $\mathbb{R}$.

Set		Reason
$\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Z}^+ \times \mathbb{Z}^+$, $\mathbb{Q} \times \mathbb{Q}$ (points with rational coordinates)	countable	sequences above; product theorem
all functions $\{0,1\} \to \mathbb{N}$	countable	same as $\mathbb{N} \times \mathbb{N}$
all 2-element subsets of $\mathbb{N}$; all finite strings over a finite alphabet	countable	inject into $\mathbb{N} \times \mathbb{N}$; countable union of finite sets
any subset of $\mathbb{Q}$; any partition of $\mathbb{N}$ (as a set of components)	countable	Theorem 7.4.3; send each component to its least element
$\mathbb{Z}^*$: all finite strings over the countably infinite alphabet $\mathbb{Z}$ (AY22/23 Q17)	countable	$\mathbb{Z}^* = \mathbb{Z}^0 \cup \mathbb{Z}^1 \cup \mathbb{Z}^2 \cup \cdots$; each $\mathbb{Z}^n$ is countable (general product); countable union (T8 Q5)
$\mathbb{Q}$; the rationals strictly between 0 and 1 (AY22/23 Q17)	countable	$\mathbb{Q}$ from $\mathbb{Q}^+$ (§37); subset of $\mathbb{Q}$
$(0,1)$, $\mathbb{R}$, $\mathbb{R} \setminus \mathbb{Q}$	uncountable	Theorem 7.4.2 and Corollary 7.4.4
$\mathcal{P}(\mathbb{N})$; all functions $\mathbb{N} \to \{0,1\}$; all infinite sequences of integers	uncountable	T8 Q7; diagonalization
all relations on $\mathbb{Z}$ (AY22/23 Q17)	uncountable	a relation on $\mathbb{Z}$ is a subset of $\mathbb{Z} \times \mathbb{Z}$, so the set is $\mathcal{P}(\mathbb{Z} \times \mathbb{Z})$ with $\mathbb{Z} \times \mathbb{Z}$ countably infinite (T8 Q7)
the set of all partitions of $\mathbb{N}$	uncountable	$\{A, \mathbb{N} \setminus A\}$ for each A gives uncountably many

AY22/23 Q17. Countable among (i) relations on $\mathbb{Z}$, (ii) rationals in $(0,1)$, (iii) $\mathbb{Z}^*$: only (ii) and (iii), option D.
AY25 Q17. A countably infinite, B uncountable: $A \cup B$ is uncountable (it contains B) and $A \cap B$ is countable (a subset of A). Both true.
AY25 Q13. $(\mathbb{Z} \times \mathbb{Z}^+)/\sim$ with $(a,b) \sim (c,d) \leftrightarrow ad = bc$ is in bijection with $\mathbb{Q}$, hence with $\mathbb{Z}$ and with $\mathbb{Z} \times \mathbb{Z}^+$: all countably infinite.

Part IX · Counting and Probability

39 · Sample spaces, the multiplication rule, permutations

DEFINITION **Sample space, event (L10 s8) · Equally likely probability (L10 s9)**
A **sample space** is the set of all possible outcomes of a random process or experiment. An **event** is a subset of a sample space.
If S is a **finite** sample space in which **all outcomes are equally likely** and E is an event in S, then

$$P(E) = \frac{|E|}{|S|} = \text{the number of outcomes in } \frac{E}{\text{the total number of outcomes in}} S.$$

THEOREM **Theorem 9.1.1 · Number of elements in a list (L10 s12)** **Theorem 9.2.1 · Multiplication rule (L10 s21)**
- If $m \le n$ are integers, there are $n - m + 1$ integers from m to n inclusive.
- If an operation consists of k steps, where step i can be performed in n_i ways **regardless of how the preceding steps were performed**, then the whole operation can be performed in $n_1 n_2 \cdots n_k$ ways.
- **Theorem 5.2.4 (L10 s24).** A finite set A has $|\mathcal{P}(A)| = 2^{|A|}$ subsets: each element is picked or dropped.

WORKED EXAMPLE **L10 s13, Ex. 1–5 · Lists, possibility trees, the multiplication rule**
- **Multiples of 5 from 100 to 999:** $5 \cdot 20, \ldots, 5 \cdot 199$, so $199 - 20 + 1 = 180$; probability $180/900 = 1/5$.
- **Tournament** (play until one team wins two in a row or three in total): the possibility tree has **10** leaves; 4 of them need five games, so $P = 4/10 = 2/5$ if all are equally likely.
- **PINs** of 4 symbols from 36: $36^4 = 1\,679\,616$ with repetition; $36 \cdot 35 \cdot 34 \cdot 33 = 1\,413\,720$ without; $P(\text{no repeat}) \approx 0.8417$.

- **Officers** (president, treasurer, secretary from Ann, Bob, Cyd, Dan; Ann is not president; Cyd or Dan is secretary). $3 \cdot 3 \cdot 2 = 18$ is **wrong**: the number of choices for secretary depends on earlier choices. Reorder the steps: secretary (2 ways), then president (2), then treasurer (2): $2 \cdot 2 \cdot 2 = 8$.
- **ICANDOIT** from 26 tiles: without replacement the probability is 0 (the letter I is needed twice); with replacement it is $(1/26)^8$.

DEFINITION **Permutation (L10 s32) · r-permutation, $P(n, r)$ (L10 s36)**
A **permutation** of a set of objects is an ordering of the objects in a row. An ***r*-permutation** of a set of n elements is an ordered selection of r of its elements; their number is $P(n, r)$.

THEOREM **Theorem 9.2.2 · Theorem 9.2.3 (L10 s34, s37) · Circular permutations (T9 Q5)**
- A set with $n \geq 1$ elements has $n!$ permutations; $0! = 1$.
- For $1 \leq r \leq n$: $P(n, r) = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}$. $P(5, 2) = 20$, $P(7, 4) = 840$, $P(5, 5) = 120$.
- n distinct objects can be arranged around a circle in $(n-1)!$ ways (rotations are the same arrangement).

WORKED EXAMPLE **L10 Ex. 6 · T9 Q1, Q2, Q5**
- **COMPUTER:** $8! = 40\,320$ orderings; with CO kept together in order, treat CO as one unit: $7! = 5040$; probability $7!/8! = 1/8$.
- **T9 Q1.** 5-card sequences with at least one picture card (12 picture cards, 40 others): with replacement $52^5 - 40^5 = 277\,804\,032$; without replacement $P(52, 5) - P(40, 5) = 232\,914\,240$.
- **T9 Q2 (AY23/24 exam).** Five-digit cheque numbers readable upside down use digits $0, 1, 6, 8, 9$: 5^5. Those that read the same after rotation (the middle digit is 0, 1 or 8; the last two digits are forced): $5 \cdot 5 \cdot 3 = 75$. Confusable numbers: $5^5 - 75 = 3050$.
- **T9 Q5.** (a) 8 boys and 4 girls at a round table, no two girls adjacent: seat the boys (7!), then put the girls into 4 of the 8 gaps ($P(8, 4)$): $7! \cdot 1680 = 8\,467\,200$. (b) 6 people, Eric not next to Freddy: $5! - 2 \cdot 4! = 72$. (c) $n-1$ people at n chairs: treat the empty chair as a person: $(n-1)!$.

METHOD **Arrangements in a row and round a table (L10 Ex. 6 · Week 10 s16, s74–76 · T9 Q5)**
- **Kept together (glue).** Treat the block as one object, arrange the objects, then order the inside of each block. 12 distinct cards (4 suits, 3 ranks each) in a row with each suit together: $4! \cdot (3!)^4 = 31\,104$. Identical letters need no inner ordering: MISSISSIPPI with the four S's consecutive is the block (SSSS) with M, I, I, I, I, P, P: $8!/(4!\,2!) = 840$.
- **Not adjacent.** Either total minus glued, or the **gap method**: arrange the other objects first, then drop the separated ones into distinct gaps. 5 people in a row with X and Y apart: $5! - 2 \cdot 4! = 72 = 3! \cdot P(4, 2)$. A row of m objects has $m+1$ gaps; a circle of m objects has m gaps (T9 Q5(a)).
- **Alternating in a row.** 3 red and 3 black cards, all distinct: the pattern is RBRBRB or BRBRBR, each filled in $3! \cdot 3!$ ways: $2 \cdot 36 = 72$. Round a table the two patterns coincide: $(n-1)!\,n!$.
- **Relative order.** 6 distinct cards in a row with A somewhere to the left of B: swapping A and B pairs the arrangements off, so exactly half qualify: $6!/2 = 360$.
- **Round a table.** Fix one person's seat to remove the rotations, then count the rest as a row.

WORKED EXAMPLE **Case split when a step's count depends on earlier choices (Week 10 s77)**
4-digit even numbers with no repeated digit. Choose the last digit first; the number of choices for the first digit then depends on whether 0 has been used, so split.
- **Last digit 0:** first digit 9 ways, then 8, then 7: $9 \cdot 8 \cdot 7 = 504$.
- **Last digit 2, 4, 6 or 8:** first digit 8 ways (not 0, not the last digit), second 8, third 7: $4 \cdot 8 \cdot 8 \cdot 7 = 1792$.

Total $504 + 1792 = 2296$ *(addition rule on disjoint cases)*.

PAST PAPER **AY20/21 Q20(d) · AY23/24 Q22, Q25(f)**
AY20/21 Q20(d). Directed graphs on vertices a, b, c without parallel edges: each of the $3 \cdot 3 = 9$ ordered pairs (loops included) is an edge or not: $2^9 = 512$. With at least one loop: subtract the graphs with no loop, $2^9 - 2^6 = 448$.
AY23/24 Q22. 8 people round a table, the three roommates not **all three** together: all seatings 7!; glue the three into one block (6 units round the table: 5!, inner orders 3!): $7! - 5! \cdot 3! = 4320$ (option B).
AY23/24 Q25(f). Five-digit cheque numbers that can be read both ways up: each digit is one of $0, 1, 6, 8, 9$: $5^5 = 3125$. Excluding the 75 that read the same after rotation (the T9 Q2 version) leaves 3050.

40 · Addition, difference, inclusion and exclusion

THEOREM **Theorems 9.3.1–9.3.3 (L10 s40, s43, s50) · Complement (L10 s48)**

- **Addition rule.** If a finite set A is the union of mutually disjoint subsets $A_1, ..., A_k$, then $|A| = |A_1| + |A_2| + \cdots + |A_k|$.
- **Difference rule.** If A is finite and $B \subseteq A$, then $|A \setminus B| = |A| - |B|$. (Proof: B and $A \setminus B$ are disjoint with union A; apply the addition rule.)
- **Inclusion/exclusion.** For finite A, B, C: $|A \cup B| = |A| + |B| - |A \cap B|$ and $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.
- **Complement.** For an event A in a finite sample space S: $P(\overline{A}) = 1 - P(A)$.

WORKED EXAMPLE **L10 Ex. 7–9 · T9 Q3**

- **Passwords** of 1 to 3 letters: $26 + 26^2 + 26^3 = 18\,278$ (addition rule over the lengths).
- **PINs with a repeated symbol:** $36^4 - 36 \cdot 35 \cdot 34 \cdot 33 = 265\,896$; probability $\approx 0.158 = 1 - 0.842$.
- **Multiples of 3 or 5 from 1 to 1000:** $333 + 200 - 66 = 467$ (multiples of both are the multiples of 15). **Neither:** $1000 - 467 = 533$, using De Morgan: $\overline{A} \cap \overline{B} = \overline{A \cup B}$.
- **T9 Q3.** Permutations of $1, ..., n$ ($n \geq 3$) with 1, 2 or 3 in its correct position: $|P_k| = (n-1)!$, $|P_i \cap P_j| = (n-2)!$, $|P_1 \cap P_2 \cap P_3| = (n-3)!$, so the count is $3(n-1)! - 3(n-2)! + (n-3)! = (3n^2 - 12n + 13)(n-3)!$.

METHOD **"At least one" and "at most"**

"At least one" is usually shortest by the difference rule: total minus "none". "At most one" is "none" plus "exactly one" by the addition rule. Overlapping conditions joined by "or" need inclusion/exclusion. Before multiplying, check that each step's count is independent of earlier choices; if not, reorder the steps or split into cases.

METHOD **Each at least once: complement, then inclusion/exclusion (T10 D3 · Week 10 s13–14 · AY23/24 Q25(g) · AY24/25 Q24(a))**

"Every value is used" is the complement of "some value is missed". Let A_i be the set of outcomes that miss value i and subtract $|A_1 \cup \cdots \cup A_m|$ from the total.

- **Two values:** $2^n - 2$ (remove the two constant strings). **Three values:** $3^n - 3 \cdot 2^n + 3$. These are the surjections from an n-set onto a 2-set and a 3-set (AY23/24 Q25(g)); the general formula is in §43.
- **Week 10 Ex. 3.** 7 balls in a row, balls of one colour identical, exactly 3 red and at least one blue and one green: place the reds, then fill 4 places with blue or green but not all one colour: $\binom{7}{3}(2^4 - 2) = 35 \cdot 14 = 490$.
- **Week 10 Ex. 2.** 7 books from 6 maths, 5 CS, 4 physics (all distinct), at least 2 per subject: split by distribution $(3,2,2)$, $(2,3,2)$, $(2,2,3)$: $\binom{6}{3}\binom{5}{2}\binom{4}{2} + \binom{6}{2}\binom{5}{3}\binom{4}{2} + \binom{6}{2}\binom{5}{2}\binom{4}{3} = 1200 + 900 + 600 = 2700$.
- **AY24/25 Q24(a).** 5 students each pick one of 5 GPUs at random. $P(\text{some GPU has more than one student}) = 1 - P(\text{all different}) = 1 - 5!/5^5 = 1 - 120/3125 = 0.9616$.

TRAP **Reserve first, then choose freely**

"Pick 2 books from each subject, then any 1 of the remaining 9" gives $15 \cdot 10 \cdot 6 \cdot 9 = 8100$, not 2700: a selection with 3 maths books is counted once for each choice of which maths book was the "extra". Reserving is safe only for **identical** objects (stars and bars); for distinct objects split into cases or use the complement.

41 · The pigeonhole principle in counting

THEOREM **Pigeonhole Principle · Generalized PHP and its contrapositive (L10 s55, s62, s65)**

- **PHP.** A function from one finite set to a smaller finite set cannot be one-to-one: at least two elements of the domain have the same image.
- **Generalized PHP.** For $f : X \to Y$ with $|X| = n$, $|Y| = m$ and a positive integer k: if $k < n/m$, then some $y \in Y$ is the image of at least $k + 1$ distinct elements of X.
- **Contrapositive form.** If for each $y \in Y$ the set $f^{-1}(\{y\})$ has at most k elements, then $n \leq km$.

Equivalently: among n pigeons in m holes, some hole holds at least $\lceil n/m \rceil$ pigeons.

METHOD **Writing a pigeonhole argument**

State (1) what the pigeons are, (2) what the holes are, (3) the rule sending each pigeon to a hole, (4) the two counts, then conclude with the principle by name. For "how many must be drawn to guarantee ...", find the **largest** selection that avoids the target and add 1.

WORKED EXAMPLE **L10 Ex. 10–12 · Decimal expansions (L10 s59–61)**

- 13 people: two share a birth month ($13 > 12$). Singapore: two people have the same number of hairs (5.47 million people, at most 300 001 possible counts).
- **85 people, 26 initials:** $3 < 85/26$, so at least 4 share a last initial. By the contrapositive: if at most 3 shared each initial there would be at most $3 \cdot 26 = 78$ people.
- **Rational numbers have terminating or repeating decimals.** In the long division of a by b every remainder lies in $\{0, ..., b-1\}$ *(Theorem 4.4.1)*. If no remainder is 0, then among $b+1$ successive remainders two are equal, $r_j = r_k$ with $j < k$ *(PHP)*, and the digits repeat from there. $3/14 = 0.2\overline{142857}$: remainders $3, 2, 6, 4, 12, 8, 10, 2, ...$
- **42 students, 12 computers, at most 6 per computer: at least 5 computers are used by 3 or more students.** Let k be the number of such computers. They serve at most $6k$ students; the other $12-k$ serve at most $2(12-k)$. So $42 \le 6k + 24 - 2k = 4k + 24$, giving $k \ge 4.5$, hence $k \ge 5$.

PROOF **T9 Q8 (AY21/22 exam) · Among any 5 distinct non-negative integers, two differ by a multiple of 4 T9 Q9 · Chess master**

Q8.

1. Let the pigeonholes be the classes $[0], [1], [2], [3]$ of congruence mod 4, and send each of the 5 integers to its class.
2. Two of the integers, x and y, lie in the same class $[i]$ *(PHP: $5 > 4$)*. So $x = 4k + i$ and $y = 4l + i$ for some $k, l \in \mathbb{Z}$.
3. Then $x - y = 4(k-l)$ with $k - l \in \mathbb{Z}$ *(closure)*, so $4 \mid (x - y)$. ■

Q9. 77 days, at least one game a day, at most 12 a week. Let P_i be the number of games played up to day i: $1 \le P_1 < P_2 < \cdots < P_{77} \le 132$. Let $Q_i = P_i + 21 \le 153$. The 154 numbers $P_1, ..., P_{77}, Q_1, ..., Q_{77}$ take values in $\{1, ..., 153\}$, so two are equal *(PHP)*. The P_i are distinct and so are the Q_i, hence $P_j = Q_i = P_i + 21$ for some i, j: exactly 21 games were played on days $i+1$ to j. ■

PAST PAPER **T10 Q10 · T10 Q12 · AY25 final Q24(f)**

T10 Q10. At a party of $n \ge 2$ people where the host shakes hands with everyone, two people shook the same number of hands: every degree lies in $\{1, ..., n-1\}$ (each person shook the host's hand), so n people share $n-1$ possible values *(PHP)*.

T10 Q12. Any simple graph G on 6 vertices, or its complement, contains a triangle. Colour the edges of K_6 black (in G) or red (in $\overline{G}$). Fix a vertex v: of its 5 edges, at least 3 have one colour *(generalized PHP, $2 < 5/2$)*, say red, to u_1, u_2, u_3. If some edge $\{u_x, u_y\}$ is red, then v, u_x, u_y form a red triangle; otherwise u_1, u_2, u_3 form a black triangle.

AY25 Q24(f). Socks: 8 red, 7 blue, 6 green, 5 black, 4 white. To avoid both "4 of one colour" and "3 each of two colours", take 3 of one colour and 2 of each of the other four: 11 socks. So **12** guarantee one of the two.

PROOF **AY20/21 Q20(f) · 21 students, non-negative integer scores with sum 200: two students have the same score**

1. Suppose not: the 21 scores are distinct. *(for contradiction)*
2. List them in increasing order, $s_1 < s_2 < \cdots < s_{21}$. Then $s_1 \ge 0$ and $s_{i+1} \ge s_i + 1$, so $s_i \ge i - 1$ for every i *(scores are non-negative integers)*.
3. So $s_1 + \cdots + s_{21} \ge 0 + 1 + \cdots + 20 = (20 \cdot 21)/2 = 210 > 200$, contradicting the sum 200.
4. Hence two students have the same score. ■

The pattern: if every hole holds at most one pigeon, the pigeons are forced into the cheapest distinct holes, and even that total is too large.

PAST PAPER **AY22/23 Q22(d) · AY24/25 Q21 · AY23/24 Q21 (divisibility chains as holes)**

AY22/23 Q22(d). 838 students, 12 birth months: some month has at least $\lceil 838/12 \rceil = 70$ students ($69 \cdot 12 = 828 < 838$).

AY24/25 Q21. 25 labelled simple graphs on 3 vertices. Holes are the isomorphism classes, fixed by the number of edges $0, 1, 2, 3$: 4 holes, so at least $\lceil 25/4 \rceil = 7$ are isomorphic to one another (option C).

AY23/24 Q21. $S = \{1, 3, 5, ..., 21\}$; smallest n such that any n numbers from S contain two with one dividing the other. Holes are **chains** in which every two members are comparable under "divides":

$$\{1, 3, 9\}, \quad \{5, 15\}, \quad \{7, 21\}, \quad \{11\}, \quad \{13\}, \quad \{17\}, \quad \{19\}.$$

These 7 chains cover S, so any 8 numbers put two in one chain *(PHP)*. Seven do not suffice: $\{9, 11, 13, 15, 17, 19, 21\}$ has no dividing pair. So $n = 8$ (option C).

Same idea. Among any $n+1$ numbers from $\{1, ..., 2n\}$ one divides another: write each as $2^k m$ with m odd; there are only n odd parts, and two numbers with the same odd part form a dividing pair.

42 · Combinations and multisets

DEFINITION ***r*-combination, $\binom{n}{r}$ (L11 s4) · Multiset (L11 s18)**

For integers $0 \le r \le n$, an ***r*-combination** of a set of n elements is a **subset** of r of them. $\binom{n}{r}$, read "n choose r", is the number of r-combinations (also written $C(n,r)$, ${}_nC_r$).

An ***r*-combination with repetition allowed**, or **multiset of size *r***, chosen from a set X of n elements is an unordered selection of r elements of X with repetition allowed, written $\left[x_{i_1}, x_{i_2}, ..., x_{i_r}\right]$.

THEOREM **Theorems 9.5.1, 9.5.2, 9.6.1 (L11 s10, s15, s21)**

- **9.5.1.** $\binom{n}{r} = \frac{P(n,r)}{r!} = \frac{n!}{r!(n-r)!}$, because an r-permutation is an r-subset followed by one of its $r!$ orderings: $P(n,r) = \binom{n}{r} \cdot r!$.
- **9.5.2 Permutations with indistinguishable objects.** If n objects consist of n_1 of type 1, ..., n_k of type k, with $n_1 + \cdots + n_k = n$, the number of distinguishable orderings is
$$\binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\cdots-n_{k-1}}{n_k} = \frac{n!}{n_1!\,n_2!\cdots n_k!}.$$
- **9.6.1 Multisets.** The number of multisets of size r from n elements, equivalently the number of ways to select r objects from n categories with repetition, is $\binom{r+n-1}{r}$. (Stars and bars: a string of r crosses and $n-1$ bars.)

Choosing k from n (L11 s23)	**Order matters**	**Order does not matter**
Repetition allowed	n^k	$\binom{k+n-1}{k}$
Repetition not allowed	$P(n,k) = n!/(n-k)!$	$\binom{n}{k} = n!/(k!(n-k)!)$

METHOD **Stars and bars (L11 s18–21)**

Line up r identical stars and cut them into n groups with $n-1$ bars: $\star\star\,|\,|\,\star\,|\,\star\star$ is the selection $(2,0,1,2)$. Every string of r stars and $n-1$ bars is one selection, so the count is $\binom{r+n-1}{r} = \binom{r+n-1}{n-1}$. The same count answers three questions: multisets of size r from n types; ways to put r **identical** objects into n **distinct** boxes; non-negative integer solutions of $x_1 + \cdots + x_n = r$.

- **Every $x_i \ge 1$.** Hand one to each box first, or place the bars in the $r-1$ gaps between stars: $\binom{r-1}{n-1}$.
- **Lower bounds $x_i \ge a_i$.** Substitute $y_i = x_i - a_i$, then solve $y_1 + \cdots + y_n = r - \sum a_i$.
- **Inequality $x_1 + \cdots + x_n \le r$.** Add a slack variable $x_{n+1} \ge 0$ and solve with equality: $\binom{r+n}{n}$.
- **Upper bound $x_1 \le c$.** Count all solutions and subtract those with $x_1 \ge c+1$; with several upper bounds use inclusion and exclusion.
- **Parity or multiples.** If every x_i must be even, substitute $x_i = 2y_i$ and solve $y_1 + \cdots + y_n = r/2$ (no solutions when r is odd). For x_i odd use $x_i = 2y_i + 1$.
- **Bounded both ways** (dice, $1 \le x_i \le 6$): shift to $y_i = x_i - 1 \in \{0, ..., 5\}$, then remove the solutions with some $y_i \ge 6$.
- **Does not apply** when the objects are distinct (n^r ways to place them) or when order within the selection matters.

Checks. $x_1 + x_2 + x_3 = 10$ with $x_1 \le 4$: $\binom{12}{2} - \binom{7}{2} = 66 - 21 = 45$. 12 identical sweets to 4 children, each getting at least 2: $y_1 + \cdots + y_4 = 4$, so $\binom{7}{3} = 35$.

WORKED EXAMPLE **L11 Ex. 1–7**

- **Committees.** 3-person committees from 4 people: $\binom{4}{3} = 4$. $P(4,2) = 12 = \binom{4}{2} \cdot 2!$, so $\binom{4}{2} = 6$.
- **Teams of 5 from 5 men and 7 women.** 3 men and 2 women: $\binom{5}{3}\binom{7}{2} = 10 \cdot 21 = 210$. At least one man: $\binom{12}{5} - \binom{7}{5} = 792 - 21 = 771$. At most one man: $\binom{7}{5} + \binom{5}{1}\binom{7}{4} = 21 + 175 = 196$.
- **MISSISSIPPI** (4 S, 4 I, 2 P, 1 M): $\binom{11}{4}\binom{7}{4}\binom{3}{2}\binom{1}{1} = \frac{11!}{4!\,4!\,2!\,1!} = 34\,650$.
- **Multisets of size 3 from $\{1,2,3,4\}$:** $\binom{3+4-1}{3} = \binom{6}{3} = 20$; e.g. $[1,3,4]$ is the string $\times\,|\,|\,\times\,|\,\times$.
- **Lunch for 20 guests from 6 stalls:** $\binom{20+6-1}{20} = \binom{25}{20} = 53\,130$.
- **$x_1 + x_2 + x_3 = 20$:** non-negative integer solutions $\binom{22}{20} = 231$. Positive solutions: put $y_i = x_i - 1$, so $y_1 + y_2 + y_3 = 17$: $\binom{19}{17} = 171$.

PAST PAPER **T9 Q4, Q6, Q7 · T10 Q2 · AY25 final Q24(a)–(e), (g)**

T9 Q6. 5 from 7 men and 6 women with at least 3 men: $\binom{7}{3}\binom{6}{2}+\binom{7}{4}\binom{6}{1}+\binom{7}{5}=525+210+21=756$; probability $756/\binom{13}{5}=756/1287\approx 0.5874$.

T9 Q7 (AY21/22). \$50 000 over 4 properties in \$1000 units: $x_1+\cdots+x_4=50$ has $\binom{53}{50}=23\,426$ solutions. Not necessarily all invested: add a slack variable x_5: $\binom{54}{50}=316\,251$.

T9 Q4, T10 Q2. n boxes, white balls in a non-empty consecutive block: $\sum_{k=1}^{n}(n-k+1)=n(n+1)/2$; or choose the two ends of the block among $n+1$ gaps: $\binom{n+1}{2}$.

AY25 Q24. (a) 3 red, 4 blue, 5 green balls in a row: $12!/(3!\,4!\,5!)=27\,720$. (b) Committee of 5 from 6 men and 5 women with at least 3 women: $\binom{5}{3}\binom{6}{2}+\binom{5}{4}\binom{6}{1}+\binom{5}{5}=150+30+1=181$. (c) 6 distinct letters with exactly 2 vowels: choose the 2 vowel positions, fill them, fill the rest: $\binom{6}{2}\cdot P(5,2)\cdot P(21,4)=15\cdot 20\cdot 143\,640$ $=43\,092\,000$. (d) n couples at a round table, men and women alternating: $(n-1)!\,n!$. (e) Every woman next to her husband: seat the n couples as units, then order each couple: $(n-1)!\,2^n$. (g) Three distinguishable dice with sum 10: $x_1+x_2+x_3=10$ with $1\le x_i\le 6$. Put $y_i=x_i-1$: $y_1+y_2+y_3=7$ with $0\le y_i\le 5$. Unrestricted: $\binom{9}{2}$ $=36$. Bad solutions with a fixed $y_i\ge 6$: give that die 6 first, leaving sum 1 over three variables, $\binom{3}{2}=3$; two dice cannot both be bad ($12>7$). So $36-3\cdot 3=27$.

PAST PAPER **Stars and bars in the finals · AY20/21 Q20(a) · AY22/23 Q22(a) · AY23/24 Q25(d) · AY24/25 Q24(c)**

AY20/21 Q20(a). $x+y+z=88$ with $x,y,z\ge 3$: subtract 3 from each, sum 79 over 3 variables: $\binom{81}{2}=3240$.

AY22/23 Q22(a). $x_1+x_2+x_3=10$ with $x_i\ge i$: put $y_i=x_i-i$, sum $10-6=4$: $\binom{6}{2}=15$.

AY23/24 Q25(d). 16 identical tasks to 4 distinct processors, each getting an even number: $x_i=2y_i$, $y_1+\cdots+y_4=8$: $\binom{11}{3}=165$.

AY24/25 Q24(c). $x_1+x_2+x_3+x_4=12$, $x_i\ge 0$: (i) $\binom{15}{3}=455$. (ii) With $x_4\le 3$: remove the solutions with $x_4\ge 4$ (sum 8 after giving x_4 four): $455-\binom{11}{3}=455-165=290$.

PAST PAPER **Selections and arrangements · Week 10 Ex. 1, 4 · AY20/21 Q20(b) · AY23/24 Q25(c) · AY24/25 Q24(b)**

Week 10 Ex. 1. 5 distinct integers from $\{1,\ldots,12\}$: exactly 2 even, $\binom{6}{2}\binom{6}{3}=300$; at least one multiple of 3, $\binom{12}{5}-\binom{8}{5}=792-56=736$.

Week 10 Ex. 4. BANANA with A first and last: arrange the remaining A, B, N, N: $4!/2!=12$.

AY20/21 Q20(b) (a repeated letter). 4 tiles from I, C, A, N, D, O, I, T, order ignored. Split on the repeated letter: with a duplicate, take both I's and 2 of the other 6 letters, $\binom{6}{2}=15$; with no duplicate, take 4 of the 7 distinct letters, $\binom{7}{4}=35$. (This counts selections of letters; do not use $\binom{8}{4}=70$, which treats the two I tiles as different.)

AY23/24 Q25(c). Morse messages with five dashes and three dots: choose the positions of the dots, $\binom{8}{3}$ $=8!/(5!\,3!)=56$.

AY24/25 Q24(b). $\binom{n}{3}=k$ and $\binom{n-1}{2}=k/2$. Since $\binom{n}{3}=\frac{n}{3}\binom{n-1}{2}$, we need $n/3=2$: $n=6$ (check: 20 and 10).

43 · Pascal's formula, the binomial theorem, combinatorial proofs

THEOREM **Theorem 9.7.1 Pascal's formula (L11 s26) · Theorem 9.7.2 Binomial theorem (L11 s34) · Identities (L11 s29–33)**

- **Pascal.** For positive integers $r\le n$: $\binom{n+1}{r}=\binom{n}{r-1}+\binom{n}{r}$.
- **Binomial theorem.** For real a,b and integer $n\ge 0$: $(a+b)^n=\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$. $\binom{n}{r}$ is a **binomial coefficient**.
- $\binom{n}{r}=\binom{n}{n-r}$; $k\binom{n}{k}=n\binom{n-1}{k-1}$; $\binom{n}{0}+\binom{n}{1}+\cdots+\binom{n}{n}=2^n$.

DEFINITION **Combinatorial proof (L11 s27)**

A **combinatorial proof** uses counting as the basis of the proof. Two kinds: a **bijective proof** (a bijection between two sets shows they have the same size) and a **proof by double counting** (count one set in two ways; the two expressions are equal).

PROOF **Pascal's formula, algebraic and combinatorial (L11 s28) · Three double-counting proofs (L11 s30–33)**

Algebraic. $\binom{n}{r-1}+\binom{n}{r}=\frac{n!}{(n-r+1)!(r-1)!}+\frac{n!}{(n-r)!\,r!}=\frac{n!\,r+n!(n-r+1)}{(n-r+1)!\,r!}=\frac{(n+1)!}{(n+1-r)!\,r!}=\binom{n+1}{r}$.

Combinatorial.

1. $\binom{n+1}{r}$ counts the r-subsets of a set A with $n+1$ elements. Fix $x\in A$.
2. The r-subsets containing x: choose the other $r-1$ elements from the remaining n: $\binom{n}{r-1}$.

3. The r-subsets not containing x: choose all r from the remaining n: $\binom{n}{r}$.
4. The two cases are disjoint and exhaustive, so $\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$ *(addition rule)*. ■

$\binom{n}{k} = \binom{n}{n-k}$. Choosing the k pupils who go is the same as choosing the $n-k$ who stay.
$k\binom{n}{k} = n\binom{n-1}{k-1}$. Count committees of k from n with a chairperson: choose the committee then the chair, or choose the chair then the other $k-1$ members.
$\sum_{k=0}^{n}\binom{n}{k} = 2^n$. Both count the subsets of an n-element set: by size on the left, by Theorem 6.3.1 on the right.

WORKED EXAMPLE **L11 Ex. 12, 13 · T10 Q1**

- $(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$.
- $(x-4y)^4 = x^4 - 16x^3y + 96x^2y^2 - 256xy^3 + 256y^4$ (substitute $b = -4y$ and keep the sign inside the power).
- $\sum_{k=0}^{n}\binom{n}{k}9^k = \sum_{k=0}^{n}\binom{n}{k}1^{n-k}9^k = (1+9)^n = 10^n$.
- **T10 Q1.** Term independent of x in $(2x^2 + 1/x)^9$: the general term is $\binom{9}{r}(2x^2)^{9-r}x^{-r} = \binom{9}{r}2^{9-r}x^{18-3r}$; $18-3r=0$ gives $r=6$: $\binom{9}{6}\cdot 2^3 = 84\cdot 8 = 672$.

THEOREM **Odd and even binomial sums · The k-th term (AY23/24 Q25(b) · AY20/21 Q20(c))**

- Put $a=1$, $b=-1$ in the binomial theorem: for $n \geq 1$, $\sum_{k=0}^{n}(-1)^k\binom{n}{k} = (1-1)^n = 0$, so the even-$k$ and odd-k coefficients have equal sums. Together they sum to 2^n, hence
$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots = 2^{n-1}.$$
A set with $n \geq 1$ elements has as many subsets of even size as of odd size.
- **AY23/24 Q25(b).** A fair coin tossed n times shows an odd number of heads with probability $2^{n-1}/2^n = 1/2$ for every $n \geq 1$, so $k=2$.
- **The k-th term** of $(a+b)^n$ is the one with $r = k-1$: $\binom{n}{k-1}a^{n-k+1}b^{k-1}$. The first term is $r=0$.
- **AY20/21 Q20(c).** Fifth term of $(1/(2\sqrt{x}) - 1/2)^{10}$ equal to 105: $r=4$, so $\binom{10}{4}\left(\frac{1}{2\sqrt{x}}\right)^6\left(-\frac{1}{2}\right)^4 = \frac{210}{64x^3\cdot 16} = 105$, giving $x^3 = 1/512$ and $x = 1/8$.

METHOD **Counting by recurrence (Week 12 s25–29, AY18/19 Q17)**
When a direct count is awkward, ask what the n-th object adds to the count for $n-1$ objects, write the recurrence with its base case, then solve it by iteration and prove the closed form by induction (Part VII). Lazy caterer: the n-th cut crosses the $n-1$ earlier cuts and adds n pieces, so $P(n) = P(n-1) + n$ with $P(0) = 1$, giving $P(n) = (n^2+n+2)/2$.

THEOREM **Counting functions (T9 D3 · T10 D2, D3)**
For finite sets with $|A| = m$ and $|B| = n$:

Functions $A \to B$	**Number**	$m=3, n=2$	$m=n=3$
all functions	n^m	8	27
injections ($m \leq n$)	$P(n,m) = n!/(n-m)!$	0	6
surjections ($m \geq n$)	$\sum_{i=0}^{n}(-1)^i\binom{n}{i}(n-i)^m$	6	6
bijections ($m=n$)	$n!$	0	6

Surjections from a 5-element set to a 3-element set: $3^5 - 3\cdot 2^5 + 3\cdot 1^5 = 150$ (inclusion/exclusion on the values missed). **AY25 Q10:** strictly increasing $f : \{0,1\} \to \{0,1,2,3\}$: $\binom{4}{2} = 6$; injective $g : \{0,1\} \to \{0,1,2\}$: $P(3,2) = 6$. **AY22/23 Q22(g):** injections with $|A|=2$, $|B|=3$: 6; with $|A| = k \leq n = |B|$: $n!/(n-k)!$. **AY23/24 Q25(g):** surjections from an n-set onto a 2-set: $2^n - 2$; onto a 3-set ($n \geq 2$): $3^n - 3\cdot 2^n + 3$.

44 · Probability axioms and expected value

DEFINITION **Probability axioms (L11 s39) · Expected value (L11 s43)**
Let S be a sample space. A **probability function** P from the set of all events in S to $\mathbb{R}$ satisfies, for all events A, B:
1. $0 \leq P(A) \leq 1$; 2. $P(\varnothing) = 0$ and $P(S) = 1$; 3. if $A \cap B = \varnothing$, then $P(A \cup B) = P(A) + P(B)$.
If the possible outcomes of an experiment are real numbers $a_1, \ldots, a_n$ occurring with probabilities $p_1, \ldots, p_n$, the **expected value** is
$$\sum_{k=1}^{n} a_k p_k = a_1p_1 + a_2p_2 + \cdots + a_np_n.$$

THEOREM **Complement · General union (L11 s40–41) · Linearity of expectation (L11 s48)**

- $P(\overline{A}) = 1 - P(A)$. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ for **any** events A, B.
- For random variables X, Y, **dependent or not**: $E[X+Y] = E[X] + E[Y]$; in general $E\left[\sum_{i=1}^{n} c_i X_i\right] = \sum_{i=1}^{n} c_i E[X_i]$.

WORKED EXAMPLE **L11 s41, Ex. 14–16 · T10 Q3, Q4, D1 · AY25 final Q23**

- **Heart or 7:** $\frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$.
- **Fair die:** $(1+2+\cdots+6)/6 = 3.5$. **Sum of two dice:** $3.5 + 3.5 = 7$ by linearity.
- **Lottery:** 500 000 tickets at \$5; prizes \$1 000 000 (1), \$1000 (10), \$500 (1000), \$10 (10 000). Expected net gain $= (999\,995 + 10 \cdot 995 + 1000 \cdot 495 + 10\,000 \cdot 5 - 488\,989 \cdot 5)/500\,000 = -1.78$ dollars.
- **T10 Q3 (AY20/21 S2).** Loaded die: each big number (4, 5, 6) is twice as likely as each small one, so $P(1) = P(2) = P(3) = 1/9$ and $P(4) = P(5) = P(6) = 2/9$. Maximum of two rolls: $P(\max = 1, \ldots, 6) = \frac{1}{81}, \frac{3}{81}, \frac{5}{81}, \frac{16}{81}, \frac{24}{81}, \frac{32}{81}$; expected value $398/81$.
- **T10 Q4.** Two balls from $\{1, 2, 2, 8, 8\}$: one ball has expectation $(1+2+2+8+8)/5 = 4.2$, so the sum has expectation 8.4 by linearity (the draws are dependent; linearity still applies).
- **T10 D1.** 2 bulbs from 8 with 3 defective: $E[X] = 2 \cdot 3/8 = 0.75$.
- **AY25 Q23.** Sundaes: 20% of customers choose 3 of 10 flavours, 80% choose 2: $0.2\binom{10}{3} + 0.8\binom{10}{2} = 24 + 36 = 60$.

METHOD **Expected count by indicator variables (L11 s48)**

To find the expected **number** of objects with some property, do not find the distribution of the count. For each object i let $X_i = 1$ if it has the property and $X_i = 0$ otherwise, so $E[X_i] = P(\text{object } i \text{ has the property})$. The count is $X = X_1 + \cdots + X_n$, and by linearity

$$E[X] = \sum_{i=1}^{n} P(\text{object } i \text{ has the property}),$$

whether or not the X_i are independent. If all n probabilities equal p, then $E[X] = np$.

METHOD **Symmetry (Week 10 s72, s74)**

If swapping two roles maps outcomes to equally likely outcomes, the two events have the same probability. For two rolls of a fair die, $P(\text{second} > \text{first}) = P(\text{first} > \text{second})$, and with $P(\text{tie}) = 6/36$ the three cases sum to 1:

$$P(\text{second} > \text{first}) = \frac{1 - P(\text{tie})}{2} = \frac{1}{2} \cdot \frac{5}{6} = \frac{5}{12}.$$

Direct check: $(5+4+3+2+1+0)/36 = 15/36$. The same argument gives "A to the left of B" in half of all arrangements (§39) and $P(\text{second draw is red}) = P(\text{first draw is red})$ without replacement (§45).

PAST PAPER **AY20/21 Q10, Q11 · AY22/23 Q22(c), (e) · AY24/25 Q24(d)**

AY20/21 Q10. Expected sum of three fair dice: $3 \cdot 3.5 = 10.5$ (option D).

AY20/21 Q11. $(\sim b \wedge c \wedge d) \vee ((\sim a \vee b \vee \sim e \vee f) \wedge c) \vee \sim(d \wedge e) \vee c$ with six variables set at random. The statement is false iff every disjunct is false: c false (which also kills the first two disjuncts) and d, e both true; a, b, f are free: $2^3/2^6 = 1/8$ (option C).

AY22/23 Q22(c). Second roll higher than the first: 5/12 by symmetry.

AY22/23 Q22(e). Two cards without replacement, scores 1 to 13: each card has expected score $(1+\cdots+13)/13 = 7$, so the combined score has expectation 14 (linearity; the draws are dependent).

AY24/25 Q24(d). Ten couples; 4 of the 20 persons leave, every set of 4 equally likely. (i) A fixed couple stays: all 4 leavers come from the other 18 persons, $\binom{18}{4}/\binom{20}{4} = 3060/4845 = 12/19$ (or $\frac{16}{20} \cdot \frac{15}{19}$: both of the couple are among the 16 who stay). (ii) Let $X_i = 1$ if couple i stays. $E[X] = 10 \cdot 12/19 = 120/19 \approx 6.32$.

45 · Conditional probability, Bayes' theorem, independence

DEFINITION Conditional probability (L11 s53) · Independent events (L11 s73) · Pairwise and mutually independent (L11 s79–80)

For events A, B in S with $P(A) \neq 0$, the **conditional probability of B given A** is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}, \quad \text{so} \quad P(A \cap B) = P(B \mid A) \cdot P(A).$$

A and B are **independent** iff $P(A \cap B) = P(A) \cdot P(B)$.

A, B, C are **pairwise independent** iff (1) $P(A \cap B) = P(A)P(B)$, (2) $P(A \cap C) = P(A)P(C)$, (3) $P(B \cap C) = P(B)P(C)$; they are **mutually independent** iff also (4) $P(A \cap B \cap C) = P(A)P(B)P(C)$. Events $A_1, ..., A_n$ are mutually independent iff the probability of the intersection of **any subset** of them is the product of their probabilities.

THEOREM Law of total probability (L11 s59, s64) · Independence in conditional form (L11 s72)

- **Total probability.** If S is the union of mutually disjoint events $B_1, ..., B_n$ of non-zero probability, then for every event A

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \cdots + P(A \mid B_n)P(B_n),$$

  because A is the disjoint union of the $A \cap B_i$ *(axiom 3)* and $P(A \cap B_i) = P(A \mid B_i)P(B_i)$. Two cases: $P(A) = P(A \mid B)P(B) + P\big(A \mid \overline{B}\big)P\big(\overline{B}\big)$. This is the sum over the branches of the tree that end in A.
- **Conditional form of independence.** If $P(B) \neq 0$: A and B are independent iff $P(A \mid B) = P(A)$. If $P(A) \neq 0$: iff $P(B \mid A) = P(B)$. Knowing one event does not change the probability of the other.
- **Union of independent events.** $P(A \cup B) = P(A) + P(B) - P(A)P(B)$.
- **Complements.** If A and B are independent, so are A and $\overline{B}$, since $P\big(A \cap \overline{B}\big) = P(A) - P(A \cap B) = P(A)(1 - P(B))$; likewise $\overline{A}$, B and $\overline{A}$, $\overline{B}$.

THEOREM Theorem 9.9.1 · Bayes' theorem (L11 s66)

Suppose S is the union of mutually disjoint events $B_1, ..., B_n$, and A and all B_i have non-zero probability. Then for $1 \leq k \leq n$:

$$P(B_k \mid A) = \frac{P(A \mid B_k) \cdot P(B_k)}{P(A \mid B_1) \cdot P(B_1) + P(A \mid B_2) \cdot P(B_2) + \cdots + P(A \mid B_n) \cdot P(B_n)}.$$

The denominator is $P(A)$, by the law of total probability.

TRAP Conditional probability and independence

- $P(B \mid A)$ and $P(A \mid B)$ are different numbers; Bayes converts one into the other.
- "At least one is a boy" is not "the first is a boy": of BB, BG, GB, GG the condition leaves three outcomes, so $P(\text{both boys} \mid \text{at least one boy}) = 1/3$.
- Independent is not disjoint: disjoint events with positive probabilities are **dependent**.
- Pairwise independence does not give (4), and (4) does not give pairwise independence. Two fair coins with A = first is heads, B = second is heads, C = both the same: every pair is independent, but $P(A \cap B \cap C) = 1/4 \neq 1/8$.

METHOD With and without replacement

Counting k draws from n distinct objects. With replacement the object goes back, so each draw has n choices: n^k ordered outcomes. Without replacement: $P(n, k)$ ordered outcomes, $\binom{n}{k}$ unordered ones. Unordered with replacement gives $\binom{k+n-1}{k}$ multisets, but these are **not equally likely**, so never use them as a sample space.

Probability, urn with b blue and g grey balls, $N = b + g$, k draws.

- **With replacement:** the draws are independent and each is blue with probability b/N. $P(\text{exactly } j \text{ blue}) = \binom{k}{j}\left(\frac{b}{N}\right)^j\left(\frac{g}{N}\right)^{k-j}$.
- **Without replacement:** the draws are dependent. Either count unordered hands, $P(\text{exactly } j \text{ blue}) = \binom{b}{j}\binom{g}{k-j}/\binom{N}{k}$, or multiply conditional probabilities along the sequence: $P(\text{blue then blue}) = \frac{b}{N} \cdot \frac{b-1}{N-1}$.
- In both cases $P(\text{the } i\text{-th draw is blue}) = b/N$ for every i, by symmetry.
- **At least one:** use the complement, $1 - P(\text{none})$.

Check, 5 blue and 7 grey, 3 draws. Without replacement: $P(\text{exactly 2 blue}) = \binom{5}{2}\binom{7}{1}/\binom{12}{3} = 70/220 = 7/22$ and $P(\text{at least one blue}) = 1 - \binom{7}{3}/\binom{12}{3} = 37/44$. With replacement: $P(\text{exactly 2 blue}) = 3 \cdot (5/12)^2(7/12) = 175/576$.

WORKED EXAMPLE **L11 Ex. 17 · Urn with 5 blue and 7 grey balls, two drawn without replacement**
$P(B_1)=\frac{5}{12}$, $P(B_2 \mid B_1)=\frac{4}{11}$, $P\left(B_2 \mid \overline{B_1}\right)=\frac{5}{11}$. Multiply along the branches of the tree:

$$P(B_1\cap B_2)=\frac{20}{132},\quad P\left(B_1\cap\overline{B_2}\right)=\frac{35}{132},\quad P\left(\overline{B_1}\cap B_2\right)=\frac{35}{132},\quad P\left(\overline{B_1}\cap\overline{B_2}\right)=\frac{42}{132}.$$

- $P(B_2)=\frac{20}{132}+\frac{35}{132}=\frac{55}{132}=\frac{5}{12}$ *(axiom 3 on the disjoint cases)*.
- $P(B_1\cup B_2)=\frac{5}{12}+\frac{5}{12}-\frac{20}{132}=\frac{90}{132}=\frac{15}{22}$; so $P(\text{no blue})=\frac{7}{22}$.
- Expected number of blue balls: $0\cdot\frac{42}{132}+1\cdot\frac{70}{132}+2\cdot\frac{20}{132}=\frac{110}{132}=\frac{5}{6}\approx 0.83$.

WORKED EXAMPLE **Bayes · L11 s63–65, Ex. 18 · T10 Q5 · AY25 final Q24(h)**

- **Two urns** (3 blue, 4 grey; 5 blue, 3 grey), one chosen at random: $P(B)=\frac{3}{7}\cdot\frac{1}{2}+\frac{5}{8}\cdot\frac{1}{2}=\frac{59}{112}$ and $P(U_1 \mid B)=(3/14)/(59/112)=\frac{24}{59}\approx 40.7\%$.
- **Medical test.** Disease in 5 per 1000; false positive rate 3%; false negative rate 1%.
  $P(\text{disease}\mid +)=\frac{0.99\cdot 0.005}{0.99\cdot 0.005+0.03\cdot 0.995}\approx 14.2\%$. $P(\text{no disease}\mid -)=\frac{0.97\cdot 0.995}{0.01\cdot 0.005+0.97\cdot 0.995}\approx 99.995\%$.
- **T10 Q5 (AY21/22 S2).** $P(D)=0.001$, $P(+\mid D)=0.85$, $P(+)=0.1$. $P(D\mid +)=(0.85\cdot 0.001)/0.1=0.0085$. False positive: $P\left(+\mid\overline{D}\right)=\frac{P\left(\overline{D}\mid +\right)\cdot P(+)}{P\left(\overline{D}\right)}=(0.9915\cdot 0.1)/0.999\approx 0.0992$.
- **AY25 Q24(h).** $P(S)=0.2$, $P(O\mid S)=0.5$, $P(O\mid H)=0.08$: $P(S\mid O)=0.1/(0.1+0.064)\approx 0.61$. $P(U\mid S)=0.35$, $P(U\mid H)=0.04$: $P(S\mid U)=0.07/(0.07+0.032)\approx 0.69$. "Urgent" is the stronger signal.

WORKED EXAMPLE **Independence · L11 Ex. 19, 20 · T9 D1 · T10 Q6**

- **Loaded coin, $P(H)=0.6$, two tosses:** $P(\text{two heads})=0.36$; $P(\text{one head})=0.6\cdot 0.4+0.4\cdot 0.6=0.48$; $P(\text{no heads})=0.16$; $P(\text{at least one})=1-0.16=0.84$.
- **Ten tosses:** $P(\text{eight heads})=\binom{10}{8}(0.6)^8(0.4)^2\approx 0.121$; $P(\text{at least eight})=\binom{10}{8}(0.6)^8(0.4)^2+\binom{10}{9}(0.6)^9(0.4)+(0.6)^{10}\approx 0.167$. Probabilities of the form $\binom{n}{k}p^k(1-p)^{n-k}$ are **binomial probabilities**.
- **T9 D1.** 3 blue and 7 white balls, two draws with replacement: $P(\text{blue then white})=0.21$; $P(\text{both white})=0.49$; $P(\text{second blue})=0.3$.
- **T10 Q6 (AY15/16).** A random relation on a 4-element set is reflexive with probability $2^{12}/2^{16}=1/16$ and symmetric with probability $2^{10}/2^{16}=1/64$. For $|A|=n$: $1/2^n$ and $2^{(n^2+n)/2}/2^{n^2}=1/2^{(n^2-n)/2}$.
- **Birthday problem (T9 D4):** $P(\text{some shared birthday among } n)=1-(365\cdot 364\cdots(365-n+1))/365^n$, which first reaches 50% at $n=23$.

PAST PAPER **Total probability and Bayes in the finals · AY22/23 Q22(f) · AY23/24 Q25(e) · AY20/21 Q20(e)**

AY22/23 Q22(f). $P(\text{spam})=0.25$, $P(F\mid\text{spam})=0.6$, $P(F\mid\text{not spam})=0.3$, where $F=$ "free gifts" appears. (i) $P(F)=0.6\cdot 0.25+0.3\cdot 0.75=0.375$. (ii) $P(\text{spam}\mid F)=0.15/0.375=0.4$.

AY23/24 Q25(e). $P(G)=0.85$, $P(\text{bonus}\mid G)=0.8$, $P(\text{bonus}\mid\text{Apple})=0.95$: $P(G\mid\text{bonus})=0.68/(0.68+0.1425)=0.68/0.8225\approx 0.8267$.

AY20/21 Q20(e), joint hypotheses. One M&M from each of two bags (one 1994, one 1996); one is yellow, one is green. A hypothesis must describe the **whole** set-up: $A=$ "bag 1 is the 1994 bag and bag 2 the 1996 bag", $B=$ the reverse, $P(A)=P(B)=1/2$. Evidence $E=$ "yellow from bag 1 and green from bag 2". Multiply the two colour probabilities under each hypothesis:

$$P(E\mid A)=0.20\cdot 0.16=0.032,\quad P(E\mid B)=0.14\cdot 0.10=0.014,\quad P(A\mid E)=\frac{0.032}{0.032+0.014}=\frac{16}{23}\approx 0.696.$$

Using the yellow alone ($0.20/0.34=10/17$) ignores half of the evidence.

PAST PAPER **Independence in the finals · AY22/23 Q13, Q22(b) · AY23/24 Q25(a) · AY24/25 Q23, Q24(e)**

AY22/23 Q22(b). A, B independent, $P(A)=0.2$, $P(A\cup B)=0.6$: $0.6=0.2+P(B)-0.2P(B)$, so $P(B)=0.5$.

AY22/23 Q13, restricted pattern. Ten tosses, $P(H)=p$; exactly 3 heads, all consecutive. Every sequence with 3 heads has probability $p^3(1-p)^7$; count only the sequences allowed: the block HHH starts at toss $1,\ldots,8$. Answer $8p^3(1-p)^7$ (option D), not $\binom{10}{3}p^3(1-p)^7$.

AY23/24 Q25(a). $P(H)=0.6$, five tosses, three heads: $\binom{5}{3}(0.6)^3(0.4)^2=0.3456$.

AY24/25 Q23. One ball from $\{1,2,3,4\}$; $A=\{1,2\}$, $B=\{1,3\}$, $C=\{1,4\}$. Each has probability $1/2$ and each pairwise intersection is $\{1\}$ with probability $1/4=\frac{1}{2}\cdot\frac{1}{2}$, but $P(A\cap B\cap C)=1/4\neq 1/8$: pairwise independent, not mutually independent (option C).

AY24/25 Q24(e). b blue and r red markers, two drawn without replacement. (i) $P(R_1)=r/(b+r)$. (ii) True: $P(R_2)=\frac{r}{b+r}\cdot\frac{r-1}{b+r-1}+\frac{b}{b+r}\cdot\frac{r}{b+r-1}=\frac{r}{b+r}$ *(total probability)*. (iii) $P(\text{different colours})=\frac{2br}{(b+r)(b+r-1)}$. (iv) $b+r=25$ and $P(\text{same})=P(\text{different})=1/2$: $2br=\frac{1}{2}\cdot 25\cdot 24=300$, so $br=150$ with $r>b$: $r=15$, $b=10$.

Part X · Graphs and Trees

46 · Graphs, special graphs, degree

DEFINITION **Undirected graph (L12 s7) · Directed graph (L12 s8)**

An **undirected graph** $G=(V,E)$ consists of two finite sets: a non-empty set V of **vertices** and a set E of **edges**, where each edge is associated with a set of one or two vertices called its **endpoints**. Write $e=\{v,w\}$.

An edge **connects** its endpoints and is **incident on** each of them; vertices joined by an edge are **adjacent**; an edge with a single endpoint is a **loop** (its vertex is adjacent to itself); edges with the same endpoints are **parallel**; edges sharing an endpoint are **adjacent edges**.

A **directed graph** (digraph) has directed edges, each associated with an **ordered pair** of vertices: $e=(v,w)$ goes from v to w.

DEFINITION **Simple graph (L12 s19) · Complete graph K_n (L12 s20) · Bipartite, $K_{m,n}$ (L12 s21) · Subgraph (L12 s22)**

- A **simple graph** is an undirected graph with no loops and no parallel edges.
- The **complete graph** K_n ($n>0$) is the simple graph on n vertices with exactly one edge between each pair of distinct vertices. It has $\binom{n}{2}=n(n-1)/2$ edges.
- A **bipartite graph** is a simple graph whose vertices can be divided into two disjoint sets U and V such that every edge connects a vertex in U to one in V. In the **complete bipartite graph** $K_{m,n}$, with $|U|=m$ and $|V|=n$, every vertex of U is joined to every vertex of V; it has mn edges.
- H is a **subgraph** of G iff every vertex of H is a vertex of G, every edge of H is an edge of G, and every edge of H has the same endpoints as in G.

DEFINITION **Degree, total degree (L12 s23) · Indegree, outdegree (L12 s26) · Complement, self-complementary, triangle (T10)**

The **degree** $\deg(v)$ is the number of edges incident on v, **with a loop counted twice**. The **total degree** of G is the sum of the degrees of all its vertices.

In a digraph, the **indegree** $\deg^-(v)$ is the number of edges ending at v and the **outdegree** $\deg^+(v)$ the number starting at v; $\sum_{v\in V}\deg^-(v)=\sum_{v\in V}\deg^+(v)=|E|$.

The **complement** $\overline{G}$ of a simple graph G has the same vertices; distinct v,w are adjacent in $\overline{G}$ iff they are not adjacent in G. G is **self-complementary** iff $G\cong\overline{G}$. A **triangle** is a simple circuit of length 3.

A **vertex colouring** assigns colours to vertices so that adjacent vertices get different colours (L12 s13).

THEOREM **Theorem 10.1.1 Handshake theorem · Corollary 10.1.2 · Proposition 10.1.3 (L12 s25)**

- For any graph G with vertices $v_1, ..., v_n$: $\deg(v_1) + \deg(v_2) + \cdots + \deg(v_n) = 2 \cdot |E|$.
- The total degree of a graph is even.
- In any graph there is an even number of vertices of odd degree.

PROOF **Handshake theorem · Proposition 10.1.3**

1. Let e be an edge of G with endpoints v_i and v_j. If $i \neq j$, then e contributes 1 to $\deg(v_i)$ and 1 to $\deg(v_j)$; if $i = j$ (a loop), it contributes 2 to $\deg(v_i)$ *(definition of degree)*.
2. So every edge contributes exactly 2 to the total degree, which is therefore $2\,|E|$. ■

Proposition 10.1.3. The total degree T is even. Let E be the sum of the even degrees and O the sum of the odd degrees, so $O = T - E$ is even *(difference of even integers)*. A sum of odd integers is even only if the number of terms is even. ■

WORKED EXAMPLE **Degree arguments · T11 D1, D2 · T10 Q8**

- **No graph has degrees 1, 1, 2, 3:** the total would be 7, which is odd.
- **No simple graph has degrees 1, 1, 3, 3 on 4 vertices:** each vertex of degree 3 is adjacent to all three others, so every other vertex has degree at least 2.
- **T11 D1.** A connected simple graph on n vertices has between $n-1$ edges (a tree) and $n(n-1)/2$ edges (K_n).
- **T11 D2.** Simple graphs on n labelled vertices: each of the $\binom{n}{2}$ possible edges is present or absent, so $2^{n(n-1)/2}$; for $n = 3$ there are 8 (in 4 isomorphism classes).
- **T10 Q8.** Non-isomorphic undirected graphs with two vertices and two edges: **4** (two parallel edges; one edge and one loop; two loops at one vertex; one loop at each vertex). The question does not say "simple".
- **K_n and $K_{m,n}$:** every vertex of K_n has degree $n-1$; in $K_{m,n}$ the degrees are n (in U) and m (in V). Check: $n(n-1) = 2 \cdot \binom{n}{2}$ and $mn + nm = 2mn$.

Colouring: chromatic number and chromatic index

DEFINITION **Chromatic number $\chi(G)$ · Chromatic index $\chi'(G)$ (AY24/25 Q25 · W12 s52)**

The **chromatic number** $\chi(G)$ is the minimum number of colours needed to colour the **vertices** of G so that no two adjacent vertices have the same colour.

The **chromatic index** $\chi'(G)$ is the minimum number of colours needed to colour the **edges** of G so that no two adjacent edges (edges sharing an endpoint) have the same colour.

$\Delta(G)$ denotes the maximum degree of G.

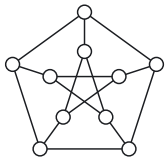
THEOREM **Values and bounds for χ and χ' (W12 s53–54, s58)**

Graph	χ	χ'	Reason
K_n	n	$n-1$ (n even), n (n odd)	all vertices mutually adjacent; for odd n a colour covers at most $(n-1)/2$ of the $n(n-1)/2$ edges
C_n	2 (n even), 3 (n odd)	2 (n even), 3 (n odd)	alternate two colours round the cycle; an odd cycle needs a third
bipartite with an edge: $K_{m,n}$, trees	2	Δ ($\max(m,n)$ for $K_{m,n}$)	colour by side; König's theorem for χ'
Petersen graph	3	4	has odd cycles; $\chi' = \Delta + 1$
friendship graph F_n, $n \geq 2$	3	$2n$	triangles; the centre has degree $2n$

- **Lower bounds:** if G contains K_k as a subgraph then $\chi(G) \geq k$ (a triangle, or any odd cycle, forces 3). $\chi'(G) \geq \Delta(G)$, since the edges at one vertex all need different colours.
- **Upper bounds:** $\chi(G) \leq \Delta(G) + 1$ (colour the vertices one at a time; at most Δ colours are blocked). For a simple graph $\chi'(G)$ is Δ or $\Delta + 1$.
- $\chi(G) = 1$ iff G has no edges; $\chi(G) \leq 2$ iff G is bipartite (§47).

DEFINITION **Petersen graph (AY24/25 Q25 · W12 s53–55)**

An outer 5-cycle, an inner 5-cycle drawn as a pentagram, and five spokes joining them.



- **10 vertices, 15 edges**, every vertex of degree 3; the shortest cycle has length 5 (no triangles, not bipartite).
- $\chi = 3$ and $\chi' = 4$, although $\Delta = 3$.
- **Not Eulerian** (every degree is odd). **Not Hamiltonian**, although a path through all 10 vertices exists.
- **Not planar:** every face would have at least 5 edges, so $5f \leq 2e = 30$ and $f \leq 6$; but Euler's formula (§49) needs $f = 15 - 10 + 2 = 7$.

METHOD **Modelling a conflict or scheduling problem as vertex colouring (L12 s13–18)**

1. **Vertices** are the objects to be assigned: items, guests, classes, radio stations, traffic signals.
2. **Edges** join two objects that **conflict** (cannot share a container, table, time slot, frequency).
3. **Colours** are the resources. Vertices of the same colour are objects that may share one resource.
4. The minimum number of resources is $\chi(G)$. **Lower bound:** find a triangle or a larger K_k. **Upper bound:** exhibit a colouring, taking vertices of high degree first. When the bounds meet, the answer is proved.

Wedding planner (L12 s16–17): guests B, C, D are in mutual conflict, so 3 tables are needed; $\{A, B, E\}$, $\{C, F\}$, $\{D, G, H\}$ works. **3 tables.**

PAST PAPER **AY22/23 Q23(f)–(i) · Volatile item packing as graph colouring**

Items $A, ..., K$ (no I); conflicts A: B, C; B: A, C, D; C: A, B, D; D: B, C, E, F; E: D, F, G, H; F: D, E, J, K; G: E, H; H: E, G; J: F, K; K: F, J.

(f) Vertices are items; two vertices are adjacent iff the items conflict; vertices of the same colour go into the same container.

(g) 14 edges: $AB, AC, BC, BD, CD, DE, DF, EF, EG, EH, GH, FJ, FK, JK$.

(h) The triangle ABC forces 3 colours, and 3 suffice: colour 1: A, D, G, J; colour 2: B, E, K; colour 3: C, F, H.

(i) 3 containers: $\{A, D, G, J\}$, $\{B, E, K\}$, $\{C, F, H\}$.

PAST PAPER **AY24/25 Q25(a)–(c), (f) · Chromatic number and index**

(a) $\chi(K_{1231}) = 1231$; $\chi(K_{1231,1231}) = 2$; $\chi(C_{1231}) = 3$ (odd cycle); $\chi(\text{Petersen}) = 3$.

(b) $\chi'(K_{100}) = 99$ (n even); $\chi'(K_{1231,1231}) = 1231$ (bipartite, $\Delta = 1231$); $\chi'(\text{Petersen}) = 4$.

(c) The Petersen graph is not Eulerian and not Hamiltonian.

(f) The graph in (e) is the **friendship graph** (T10 Q7), here F_6. With $2n + 1$ vertices and $n > 1$: $\chi' = 2n$. The $2n$ edges at the centre need $2n$ colours; each rim edge then takes the colour of a spoke of another triangle. ($F_1 = K_3$ needs 3, hence $n > 1$.)

Matchings and dominating sets

DEFINITION **Matching · Maximal, maximum, perfect matching · Neighbours $N(S)$ (AY25/26 Q25)**

A **matching** in $G = (V, E)$ is a subset $F \subseteq E$ in which no two edges share a vertex.

- **Maximal matching:** cannot be extended by adding another edge (every other edge meets an edge of F).
- **Maximum matching:** contains the largest possible number of edges.
- **Perfect matching:** covers every vertex of the graph.

For a set S of vertices, $N(S)$ is the set of vertices adjacent to at least one vertex of S.

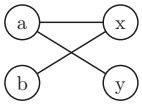
Every maximum matching is maximal, **not conversely**. Every perfect matching is maximum.

THEOREM **Hall's Marriage Theorem (AY25/26 Q25)**

A bipartite graph $G = (U, V, E)$ has a matching that covers all vertices in U **iff** for every subset $S \subseteq U$: $|N(S)| \geq |S|$.

METHOD **Using Hall's theorem in both directions**
To show a matching covering U exists: write one down. Start with the vertices of degree 1 (their edge is forced), delete the matched vertices, repeat. Checking Hall's condition directly means all $2^{|U|}$ subsets.
To show none exists: give one set $S \subseteq U$ with $|N(S)| < |S|$. Look for several vertices of U whose neighbours all lie in a small part of V; a vertex of degree 0 and the case $|U| > |V|$ ($S = U$) are instant.
Perfect matching: a matching covering U has $|U|$ edges and meets $|U|$ vertices of V, so it is perfect iff also $|U| = |V|$.

PAST PAPER **AY25/26 Q25(e)–(h) · Matchings**
(e) The additional condition is $|U| = |V|$.
(f) $U = \{a, b\}$, $V = \{x, y\}$, $E = \{ax, ay, bx\}$ (drawn on the right). $\{ax\}$ is **maximal** (ay shares a, bx shares x) but has 1 edge; $\{ay, bx\}$ is **maximum** with 2 edges.
(g) G_1: a has only x and c has only y, so ax and cy are forced; then bw and dz. **Yes:** $\{(a, x), (b, w), (c, y), (d, z)\}$.
(h) G_2: take $S = \{a, b, c\}$. $N(S) = \{w, x\}$, so $|N(S)| = 2 < 3 = |S|$. **No perfect matching.**



PAST PAPER **T10 Q11 (AY21/22 exam) · Dominating sets**
$D \subseteq V$ is a **dominating set** of a simple graph iff every vertex not in D is adjacent to at least one vertex in D; it is **minimal** iff no proper subset of D is dominating.
(a) Size one (one vertex adjacent to all others), on four vertices: K_4 with any vertex; the star $K_{1,3}$ with its centre.
(b) Size three, on four vertices: one edge plus two isolated vertices (the two isolated vertices and one end of the edge); the star $K_{1,3}$ with its three leaves.
Minimal is not minimum: the star has minimal dominating sets of sizes 1 and 3.

47 · Walks, connectedness, Euler and Hamiltonian circuits

DEFINITION **Walk, trail, path, closed walk, circuit, simple circuit (L12 s32–33)**
A **walk** from v to w is a finite alternating sequence $v_0 e_1 v_1 e_2 \cdots v_{n-1} e_n v_n$ of adjacent vertices and edges with $v_0 = v$, $v_n = w$, and v_{i-1}, v_i the endpoints of e_i. Its **length** is n, the number of edges. The **trivial walk** from v to v is the single vertex v.

	Repeated edge?	Repeated vertex?	Closed?	Notes
Walk	allowed	allowed	allowed	
Trail	no	allowed	allowed	a walk with no repeated edge
Path	no	no	no	a trail with no repeated vertex
Closed walk	allowed	allowed	yes	
Circuit (cycle)	no	allowed	yes	closed walk of length at least 3, no repeated edge
Simple circuit	no	first and last only	yes	

An undirected graph is **cyclic** if it contains a loop or a cycle; otherwise it is **acyclic**. (Other books say "path" for trail and "cycle" for simple circuit; the course follows Epp.)

DEFINITION **Connected (L12 s35) · Connected component (L12 s38)**
Vertices v and w are **connected** iff there is a walk from v to w. G is **connected** iff $\forall v, w \in V$ there is a walk from v to w.
H is a **connected component** of G iff (1) H is a subgraph of G, (2) H is connected, and (3) no connected subgraph of G has H as a subgraph and contains vertices or edges not in H. A connected component is a connected subgraph of largest possible size.

THEOREM **Lemma 10.2.1 (L12 s37)**
Let G be a graph.
- (a) If G is connected, then any two distinct vertices of G can be connected by a **path**.
- (b) If v and w are part of a circuit and one edge is removed from the circuit, there is still a trail from v to w.
- (c) If G is connected and contains a circuit, then an edge of the circuit can be removed without disconnecting G.

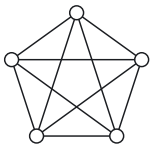
DEFINITION **Euler circuit, Eulerian graph (L12 s41) · Euler trail (L12 s43) · Hamiltonian circuit, Hamiltonian graph (L12 s49)**
An **Euler circuit** for G is a circuit that contains every vertex and traverses **every edge exactly once**; a graph with one is **Eulerian**.
An **Euler trail** from v to w ($v \neq w$) starts at v, ends at w, passes through every vertex at least once and traverses every edge exactly once.
A **Hamiltonian circuit** for G is a simple circuit that includes **every vertex** (each exactly once, except that the first and last are the same); a graph with one is **Hamiltonian**.

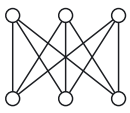
THEOREM **Theorems 10.2.2–10.2.4 · Corollary 10.2.5 · Proposition 10.2.6 (L12 s41–43, s52)**
- **10.2.2.** If a graph has an Euler circuit, then every vertex has positive even degree. (Contrapositive: an odd-degree vertex rules out an Euler circuit.)
- **10.2.3.** If G is connected and every vertex has positive even degree, then G has an Euler circuit.
- **10.2.4.** G has an Euler circuit **iff** G is connected and every vertex has positive even degree.
- **10.2.5.** There is an Euler trail from v to w iff G is connected, v and w have odd degree, and every other vertex has positive even degree.
- **10.2.6.** If G has a Hamiltonian circuit, then G has a subgraph H that (1) contains every vertex of G, (2) is connected, (3) has as many edges as vertices, and (4) has every vertex of degree 2.

There is no simple criterion for Hamiltonian circuits. To show none exists, use the contrapositive of 10.2.6.

PROOF **Theorem 10.2.2 · An Euler circuit forces every degree to be positive and even**
1. Let C be an Euler circuit of G and v any vertex. C contains v, so $\deg(v) > 0$.
2. Follow C once around. Each time it passes through v it arrives on one edge and leaves on another, using two edge-ends at v; if v is the starting vertex, the first departure pairs with the final arrival.
3. C uses every edge exactly once, so the edge-ends at v are split into pairs. Hence $\deg(v)$ is even. ■

WORKED EXAMPLE **Euler and Hamilton · Königsberg · T10 Q7 · AY25 final Q25(b)–(d)**
- **Königsberg** (4 land masses, 7 bridges): degrees $5, 3, 3, 3$. No Euler circuit; no Euler trail either, since four vertices are odd.
- "Every vertex has positive even degree" alone is not enough: two disjoint triangles are not connected.
- $\boldsymbol{K_n}$ ($n \geq 3$) is Hamiltonian; it is Eulerian iff n is odd.
- $\boldsymbol{K_{m,n}}$ ($m, n \geq 2$) is Eulerian iff m and n are both even; it is Hamiltonian iff $m = n$ (a circuit alternates between the two sides).
- **Cycle graph $\boldsymbol{C_n}$** is bipartite iff n is even: C_{12} and C_{8888} are; C_{57}, C_{123}, C_{1231}, C_{12345} are not.
- **Friendship graph $\boldsymbol{F_n}$** (n triangles sharing a centre, T10 Q7): Eulerian, since the degrees are 2 and $2n$. Not Hamiltonian for $n \geq 2$: passing from one triangle to another repeats the centre.


K_5


$K_{3,3}$

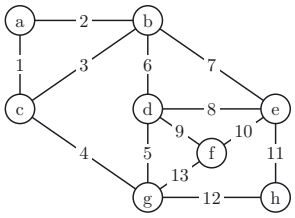
PAST PAPER **AY25 final Q9 · Relations from walks**
On the vertex set of a graph let xRy iff there is a walk from x to y.
Undirected: R is an equivalence relation (reflexive by the trivial walk; symmetric by reversing a walk; transitive by joining walks). Its classes are the vertex sets of the connected components.
Directed: R need not be a partial order. Even with no pair of opposite edges, the directed cycle $a \to b \to c \to a$ gives aRb and bRa with $a \neq b$, so antisymmetry fails. Only (i) is true.

PAST PAPER **AY22/23 Q23(a), (b) · AY23/24 Q24(c)(i) · Where an Euler trail starts and ends**

By Corollary 10.2.5 an Euler trail exists iff the graph is connected with **exactly two** odd vertices, and every Euler trail **starts at one of them and ends at the other**. No odd vertex: an Euler circuit, which may start anywhere. More than two: neither.

AY22/23 Q23. Degrees: a 2, b 4, c 3, d 4, e 4, f 3, g 4, h 2. **(a)** One trail: $c, a, b, c, g, d, b, e, d, f, e, h, g, f$ (the edge labels on the right give the order). **(b)** It may start at $\boldsymbol{c}$ **or** $\boldsymbol{f}$ only.

AY23/24 Q24(c)(i). Remove an edge $\{u, v\}$, $u \neq v$, from a graph with an Euler circuit: the edge lies on a circuit, so G' stays connected (Lemma 10.2.1(c)), and exactly u and v become odd. G' has an Euler trail from u to v: **true**. (If the removed edge were a loop, G' would keep an Euler circuit instead.)



Bipartite graphs and odd cycles

THEOREM **A graph is bipartite iff it has no cycle of odd length (AY24/25 Q22 · AY22/23 Q19)**

Equivalently: G is bipartite iff its vertices can be coloured with at most 2 colours ($\chi(G) \leq 2$).

So $K_{m,n}$, trees, forests and even cycles are bipartite; a graph containing a triangle, or any odd cycle, is not.

PROOF **Bipartite ⇔ no odd cycle**

1. ($\Rightarrow$) Let G be bipartite with sides U, V, and let $v_0 v_1 \cdots v_k$ with $v_k = v_0$ be a cycle. Every edge joins U to V, so the vertices alternate sides *(definition of bipartite)*.
2. After k steps the cycle is back on the side of v_0, so k is even.
3. ($\Leftarrow$) Suppose G has no odd cycle. Treat each connected component separately; pick a vertex r in it, and let $d(x)$ be the length of a shortest path from r to x *(Lemma 10.2.1(a))*.
4. Put x in U if $d(x)$ is even and in V if $d(x)$ is odd.
5. Suppose an edge $\{x, y\}$ had both ends on the same side. The shortest paths from r to x and from r to y, with the edge $\{x, y\}$, form a closed walk of length $d(x) + d(y) + 1$, which is odd.
6. A closed walk of odd length contains a cycle of odd length: contradiction. So every edge joins U to V. ■

PAST PAPER **AY24/25 Q22 · AY22/23 Q19 · True or false**

AY24/25 Q22. (i) Bipartite iff no odd cycle: **true**. (ii) Bipartite iff the sum of all degrees is even: **false**, since that sum is even for every graph (handshake theorem), for example K_3. (iii) $K_{2,n}$ is planar for $n > 3$: **true** (§49). Answer: only (i) and (iii).

AY22/23 Q19. (i) A simple graph with 6 vertices, 5 edges and no cycles cannot have exactly one vertex of degree 1: **true**. A forest with c components has $v - c$ edges, so $c = 1$: it is a tree, with at least two leaves. (ii) Not connected $\Rightarrow e < n - 1$: **false** (K_4 plus an isolated vertex: $n = 5$, $e = 6$). (iii) All cycles in $K_{3,4}$ have an even number of edges: **true** (bipartite). Answer: only (i) and (iii).

Cut-points, bridges, vertex cut sets

DEFINITION **$G - v$, cut-point · $G - e$, bridge (AY22/23 Q23) · Vertex cut set (AY24/25 Q25)**

Let $G = (V, E)$ be a connected undirected graph.

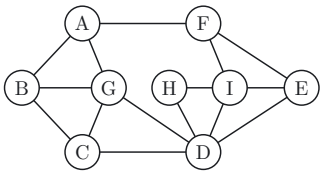
$G - v$ is G with the vertex v and all edges incident to v deleted; v is a **cut-point** (cut vertex) iff $G - v$ is disconnected.

$G - e$ is G with the edge e deleted (its endpoints stay); e is a **bridge** iff $G - e$ is disconnected.

A **vertex cut set** is a set $S \subseteq V$ whose removal, with all incident edges, disconnects G. A **minimum vertex cut set** is one of the smallest possible size.

THEOREM **Bridges and cut-points · AY22/23 Q23(c), (d)**

- An edge is a bridge **iff it lies on no circuit**. (On a circuit: removing it keeps G connected, Lemma 10.5.3. On no circuit: if $G-e$ were connected, a path between the endpoints of e together with e would be a circuit.)
- **In a tree with 2 or more vertices every edge is a bridge** (a tree has no circuits): Q23(d) **true**.
- **In a tree every vertex of degree greater than 1 is a cut-point**: Q23(c) **true**. Two neighbours of v are joined by the unique path through v (§50), so they are separated in $T-v$. A leaf is never a cut-point.
- K_n has no vertex cut set. In C_n ($n \geq 4$) no single vertex is a cut-point; two non-adjacent vertices form a minimum vertex cut set.
- The neighbours of a vertex v form a vertex cut set whenever other vertices remain, so a minimum vertex cut set has at most (minimum degree) vertices. Search sizes $1, 2, \ldots$ in turn.



PAST PAPER **AY24/25 Q25(d) · Minimum vertex cut set**

No single vertex disconnects the graph, so the minimum size is 2. The left part $\{A, B, C, G\}$ is joined to the rest only by AF, CD and GD.
$\boldsymbol{\{A, D\}}$ separates $\{B, C, G\}$ from $\{E, F, H, I\}$; $\boldsymbol{\{D, F\}}$ separates $\{A, B, C, G\}$ from $\{E, H, I\}$; $\boldsymbol{\{D, I\}}$ isolates H, whose only neighbours are D and I. Any one of these three is a correct answer, and there are no others of size 2.

PAST PAPER **Edge counts and components · AY23/24 Q19, Q20, Q24(c)(iii), Q24(d)**

Extremes. A simple graph with n vertices and k connected components has between $n-k$ edges (a forest) and $\binom{n-k+1}{2}$ edges (K_{n-k+1} with $k-1$ isolated vertices). So a disconnected simple graph has at most $\binom{n-1}{2}$ edges, and $\binom{n-1}{2}+1$ edges force connectedness.
Q24(d). G is connected on n vertices (n even); removing one edge leaves two complete graphs K_a and K_{n-a}. $|E| = \binom{a}{2} + \binom{n-a}{2} + 1$, least when $a = n/2$: $\boldsymbol{n(n-2)/4+1}$ (check $n=4$: 3, the path).
Q24(c)(iii). "Every vertex of degree $n-2$ is possible for all $n > 3$": **false**. The total degree $n(n-2)$ must be even, so n must be even (then K_n minus a perfect matching works).
Q19, Q20. v_i and v_j are joined iff $5 \mid (i+j)$. Indices with residues 1 and 4 (mod 5) form one component, residues 2 and 3 another, residue 0 a third. At most 3 components; the residue-0 component has $\lfloor n/5 \rfloor$ vertices; the other two differ in size by at most 1: **all of (i), (ii), (iii)**. If $5 \mid n$ the sizes are $2n/5$, $2n/5$, $n/5$: **two components with $2n/5$ vertices each**.

48 · Adjacency matrices and counting walks

DEFINITION **Adjacency matrix (L12 s61–62) · Matrix product (L12 s64) · Identity, powers (L12 s69–70)**

For a graph G with ordered vertices $v_1, \ldots, v_n$, the **adjacency matrix** is the $n \times n$ matrix $A = (a_{ij})$ over the non-negative integers with

- **directed:** a_{ij} = the number of arrows from v_i to v_j;
- **undirected:** a_{ij} = the number of edges connecting v_i and v_j.

The matrix of an undirected graph is **symmetric**: $a_{ij} = a_{ji}$.
For an $m \times k$ matrix A and a $k \times n$ matrix B, the **product** $AB = (c_{ij})$ has $c_{ij} = \sum_{r=1}^{k} a_{ir} b_{rj}$ (row i of A times column j of B). Matrix multiplication is associative but **not commutative**.
I_n is the $n \times n$ **identity matrix** (1 on the main diagonal, 0 elsewhere). **Powers:** $A^0 = I$ and $A^n = AA^{n-1}$ for $n \geq 1$.

THEOREM **Theorem 10.3.2 · Counting walks (L12 s74)**

If A is the adjacency matrix of a graph with vertices $v_1, \ldots, v_m$, then for every positive integer n and all i, j:

the ij-th entry of A^n = the number of walks of length n from v_i to v_j.

WORKED EXAMPLE **L12 s72–73 · Walks of length 2**

G has vertices v_1, v_2, v_3 and edges $e_1 = \{v_1, v_2\}$, a loop e_2 at v_2, and parallel edges $e_3, e_4 = \{v_2, v_3\}$.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 6 & 2 \\ 2 & 2 & 4 \end{pmatrix}.$$

Entry $(2,2)$ of A^2 is $1 \cdot 1 + 1 \cdot 1 + 2 \cdot 2 = 6$: one walk via v_1 ($v_2 e_1 v_1 e_1 v_2$), one via the loop ($v_2 e_2 v_2 e_2 v_2$), and four via v_3 (choose e_3 or e_4 each way). A walk may repeat edges, and different parallel edges give different walks.

Reading a matrix: a non-zero diagonal entry is a loop; an entry greater than 1 means parallel edges; in a simple graph the row sum of v_i is $\deg(v_i)$, and the diagonal of A^2 lists the degrees.

METHOD **Walks of length 3 or 4 without computing every entry**
Rows are "from", columns are "to"; a digraph's matrix need not be symmetric, and a loop puts 1 on the diagonal. Compute A^2 in full, then only the entry wanted:

$$(A^3)_{ij} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } A^2), \qquad (A^4)_{ij} = (\text{row } i \text{ of } A^2) \cdot (\text{column } j \text{ of } A^2).$$

PAST PAPER **AY25/26 Q20 · AY24/25 Q20 · AY23/24 Q24(b) · AY20/21 Q12 · Counting walks in a digraph**
AY25/26 Q20. Arrows $a \to c$, $b \to a$, $b \to b$, $c \to a$, $c \to b$ (rows and columns in the order a, b, c):

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad A^4 = (A^2)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 3 & 3 & 2 \\ 3 & 3 & 2 \end{pmatrix}.$$

Walks of length 4: a to c: 1; b to b: 3; c to a: 3; c to b: 3. The least of all nine entries is ***a* to *c*** (option A).
AY24/25 Q20. Arrows $a \to a$, $a \to b$, $a \to c$, $b \to a$, $b \to c$, $c \to a$:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad A^4 = \begin{pmatrix} 13 & 6 & 9 \\ 9 & 4 & 6 \\ 6 & 3 & 4 \end{pmatrix}.$$

a to b: 6; b to b: 4; c to a: 6; c to b: 3, the least entry: ***c* to *b*** (option D).
AY23/24 Q24(b). $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$, $A^2 = \begin{pmatrix} 5 & 4 & 4 \\ 4 & 7 & 6 \\ 2 & 3 & 6 \end{pmatrix}$. Walks of length 4 from v_1 to v_3: $(5,4,4) \cdot (4,6,6)$ $= 20 + 24 + 24 = \mathbf{68}$.
AY20/21 Q12. Rows of A: $(1,1,0,1)$, $(0,0,1,1)$, $(0,0,1,1)$, $(0,1,0,1)$. Column 4 of A^2 is $(3,2,2,2)$, so walks of length 3 from vertex 1 to vertex 4: $(1,1,0,1) \cdot (3,2,2,2) = \mathbf{7}$.

49 · Isomorphism and planar graphs

DEFINITION **Isomorphic graphs (L12 s79)**
$G = (V_G, E_G)$ is **isomorphic** to $G' = (V_{G'}, E_{G'})$, written $G \cong G'$, iff there are bijections $g : V_G \to V_{G'}$ and $h : E_G \to E_{G'}$ that preserve the edge-endpoint functions: for all $v \in V_G$ and $e \in E_G$,

$$v \text{ is an endpoint of } e \Leftrightarrow g(v) \text{ is an endpoint of } h(e).$$

For simple graphs: $G \cong G'$ iff there is a bijection $\pi : V_G \to V_{G'}$ such that $\{u, v\} \in E_G$ $\Leftrightarrow \{\pi(u), \pi(v)\} \in E_{G'}$.

THEOREM **Theorem 10.4.1 (L12 s81) · Invariants**
Graph isomorphism is an equivalence relation on any set of graphs. (Reflexive: identity maps. Symmetric: inverse bijections. Transitive: compose the bijections.)
Isomorphic graphs agree on every structural property: number of vertices, number of edges, the list of degrees, number of connected components, number of circuits of each length, being bipartite, being Eulerian or Hamiltonian.

METHOD **Isomorphic or not?**
To prove isomorphic: write the bijection on vertices explicitly and check that every edge maps to an edge (and the edge counts agree).
To prove not isomorphic: name one invariant that differs. Compare in this order: vertices, edges, sorted degree list, then circuits of length 3 or 4, then what the vertices of a given degree are adjacent to. Equal degree lists do **not** prove isomorphism.

DEFINITION **Planar graph (L12 s82) · Subdivision (L12 s84)**
A **planar graph** is a graph that **can be** drawn on a plane without edges crossing. One drawing with crossings does not show a graph is non-planar. A planar drawing divides the plane into **faces** (regions), including the outer one.
A **subdivision** of G is obtained by inserting new vertices into edges of G (replace an edge by a path through a new vertex).

THEOREM **Kuratowski's theorem (L12 s83) · Euler's formula (L12 s86)**
- K_5 and $K_{3,3}$ are not planar. A finite graph is planar **iff** it has no subgraph that is a subdivision of K_5 or of $K_{3,3}$.
- For a **connected planar** simple graph with v vertices, e edges and f faces: $f = e - v + 2$.

Examples: K_4 is planar with $v = 4$, $e = 6$, $f = 4$. $K_{3,4}$ is not planar, since it contains $K_{3,3}$ (AY25 Q25(a)). A tree has $f = (v-1) - v + 2 = 1$ face.

THEOREM **Consequences of Euler's formula (AY22/23 Q18 · AY23/24 Q18 · AY24/25 Q22)**
Let G be a **simple connected planar** graph with $v \geq 3$ vertices and e edges.
- $e \leq 3v - 6$. If G has no triangles (for example G bipartite): $e \leq 2v - 4$.
- **Every simple planar graph has a vertex of degree at most 5.**
- K_5: $e = 10 > 9$. $K_{3,3}$: no triangles and $e = 9 > 8$. Both are non-planar. The inequalities are necessary only: the Petersen graph has $15 \leq 24$ edges and is not planar.
- **Planar:** K_n iff $n \leq 4$; $K_{m,n}$ iff $\min(m,n) \leq 2$. $\boldsymbol{K_{2,n}}$ **is planar for every** $\boldsymbol{n}$**:** put the n vertices in a column and the other two on its left and right. $\boldsymbol{K_5}$ **minus one edge is planar:** draw K_4 as a triangle with a centre and join a fifth vertex, placed outside, to the three corners; only the edge from centre to outside is absent.
- With c connected components the formula becomes $f = e - v + c + 1$.

PROOF $e \leq 3v - 6$ · $e \leq 2v - 4$ · **A vertex of degree at most 5**
1. Every face is bounded by at least 3 edges *(simple, $v \geq 3$)* and every edge borders at most 2 faces, so $3f \leq 2e$.
2. Substitute $f = e - v + 2$ *(Euler's formula)*: $3(e - v + 2) \leq 2e$, that is $e \leq 3v - 6$.
3. With no triangles every face has at least 4 edges: $4f \leq 2e$, so $2(e - v + 2) \leq e$, that is $e \leq 2v - 4$.
4. Suppose every vertex of a simple planar graph had degree at least 6; pick a connected component (it has $v \geq 7$). Then $2e \geq 6v$ *(handshake theorem)*, so $e \geq 3v > 3v - 6$: contradiction with line 2. ■

PAST PAPER **AY22/23 Q18 · AY23/24 Q18 · Planarity true or false**
AY22/23 Q18. (i) $K_5 - \{e\}$ is planar: **true**. (ii) K_5 is Eulerian and Hamiltonian: **true** (connected, every degree 4; K_n is Hamiltonian for $n \geq 3$). (iii) A simple planar graph with 6 vertices, 8 edges and 7 faces: **false**. $f = e - v + c + 1 = 3 + c$ needs $c = 4$ components, but 6 vertices in 4 components carry at most $\binom{3}{2} = 3$ edges. Answer: only (i) and (ii).
AY23/24 Q18. (i) Two spanning trees of a graph always share an edge: **false** (§52). (ii) Every simple planar graph has a vertex of degree at most 5: **true**.

PAST PAPER **T11 Q1 · Self-complementary graphs on $n = 3, 4, 5, 6$ vertices**
If $G \cong \overline{G}$ then G and $\overline{G}$ have the same number of edges, so $\binom{n}{2}$ is even. For $n = 3$ and $n = 6$, $\binom{n}{2} = 3$ and 15: **none**.
$n = 4$ (3 edges): the path on 4 vertices, and only that. $n = 5$ (5 edges): the cycle C_5, and the "bull" (a triangle with a pendant edge at two of its vertices): **two**.

THEOREM **Self-complementary graphs need $n \equiv 0$ or $1 \pmod 4$ · AY20/21 Q21(b), (c)**
A self-complementary simple graph on n vertices has exactly $n(n-1)/4$ edges. Hence $n \equiv 0$ or $1 \pmod 4$; in particular there is **no** self-complementary graph with $4k + 2$ (or $4k + 3$) vertices.

PROOF **Edge count of a self-complementary graph**
1. Let $G \cong \overline{G}$ with n vertices. Isomorphic graphs have the same number of edges, so $|E| = |\overline{E}|$ *(Theorem 10.4.1, invariants)*.
2. Each of the $\binom{n}{2}$ pairs of vertices is an edge of exactly one of G, $\overline{G}$: $|E| + |\overline{E}| = n(n-1)/2$. So $|E| = n(n-1)/4$.
3. $|E|$ is an integer, so $4 \mid n(n-1)$. One of n, $n-1$ is odd, so 4 divides the other: $n \equiv 0$ or $1 \pmod 4$.
4. For $n = 4k + 2$: $n(n-1)/2 = (2k+1)(4k+1)$ is odd, so it cannot be split into two equal parts. No such graph exists. ■

WORKED EXAMPLE **Small enumerations up to isomorphism · AY22/23 Q20 · AY24/25 Q21 · AY20/21 Q20(d)**
- **Simple graphs on $\boldsymbol{n = 1, 2, 3, 4, 5}$ vertices:** 1, 2, 4, 11, 34 classes; connected ones: 1, 1, 2, 6, 21.
- **The 11 on 4 vertices, by number of edges** $0, 1, \ldots, 6$: $1, 1, 2, 3, 2, 1, 1$ (the list is symmetric under taking complements). With 3 edges: path, star, triangle plus isolated vertex.
- **AY22/23 Q20.** Disconnected simple graphs on 4 vertices: $11 - 6 = \mathbf{5}$ (no edge; one edge; two disjoint edges; two adjacent edges plus an isolated vertex; triangle plus an isolated vertex).

- **AY24/25 Q21.** 25 labelled simple graphs on 3 vertices fall into 4 isomorphism classes (0, 1, 2 or 3 edges), so at least $\lceil 25/4 \rceil = \mathbf{7}$ are isomorphic to one another (generalized pigeonhole, §41).
- **AY20/21 Q20(d).** Directed graphs on 3 labelled vertices, no parallel edges: each of the 9 ordered pairs (loops included) is present or absent, $2^9 = \mathbf{512}$. With at least one loop: $512 - 2^6 = \mathbf{448}$.

50 · Trees

DEFINITION **Circuit-free, tree, trivial tree, forest (L13 s4) · Leaf, internal vertex (L13 s9)**

A graph is **circuit-free** iff it has no circuits. A simple graph is a **tree** iff it is **circuit-free and connected**. A **trivial tree** is a single vertex. A simple graph is a **forest** iff it is circuit-free and not connected.

In a tree with one or two vertices each vertex is a **terminal vertex** (**leaf**). In a tree with at least three vertices, a vertex of degree 1 is a **leaf** and a vertex of degree greater than 1 is an **internal vertex**.

THEOREM **Lemma 10.5.1 · Theorem 10.5.2 · Lemma 10.5.3 · Theorem 10.5.4 · Lemma 10.5.5 (L13 s7–15, T11)**

- **10.5.1.** Any non-trivial tree has at least one vertex of degree 1 (in fact at least two).
- **10.5.2.** Any tree with n vertices ($n > 0$) has $n - 1$ edges.
- **10.5.3.** If G is connected and C is a circuit in G, removing one edge of C leaves G connected.
- **10.5.4.** If G is a **connected** graph with n vertices and $n - 1$ edges, then G is a tree.
- **10.5.5 (T11).** In a simple graph, if there are two distinct paths from v to a different vertex w, then G contains a cycle.

Connectedness is needed in 10.5.4: a triangle with a pendant edge, together with one isolated vertex, has 5 vertices and 4 edges but is not a tree (L13 s17).

PROOF **Lemma 10.5.1 (L13 s7) · Theorem 10.5.2 by induction (L13 s10–11) · At least two leaves (L13 s12)**

10.5.1. Start at any vertex v and walk along edges, never reusing an edge. A vertex cannot be revisited *(T is circuit-free)*, and T has finitely many vertices, so the walk stops at a vertex with no unused edge: a vertex of degree 1.

10.5.2. Let $P(n) \equiv$ "any tree with n vertices has $n - 1$ edges".

1. $P(1)$: a tree with one vertex has no edges *(a simple graph has no loops)*.
2. Assume $P(k)$ for an arbitrary $k \geq 1$. Let T be any tree with $k + 1$ vertices.
3. T is non-trivial, so it has a vertex v of degree 1 *(Lemma 10.5.1)* let e be the edge at v.
4. Let T' be T with v and e removed. T' has k vertices, is circuit-free (a subgraph of T) and is connected (no path between other vertices passes through v, as $\deg(v) = 1$). So T' is a tree.
5. T' has $k - 1$ edges *(inductive hypothesis)*, so T has $k = (k + 1) - 1$ edges. Thus $P(k + 1)$ is true. ■

Two leaves. Let T be a tree with $n \geq 2$ vertices. Every degree is at least 1 *(connected)*. If at most one vertex had degree 1, the total degree would be at least $1 + 2(n - 1) = 2n - 1$; but it equals $2(n - 1) = 2n - 2$ *(Theorem 10.5.2 and the handshake theorem)*. Contradiction. ■

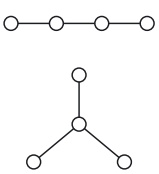
PROOF **Theorem 10.5.4 · A connected graph with n vertices and $n - 1$ edges is a tree (L13 s15–16)**

1. Let G be connected with n vertices and $n - 1$ edges. It suffices to show that G is circuit-free.
2. Suppose G has a circuit. Remove an edge of it; the graph stays connected *(Lemma 10.5.3)*. Repeat while a circuit remains.
3. The result G'' is connected and circuit-free, so it is a tree with n vertices and therefore $n - 1$ edges *(Theorem 10.5.2)*.
4. But at least one edge was removed, so G'' has at most $n - 2$ edges: contradiction.
5. Hence G is circuit-free, and G is a tree. ■

WORKED EXAMPLE **L13 s13 · T11 Q3 · Trees on few vertices**

A tree with 4 vertices has 3 edges and total degree 6, with at least two leaves: the degree lists are $1, 1, 2, 2$ (a path) and $1, 1, 1, 3$ (a star). So there are **two** non-isomorphic trees with 4 vertices.

Labelled copies (T11 Q3): $n = 1$: 1; $n = 2$: 1; $n = 3$: one tree (the path), 3 copies (choose the middle vertex); $n = 4$: the path has $4!/2 = 12$ copies and the star has 4 (choose the centre), 16 in all. In general there are n^{n-2} labelled trees.



WORKED EXAMPLE **Counting trees · W12 s51 · AY20/21 Q21(e)**

- **Non-isomorphic trees on $n = 1, ..., 6$ vertices:** $1, 1, 1, 2, 3, 6$.
- **5 vertices (W12 s51):** 4 edges, total degree 8, at least two leaves; the other three degrees sum to 6: $\{1, 1, 4\}$ (star), $\{1, 2, 3\}$, $\{2, 2, 2\}$ (path). **3 trees.**
- **Rooted trees** (an isomorphism must send root to root) on $n = 1, ..., 5$ vertices: $1, 1, 2, 4, 9$. **AY20/21 Q21(e), 5 vertices:** delete the root and list the sizes of the subtrees hanging from it. Sizes $\{4\}$: 4 trees (one per rooted tree on 4 vertices); $\{3, 1\}$: 2; $\{2, 2\}$: 1; $\{2, 1, 1\}$: 1; $\{1, 1, 1, 1\}$: 1. Total **9**.
- **Forest:** with c components, $e = v - c$. So "no cycles and $e = v - 1$" already forces a tree (AY22/23 Q19(i), §47).

PROOF **T11 Q4–Q6 · Edge bounds and the unique-path characterization**

Let $G = (V, E)$ be a simple graph.

1. **Connected $\Rightarrow |E| \geq |V| - 1$.** While G has a circuit, remove one of its edges; G stays connected *(Lemma 10.5.3)*. The process ends with a connected circuit-free graph on V, a tree with $|V| - 1$ edges *(Theorem 10.5.2)*, all of which are edges of G. **Converse false:** a triangle plus an isolated vertex has $3 \geq 4 - 1$ edges and is not connected.
2. **Acyclic $\Rightarrow |E| \leq |V| - 1$.** Each connected component is a tree. With $c \geq 1$ components of sizes $n_1, ..., n_c$, $|E| = \sum(n_i - 1) = |V| - c \leq |V| - 1$. **Converse false:** the same graph has $3 \leq 3$ edges and contains a cycle.
3. **G is a tree iff there is exactly one path between every pair of distinct vertices.**
   ($\Rightarrow$) G is connected, so any two distinct vertices are joined by a path *(Lemma 10.2.1(a))*. Two distinct paths between them would give a cycle *(Lemma 10.5.5)*, contradicting circuit-freeness.
   ($\Leftarrow$) A path between every pair makes G connected. If G had a circuit, two vertices on it would be joined by two distinct paths (the two ways round), contradicting uniqueness. So G is circuit-free and connected. ■

PAST PAPER **T11 Q7 (AY17/18 exam) · Splitting a pile of n stones**

Splitting a pile of k into k_1 and k_2 earns $k_1 k_2$ dollars. Model the stones as the vertices of K_n, where an edge means "still in the same pile". A split into k_1 and k_2 deletes exactly the $k_1 k_2$ edges between the two parts and earns one dollar per deleted edge. At the end every stone is alone, so all $\binom{n}{2}$ edges have been deleted exactly once. The total is $\boldsymbol{n(n-1)/2}$ **whatever the strategy**.

51 · Rooted trees and binary trees

DEFINITION **Rooted tree, level, height · Child, parent, sibling, ancestor, descendant (L13 s19)**

A **rooted tree** is a tree with one distinguished vertex, the **root**. The **level** of a vertex is the number of edges on the unique path between it and the root (the root has level 0). The **height** of a rooted tree is the maximum level of any vertex.

The **children** of v are the vertices adjacent to v and one level farther from the root; then v is their **parent**; distinct vertices with the same parent are **siblings**. If v lies on the unique path between w and the root ($v \neq w$), then v is an **ancestor** of w and w a **descendant** of v.

DEFINITION **Binary tree, full binary tree · Left and right subtree (L13 s22)**

A **binary tree** is a rooted tree in which every parent has at most two children, each designated a **left child** or a **right child** (not both), with at most one of each. A **full binary tree** is a binary tree in which each parent has **exactly two** children.

The **left subtree** of a parent v is the binary tree whose root is the left child of v and whose vertices are that child and all its descendants; the **right subtree** is defined analogously.

THEOREM **Theorem 10.6.1 Full binary tree theorem (L13 s27) · Theorem 10.6.2 (L13 s29)**

- **10.6.1.** A full binary tree with k internal vertices has $2k + 1$ vertices in total and $k + 1$ leaves.
- **10.6.2.** A binary tree of height h with t leaves satisfies $t \leq 2^h$; equivalently $\log_2 t \leq h$.

Hence: no full binary tree has 10 internal vertices and 13 leaves (it would have 11 leaves); no binary tree of height 5 has 38 leaves ($38 > 32$). A binary tree of height h has at most $2^{h+1} - 1$ vertices.

PROOF **Theorem 10.6.1 (L13 s27) · Theorem 10.6.2 by strong induction (L13 s30–34)**

10.6.1. Every vertex except the root has a parent. Each of the k internal vertices has exactly two children, so $2k$ vertices have a parent. Total: $2k+1$. Leaves: $(2k+1)-k=k+1$. ■

10.6.2. Let $P(h) \equiv$ "every binary tree of height h has at most 2^h leaves".

1. $P(0)$: the tree is a single vertex, a leaf: $t=1=2^0$.
2. Let $k \geq 0$ and assume $P(i)$ for $0 \leq i \leq k$. Let T have height $k+1$, root v and t leaves. Since $k+1 \geq 1$, v has a child.
3. Case 1 (v has one child): its subtree T_L has height k, so $t_L \leq 2^k$ *(inductive hypothesis)*. In the slides the root, having degree 1, is also counted, so $t = t_L + 1 \leq 2^k + 1 \leq 2^k + 2^k = 2^{k+1}$ *($2^k \geq 1$)*.
4. Case 2 (v has two children): the subtrees T_L, T_R have heights $h_L, h_R \leq k$, so $t = t_L + t_R \leq 2^{h_L} + 2^{h_R} \leq 2^k + 2^k = 2^{k+1}$ *(inductive hypothesis)*.
5. In both cases $P(k+1)$ is true. ■

DEFINITION **Tree traversal: BFS and DFS (L13 s36–39)**

Breadth-first search: visit the root, then the vertices level by level, left to right.

Depth-first search, three orders, each applied recursively:

- **Pre-order:** root, left subtree, right subtree.
- **In-order:** left subtree, root, right subtree.
- **Post-order:** left subtree, right subtree, root.

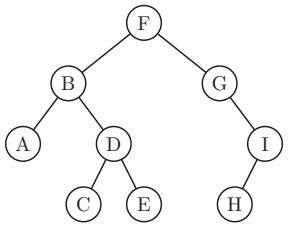
For the tree on the right (L13 s40):

BFS: $F, B, G, A, D, I, C, E, H$

pre-order: $F, B, A, D, C, E, G, I, H$

in-order: $A, B, C, D, E, F, G, H, I$

post-order: $A, C, E, D, B, H, I, G, F$



METHOD **Rebuilding a binary tree from two traversals**

Pre-order + in-order (or **post-order + in-order**) determine the tree. The root is the **first** item of the pre-order (the **last** of the post-order). Find it in the in-order: everything to its left is the left subtree, everything to its right is the right subtree. Split the other traversal into blocks of the same sizes and recurse.

Pre-order + post-order do not determine the tree: a vertex with exactly one child could have it on the left or the right, so there are 2^s trees, where s is the number of one-child vertices.

Expression trees (L13 s24–26): operators at internal vertices, operands at leaves; $((a-b)\cdot c)+(d/e)$ has root $+$.

PAST PAPER **T11 Q8, Q9, D3 · AY25 final Q19, Q21, Q22**

T11 Q8(a). In-order E A C K F H D B G, pre-order F A E K C D H G B. Root F; left part E A C K, right part H D B G. Left subtree: root A, left child E, right child K with left child C. Right subtree: root D, left child H, right child G with left child B. **Post-order: E, C, K, A, H, B, G, D, F.**

T11 Q8(b). In-order D B F E A G C H K, post-order D F E B G K H C A. Root A; left subtree: root B, left child D, right child E with left child F; right subtree: root C, left child G, right child H with right child K. **Pre-order: A, B, D, E, F, C, G, H, K.**

T11 Q9. Binary trees with in-order X Y Z: **5**; with in-order A B C D: **14** (the Catalan numbers: choose the root, then the shapes of the two sides).

AY25 Q21. Pre-order G B E D F C A, in-order B G F D C E A: root G with left child B; right subtree has root E, left child D (children F and C), right child A. Post-order: B, F, C, D, A, E, G.

AY25 Q22. Pre-order A B C D E, post-order C E D B A: A has the single child B; B has children C and D; D has the single child E. Two one-child vertices: $2^2 = 4$ trees.

AY25 Q19. Unlabelled rooted binary trees, left and right not distinguished: 2 with 3 vertices, 3 with 4 vertices, **6** with 5 vertices.

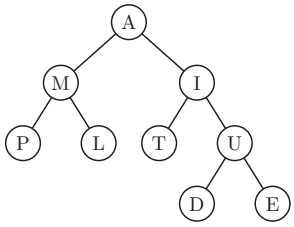
METHOD **Pre-order + post-order · When the tree is unique, and how to rebuild it**

1. The root is the **first** item of the pre-order and the **last** of the post-order.
2. If more items remain, the **second** item c of the pre-order is the root of the first subtree. Find c in the post-order: the items up to and including c form that subtree (say k items); the next k items of the pre-order after the root are the same set.
3. The items left over (before the root in the post-order) form the second subtree. Recurse on each block.
4. **Ambiguity:** if c is the second-last item of the post-order, nothing is left over: the vertex has **one** child, which may be a left or a right child. Each such vertex doubles the count (2^s trees, AY25/26 Q22).

Unique iff no vertex has exactly one child. For a **full** binary tree (given, as in AY23/24 Q24(e), or found, as in AY24/25 Q25(g)) step 4 never occurs: the first block is the left subtree and the rest the right subtree.

PAST PAPER **AY24/25 Q25(g) · AY23/24 Q24(e) · In-order from pre-order and post-order**

AY24/25 Q25(g). Pre A M P L I T U D E, post P L M T D E U I A. Root A. Next is M; in the post-order M ends the block P L M, so the left subtree is $\{M, P, L\}$ and the right is $\{I, T, U, D, E\}$.
Left (pre M P L, post P L M): M with children P, L. Right (pre I T U D E, post T D E U I): I with left child T (block T) and right subtree pre U D E, post D E U: U with children D, E. No one-child vertex, so the tree is unique. **In-order: P M L A T I D U E.**



AY23/24 Q24(e) (full binary tree). Pre S U R E C A N D O I T, post E C R A U O I D T N S. Root S; left block E C R A U (root U: left R with children E, C; right A); right block O I D T N (root N: left D with children O, I; right T). **In-order: E R C U A S O D I N T.**

PAST PAPER **AY22/23 Q23(e) · AY20/21 Q21(d) · Rebuilding with the in-order**

AY22/23 Q23(e). In-order E D B C G F A H K, pre-order C B D E A G F H K. Root C; left part E D B is the chain B, left child D, left child E. Right part G F A H K: root A; left G with right child F; right H with right child K. **Post-order: E D B F G K H A C.**

AY20/21 Q21(d). In-order U C A N D O I T Y E H, post-order C U N A T Y I H E O D. Root D. Left part U C A N: root A, left child U (with right child C), right child N. Right part O I T Y E H: root O with no left child and right child E; E has left child I and right child H; I has right child Y; Y has left child T. (Pre-order: D A U C N O E I Y T H.)

DEFINITION **Binary search tree (AY23/24 Q24(f)) · Height-balanced binary tree (W13 s7)**

A **binary search tree** (BST) is a binary tree with distinct values at its vertices such that, for every vertex with value x, all values in its left subtree are smaller than x and all values in its right subtree are greater than x. Its in-order traversal lists the values in increasing order.

A **height-balanced** binary tree is a binary tree in which, at every vertex, the heights of the left and right subtrees differ by at most 1 (an absent subtree has height -1).

THEOREM **Catalan numbers count binary trees · Minimum size of a height-balanced tree**

- The number of BSTs on n given values = the number of binary trees with a given in-order = the number of binary tree shapes with n vertices = the **Catalan number**
$$C_n = \frac{(2n)!}{(n+1)!\,n!} = \frac{1}{n+1}\binom{2n}{n}: \quad 1, 2, 5, 14, 42, 132, \ldots \quad (n = 1, 2, 3, \ldots).$$
Choosing the root leaves i values on the left and $n-1-i$ on the right: $C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$, $C_0 = 1$.
- **AY23/24 Q24(f):** BSTs with 2, 3, 4 values: **2, 5, 14** (compare T11 Q9).
- **Height-balanced, minimum number of vertices $N(h)$ for height h:** a root, a subtree of height $h-1$ and one of height $h-2$: $N(h) = 1 + N(h-1) + N(h-2)$ with $N(-1) = 0$, $N(0) = 1$, giving $1, 2, 4, 7, 12, 20$. Height 4 needs **12** vertices (AY22/23 Sem 2 Q20, W13 s7).
- **AY23/24 Q24(g)(i):** "a binary tree has k leaves and $k+1$ internal vertices" is **false** as a general claim; a full binary tree has it the other way round (k internal, $k+1$ leaves).

52 · Spanning trees and minimum spanning trees

DEFINITION Spanning tree (L13 s45) · Weighted graph, minimum spanning tree (L13 s48)

A **spanning tree** for a graph G is a subgraph of G that contains every vertex of G and is a tree.
A **weighted graph** is a graph in which each edge e has a positive real **weight** $w(e)$; the **total weight** $w(G)$ is the sum of the weights of all edges. A **minimum spanning tree** (MST) for a connected weighted graph is a spanning tree with the least possible total weight.

THEOREM Proposition 10.7.1 (L13 s45)

(1) Every connected graph has a spanning tree. (2) Any two spanning trees for a graph have the same number of edges ($n-1$ for n vertices).
Counting spanning trees: if G has exactly one circuit, of length k, removing any one of its edges gives a spanning tree: k spanning trees (L13 s46: 3). K_n has n^{n-2}.

THEOREM An edge lies in every spanning tree iff it is a bridge (AY23/24 Q24(g)(iii))

Let e be an edge of a connected graph G. Then e is contained in every spanning tree of G iff removing e disconnects G.

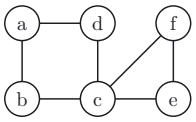
PROOF Bridges and spanning trees

1. ($\Leftarrow$) Let e be a bridge and suppose a spanning tree T does not contain e. Then T is a connected subgraph of $G-e$ containing every vertex, so $G-e$ is connected: contradiction.
2. ($\Rightarrow$, contrapositive) If e is not a bridge, $G-e$ is connected, so it has a spanning tree T *(Proposition 10.7.1)*. T contains every vertex of G, so it is a spanning tree of G that does not contain e. ■

METHOD Counting spanning trees when there is more than one circuit (T11 Q2 · L13 s46)

A spanning tree keeps $n-1$ edges: delete $e-(n-1)$ edges so that every circuit is broken and the graph stays connected. Bridges are never deleted.

- **Circuits sharing no edge** (they meet only at vertices): delete one edge from each, independently, and **multiply** the circuit lengths.
  T11 Q2 (right): the 4-circuit $adcb$ and the triangle cfe share only the vertex c: $4 \times 3 =$ **12**.
- **Two circuits sharing an edge** (or a path): split into cases. Shared edge deleted: one large circuit is left, delete any one of its edges. Shared edge kept: delete one other edge from each circuit.
  A 4-circuit and a triangle sharing an edge: $5 + 3 \times 2 = 11$. C_4 plus a chord: $4 + 2 \times 2 = 8$.
- Check on a known value: K_n has n^{n-2} (K_4: 16).



METHOD Algorithm 10.7.1 · Kruskal (L13 s50)

Input: a connected weighted graph G with n vertices.

1. Initialize T with all vertices of G and no edges.
2. Let E be the set of all edges of G; let $m = 0$.
3. While $m < n-1$:
   3.1. find an edge e in E of least weight;
   3.2. delete e from E;
   3.3. if adding e to T does not produce a circuit, add e to T and set $m = m+1$.

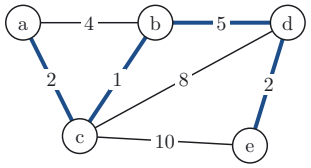
Output: T, a minimum spanning tree for G.

METHOD Algorithm 10.7.2 · Prim (L13 s55)

Input: a connected weighted graph G with n vertices.

1. Pick a vertex v; let T be the graph with this vertex only.
2. Let V be the set of all vertices of G except v.
3. For $i = 1$ to $n-1$:
   3.1. find an edge e of least weight among those connecting T to a vertex in V; let w be its endpoint in V;
   3.2. add e and w to T, and delete w from V.

Output: T, a minimum spanning tree for G.



WORKED EXAMPLE Kruskal and Prim on one graph

Edges: $ab = 4$, $ac = 2$, $bc = 1$, $bd = 5$, $cd = 8$, $ce = 10$, $de = 2$.
Kruskal: bc (1) added; ac (2) added; de (2) added; ab (4) rejected, since it closes the circuit abc; bd (5) added. Four edges: stop.
Prim from a: add ac (2), then cb (1), then bd (5), then de (2).

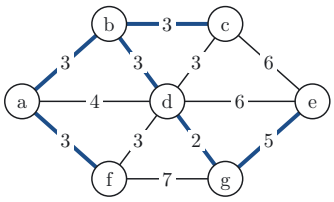
Both give the same tree, of weight $1+2+2+5=10$, with the edges added in different orders.

WORKED EXAMPLE **L13 s52, s57 · The airline graph (8 cities)**
Kruskal: Chi–Mil 74, Lou–Cin 83, Lou–Nas 151, Cin–Det 230, StL–Lou 242, StL–Chi 262 added; Chi–Lou 269, Lou–Det 306, Lou–Mil 348 **not added** (each would close a circuit); Min–Chi 355 added. Total weight **1397**.
Prim from Minneapolis: Min–Chi 355, Chi–Mil 74, Chi–StL 262, StL–Lou 242, Lou–Cin 83, Lou–Nas 151, Cin–Det 230. The same tree, total 1397.

TRAP **Minimum spanning trees**
- Kruskal may hold a **disconnected** forest midway; Prim's partial tree is always connected. Marking a Prim step that does not touch the current tree is an error.
- With repeated weights there may be several MSTs, but all have the same total weight; with all weights distinct the MST is unique. To count MSTs, look at each tie and see which choices are interchangeable.
- An MST is not a shortest-path tree. Dijkstra's algorithm (L13 s59 onward) is not examinable this semester.
- Stop after $n-1$ edges. If a question asks for the order of edges added, follow the named algorithm exactly.

METHOD **Counting minimum spanning trees when weights tie**
Run Kruskal one weight class at a time. For each weight, look at the edges of that weight that do not close a circuit with **lighter** edges, and count the ways to choose among them; ties in different parts of the graph are independent, so **multiply**. A tie where every tied edge is rejected, or every tied edge is accepted, gives no choice.
Equivalently: fix the forced edges, then count the spanning trees of what is left of the tied part.

PAST PAPER **AY24/25 Q25(e) · W12 s60 (AY20/21 Sem 2 Q21b) · Number of MSTs**
AY24/25 Q25(e). Friendship graph F_6; the triangles (spoke, spoke, rim) have weights $(3,3,3)$, $(9,9,3)$, $(10,10,7)$, $(6,3,3)$, $(9,3,6)$, $(9,3,6)$. The triangles share only the centre, so each one independently drops one heaviest edge and keeps the other two.
(i) Weight $6+12+17+6+9+9=$ **59**. (ii) Choices of the dropped edge: $3\times2\times2\times1\times1\times1=$ **12** MSTs.
W12 s60. Edges $ab=1$, $bc=1$, $ae=2$, $be=2$, $cd=2$, $bd=2$, a second $bd=3$, $de=3$. Both edges of weight 1 are taken. Then e is reached by ae or be (not both: circuit abe) and d by cd or the lighter bd (not both: circuit bcd): $2\times2=$ **4** MSTs, each of weight 6.



PAST PAPER **T11 Q10 (AY23/24 Sem 2 exam) · Weight and number of MSTs**
Vertices are labelled here for reference. **Kruskal:** dg (2) is taken. Weight 3: the six edges ab, bc, bd, cd, af, fd form a 4-circuit $abdf$ and a triangle bcd sharing bd; any spanning tree of this part (4 edges) is taken. Then ad (4) is rejected and ge (5) completes the tree; the edges of weight 6 and 7 are never used.
(a) Weight $2+4\times3+5=$ **19**.
(b) The tied part has $5+3\times2=11$ spanning trees, so there are 11 MSTs on the labelled graph. Up to isomorphism there are **5**: degree lists $1,1,1,1,2,2,4$; $1,1,1,1,2,3,3$; $1,1,2,2,2,2,2$ (a path); and two different trees with $1,1,1,2,2,2,3$ (longest path of 4 edges, or of 5 edges). One MST is shown in bold.

PAST PAPER **AY20/21 Q21(a) · AY23/24 Q24(a), Q24(g)(ii), Q18(i) · More on spanning trees**
AY20/21 Q21(a). Edges ab 62, bc 80, ad 61, bd 29, be 37, bg 37, ce 70, df 45, dg 32, eg 9, eh 13, fg 38, gh 12. Kruskal takes $\{e,g\}$ 9, $\{g,h\}$ 12, $\{b,d\}$ 29, $\{d,g\}$ 32, $\{f,g\}$ 38, $\{a,d\}$ 61, $\{c,e\}$ 70 and rejects 13, both 37s (each closes a circuit), 45, 62, 80. **Weight 251**; the MST is unique in spite of the tie.
AY23/24 Q24(a). From the weight matrix: $w_{12}=w_{14}=w_{24}=3$, $w_{23}=w_{34}=4$, every edge at vertex 5 weighs at least 5 ($w_{15}=w_{25}=w_{45}=5$), $w_{35}=6$, $w_{13}=8$. Kruskal: two edges of weight 3, one of weight 4, one of weight 5. **Weight 15.**
AY23/24 Q24(g)(ii). With all weights distinct the MST is unique, so Prim and Kruskal give the same tree: **true**.
AY23/24 Q18(i). "Two spanning trees of a graph always share an edge": **false**. In K_4 the paths $1,2,3,4$ and $2,4,1,3$ are spanning trees with no common edge.

Part XI · Exam Playbook and Locators

METHOD **Final exam format and approach (AY25/26 paper)**
Format: 2 hours, open book, calculators allowed, 100 marks. Part A: 23 MCQ at 2 marks each (46). Part B: three structured questions (54), recently Counting and Probability, Graphs and Trees, and Functions; many parts need the answer only, "as a single number".

- **Coverage is the whole module.** Roughly the first third of Part A is logic, sets and relations (§1–25); the rest is functions, induction and recursion, cardinality, counting, graphs and trees (§26–52).
- **New definitions in the question** (matchings, dominating sets, order of a function, left inverses) are routine: copy the definition onto scratch paper and test it on the smallest example.
- **Counting answers:** decide ordered or unordered, repetition or not (table in §56), then check the formula on a case small enough to list.
- **"Single number" answers** get no method marks: recompute by a second route (difference rule against direct cases).
- **Proof parts:** use the class format for induction (§32), the injective/surjective templates (§28), and name the theorem used at each step.

53 · Part A: multiple-choice questions

METHOD **Approach**

- **Judge each of (i)–(iv) on its own** and mark T/F in the margin before reading the options. Several options usually differ in only one item. "None of (A)–(D)" has been the correct answer several times (AY23 Q2, AY24 Q12, AY25 Q2).
- **"For all sets / all relations" claims:** look for a counterexample among $\varnothing$, a singleton, $\{a, b\}$, the empty relation and the identity relation. One counterexample is enough.
- **Relations on small sets:** draw the digraph. **Posets:** draw the Hasse diagram first, then read off the answers.
- **Nested $\varnothing$ / power sets:** write each layer out explicitly, especially for $\mathcal{P}$ of a set containing $\varnothing$.
- **Counting questions:** use the table in §55, or enumerate the $n = 2$ case to test an option.

Before choosing an answer: check quantifier connective and domain (§5–7), membership versus subset (§14), empty sets in products (§15), composition order (§20), antisymmetry (§21), class versus quotient (§23), and maximal versus largest (§24), setwise preimage versus inverse (§26), co-domain versus range (§28), basis value of an induction (§32), ordered versus unordered and repetition (§42), $P(A \mid B)$ versus $P(B \mid A)$ (§45), trail versus path and Euler versus Hamilton (§47).

54 · Part B: written answers

Computations: give complete set-roster answers; use "None" when appropriate. For a quotient, choose one representative per distinct class. **Proofs:** state the claim, unpack definitions, number and justify each step, then conclude (§9). For "prove or disprove", state the verdict and supply a proof or one explicit counterexample.

55 · Counting facts

Relations on an n-element set	General	$n = 2$	$n = 3$
all relations	2^{n^2}	16	512
reflexive	2^{n^2-n}	4	64
symmetric	$2^{n(n+1)/2}$	8	64
reflexive and symmetric	$2^{n(n-1)/2}$	2	8
antisymmetric	$2^n \cdot 3^{n(n-1)/2}$	12	216
asymmetric	$3^{n(n-1)/2}$	3	27
equivalence relations (Bell)	B_n	2	5
partial orders	—	3	19
total orders	$n!$	2	6

Derivation. Each of the n^2 pairs is either in the relation or not; reflexivity fixes the n diagonal pairs; symmetry decides each unordered pair once; antisymmetry allows 3 of the 4 states of each off-diagonal pair; asymmetry also forbids loops. For $n = 4$ there are 15 equivalence relations, 219 partial orders and 24 total orders. **AY25 Q4:** on a three-element set there are **5** equivalence relations. **AY25 Q12:** on $\{0, 1\}$ there are 2 equivalence relations and 3 partial orders, so $|F| + 1 = |G|$. Answer B.

Other counts	Value
$\lvert A \times B \rvert$	$\lvert A \rvert \cdot \lvert B \rvert$
$\lvert \mathcal{P}(A) \rvert$	$2^{\lvert A \rvert}$
relations from A to B	$2^{\lvert A \rvert \, \lvert B \rvert}$
partitions of an n-set	B_n : $1, 2, 5, 15, 52$

56 · Formula sheet for Lectures 7–13

Counting	Value
integers from m to n	$n-m+1$
ordered, repetition allowed	n^k
ordered, no repetition	$P(n,k)=n!/(n-k)!$
unordered, no repetition	$\binom{n}{k}=n!/(k!(n-k)!)$
unordered, repetition allowed	$\binom{k+n-1}{k}$
orderings with repeated types	$n!/(n_1!\cdots n_k!)$
circular arrangements of n	$(n-1)!$
$x_1+\cdots+x_n=r$, $x_i\geq 0$	$\binom{r+n-1}{r}$
same, $x_i\geq 1$	$\binom{r-1}{n-1}$
functions $A\to B$, $\lvert A\rvert=m$, $\lvert B\rvert=n$	n^m
injections; bijections ($m=n$)	$P(n,m)$; $n!$
subsets of an n-set	2^n
pigeons: some hole has at least	$\lceil n/m\rceil$

Sums	Closed form
$1+2+\cdots+n$	$n(n+1)/2$
$1^2+2^2+\cdots+n^2$	$n(n+1)(2n+1)/6$
$\sum_{i=0}^{n} r^i$ $(r\neq 1)$	$(r^{n+1}-1)/(r-1)$
$\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$	$(a+b)^n$
$\sum_{k=0}^{n}\binom{n}{k}$	2^n

Probability	Formula
equally likely outcomes	$P(E)=\lvert E\rvert\ /\ \lvert S\rvert$
complement	$P(\overline{A})=1-P(A)$
union	$P(A)+P(B)-P(A\cap B)$
conditional	$P(B\mid A)=P(A\cap B)/P(A)$
independent	$P(A\cap B)=P(A)P(B)$
expected value	$\sum a_k p_k$
linearity	$E[X+Y]=E[X]+E[Y]$
binomial	$\binom{n}{k}p^k(1-p)^{n-k}$

Graphs and trees	Fact
total degree	$2\lvert E\rvert$
edges of K_n; of $K_{m,n}$	$n(n-1)/2$; mn
simple graphs on n labelled vertices	$2^{n(n-1)/2}$
Euler circuit	connected, all degrees even
Euler trail v to w	connected, only v, w odd
walks of length n from v_i to v_j	entry ij of A^n
connected planar	$f=e-v+2$
tree on n vertices	$n-1$ edges, at least 2 leaves
full binary tree, k internal	$2k+1$ vertices, $k+1$ leaves
binary tree of height h	at most 2^h leaves
labelled trees on n vertices	n^{n-2}

Function properties under composition (§29). f, g injective $\Rightarrow$ $g\circ f$ injective; f, g surjective $\Rightarrow$ $g\circ f$ surjective; $g\circ f$ injective $\Rightarrow$ f injective; $g\circ f$ surjective $\Rightarrow$ g surjective; $(g\circ f)^{-1}=f^{-1}\circ g^{-1}$. **Finite sets (§35):** injection $A\to B$ gives $\lvert A\rvert\leq\lvert B\rvert$; surjection gives $\lvert A\rvert\geq\lvert B\rvert$; bijection gives $\lvert A\rvert=\lvert B\rvert$. For $\lvert A\rvert=\lvert B\rvert$ finite, injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective.

57 · Question locators

Proof methods: §9–12; set inclusion/equality: §17; relation properties: §21. Each worked example is beside its topic.

Past-paper question	Solved in
AY23 Q2, Q4, Q5, Q16	§6, §4, §7, §7
AY23 Q3, Q6, Q7, Q8, Q17	§2, §18, §16, §14, §18
AY23 Q9, Q10, Q11, Q12, Q13, Q14, Q15, Q18, Q19	§21, §21, §20, §23, §24, §24, §25, §25, §23
AY24 Q2, Q3, Q4, Q5	§5, §2, §2, §4
AY24 Q6, Q7, Q8, Q16	§15, §18, §18, §18
AY24 Q9, Q10, Q11, Q12, Q13–15, Q17	§23, §21, §20, §20, §25, §20–§25
AY25 Q1, Q2, Q4, Q6, Q7, Q8, Q9, Q10, Q12	§18, §6, §55, §8, §18, §18, §19, §19, §55
AY25 Q11, Q13, Q14, Q15, Q16, Q17	§21, §4, §21, §23, §4; Q17(a)–(e): §20, §13, §23, §23, §25

Tutorial question	Topic	Solved in
T1 Q3	One law per line; true/false	§2
T1 Q7, Q8, Q9	Fallacies; critical rows by deduction; knights and knaves	§4
T1 Q11	Critique a biconditional proof; contraposition	§9
T2 Q4	Closure under division; rational addition	§11, §12
T2 Q7	Proof by examples; wrong negation	§9
T2 Q8, Q11	Cases inside a conditional; factor bound	§11
T2 Q9, Q10	Translation and scope; universal transitivity	§5, §8
T3 Q3(c)	Cartesian product versus power set	§15
T3 Q4–6	Double inclusion; element method; set identities	§17
T3 Q7	$A\subseteq B\leftrightarrow A\cup B=B$	§18

T4 Q2	Symmetry and the inverse relation	§21
T4 Q3	Classifying $xy \geq 0$, $xy > 0$, $\lvert x - y \rvert \leq 2$	§21, §25
T4 Q4	Powers of a relation on the digraph	§20
T4 Q6, Q7, Q9	Repeated composition; associativity; parity relation	§20
T4 Q8, Q10	Classes of pairs; equivalence by sign	§23
T5 D1–D4	Empty relation; identity relation; alternative antisymmetry	§21
T5 Q1	Antisymmetry counterexample on strings	§24
T5 Q2, Q4	Linearizations with $a \preccurlyeq^* b$; listing all	§25
T5 Q5, Q8	Componentwise order; comparable versus compatible	§24
T5 Q6	The three conditions defining a closure	§22
T5 Q7	Asymmetric implies antisymmetric	§21

58 · Question locators for Tutorials 6–11 and the final

Tutorial question	Topic	Solved in
T6 D1, D3, D4; Q1, Q2	Is it a function; equality; a wrong proof; images and preimages	§26, §27, §29, §26
T6 Q3, Q6	Inverse of a composition; $g \circ f$ injective	§29
T6 Q4, Q5	Classify four functions; left and right inverses	§28
T6 Q7, Q8; T8 Q10(c)	Order of a bijection and of an input	§29
T6 Q9, Q10; T8 Q10(a), (b)	$f^{-1}(f(X))$; well-defined operations on a quotient	§26, §30
T7 Q1, D2, D3	Mistakes in induction proofs	§32
T7 Q2, Q3, Q4	Sum of squares; Bernoulli; divisibility by 2^{n+2}	§32
T7 Q5, Q6, Q7	\$3 and \$5 notes; sums of powers of 2; three-term recurrence	§33
T7 Q8, Q9; T8 Q9	Fibonacci identities; Tower of Hanoi	§33
T7 Q10, Q11	Membership of recursively defined sets	§34
T8 D1, D2, D3; Q1	Countable or not; bijection $\mathbb{N} \to \mathbb{Z}$	§37, §38
T8 Q2, Q3, Q4, Q5	Unions of countable sets; limits of induction	§37
T8 Q6, Q7, Q8	$X \cup Y \to X$; $\mathcal{P}(A)$ uncountable; $\lvert A \rvert \leq \lvert R \rvert$	§38, §38, §35
T9 D1, D4; Q1, Q2, Q5	Replacement; birthdays; cards; cheque numbers; round tables	§45, §45, §39
T9 Q3, Q4, Q6, Q7	Inclusion/exclusion; consecutive boxes; committees; investments	§40, §42
T9 Q8, Q9; T10 Q10, Q12	Pigeonhole arguments	§41
T10 D1, Q3, Q4; Q5, Q6	Expected value; Bayes; random relations	§44, §45
T10 D2, D3, Q1, Q2	Counting functions; binomial term; choosing block ends	§43, §42
T10 Q7, Q8; T11 D1, D2	Friendship graph; small graphs; edge counts	§47, §46
T11 Q1, Q3, Q4–Q7	Self-complementary graphs; trees; edge bounds; stones	§49, §50
T11 D3, Q8, Q9	Traversals and rebuilding binary trees	§51
T10 Q11; T11 Q2, Q10	Dominating sets; counting spanning trees and MSTs	§46; §52

AY25/26 final question	Solved in
Q9; Q10; Q11, Q12, Q26(b)	§47; §43; §29
Q13, Q17, Q18; Q14–Q16	§38; §34
Q19, Q21, Q22; Q23	§51; §44
Q24(a)–(e), (g); Q24(f); Q24(h)	§42; §41; §45
Q25(a); Q25(b)–(d)	§49; §47
Q20; Q25(e)–(h)	§48; §46

The folder has no official answers for the AY25/26 final; those solutions are worked here.

Earlier final, graphs and trees	Topic	Solved in
AY24/25 Q20; Q21; Q22	Walks of length 4; isomorphism classes; bipartite and planar facts	§48; §49; §47
AY24/25 Q25(a)–(c), (f); (d); (e); (g)	Chromatic number and index, Petersen graph; vertex cut set; number of MSTs; pre-order + post-order	§46; §47; §52; §51

AY23/24 Q18; Q19, Q20	Planar degree bound, disjoint spanning trees; components	§49, §52; §47
AY23/24 Q24(a), (g)(ii), (g)(iii); (b)	MST from a weight matrix, bridges; walks of length 4	§52; §48
AY23/24 Q24(c)(i), (c)(iii), (d); (e), (f), (g)(i)	Euler trail after deleting an edge, edge counts; full binary tree, BSTs	§47; §51
AY22/23 Q18; Q19; Q20	Planarity; forests and bipartite cycles; disconnected graphs on 4 vertices	§49; §47; §49
AY22/23 Q23(a), (b); (c), (d); (e); (f)–(i)	Euler trail; cut-points and bridges; traversals; packing as colouring	§47; §47; §51; §46
AY20/21 Q12; Q20(d); Q21(a)	Walks of length 3; counting digraphs; MST	§48; §49; §52
AY20/21 Q21(b), (c); (d); (e)	Self-complementary graphs; rebuilding a tree; rooted trees on 5 vertices	§49; §51; §50

AY24/25 Q25(a)–(f) follows the answers in the Week 12 recitation; the other solutions are worked here.

59 · Self-test: true or false?

Decide each statement before reading the reason.

Statement		Reason
$\sim(p \leftrightarrow q) \equiv \sim p \leftrightarrow q$	**T**	both are true exactly when p, q differ
$(p \vee q) \to r \equiv (p \to r) \vee (q \to r)$	**F**	it is $\wedge$; $p = T, q = F, r = F$ separates them
$\exists y \in \mathbb{Z}\, \forall x \in \mathbb{Z}\, (x < y)$	**F**	take $x = y$
$\exists x \in \varnothing\, P(x)$	**F**	no witness exists; $\forall x \in \varnothing\, P(x)$ is **T**
$\varnothing \in \{\varnothing\}$ and $\varnothing \subseteq \{\varnothing\}$	**T**	listed element; Theorem 6.2.4
$\{\varnothing\} \subseteq \varnothing$	**F**	$\varnothing \in \{\varnothing\}$ but $\varnothing \notin \varnothing$
$\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$	**T**	$X \subseteq A \cap B \leftrightarrow X \subseteq A \wedge X \subseteq B$
$A \times B = B \times A \Rightarrow A = B$	**F**	$A = \varnothing$, $B = \{1\}$: both products are $\varnothing$
a partition of a finite A has at most $\lvert A\rvert$ components	**T**	components are non-empty and disjoint
symmetric and transitive $\Rightarrow$ reflexive	**F**	the empty relation on $\{a\}$
$R \circ R^{-1}$ is always symmetric	**T**	$(R \circ R^{-1})^{-1} = (R^{-1})^{-1} \circ R^{-1}$
the inverse of a partial order is a partial order	**T**	each property survives reversing pairs
the union of two equivalence relations is one	**F**	$1 \sim 2$ and $2 \sim 3$ from different relations
the intersection of two equivalence relations is one	**T**	each property is preserved by $\cap$
a unique minimal element is smallest, in any poset	**F**	true only for finite posets (§24)
$(\mathbb{Z}^+, \mid)$ has a smallest element	**T**	$1 \mid n$ for all n
there are 4 relations on a one-element set	**F**	$2^{1^2} = 2$: $\varnothing$ and $\{(a, a)\}$
$a \mid b \wedge a \mid c \Rightarrow a \mid (b - c)$	**T**	$b - c = a(r - s)$

60 · Self-test for Lectures 7–13

Decide each statement before reading the reason.

Statement		Reason
$f^{-1}(B)$ is defined only when f is a bijection	**F**	the setwise preimage exists for every function (§26)
$f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 3x + 1$ is a bijection	**F**	injective, but 0 has no preimage
$g \circ f$ injective $\Rightarrow$ g injective	**F**	only f must be injective (§29)
the composition of two bijections is a bijection	**T**	Theorems 7.3.3 and 7.3.4
$f : \mathbb{Z} \to \mathbb{Z}$ and $g : \mathbb{Z} \to \mathbb{Q}$ with the same formula are equal	**F**	different co-domains (Theorem 7.1.1)
$X = f^{-1}(f(X))$ for every $X \subseteq A$	**F**	only $\subseteq$; equality needs injectivity
$[x]_3 \mapsto [x]_2$ is a function $\mathbb{Z}_3 \to \mathbb{Z}_2$	**F**	$[0]_3 = [3]_3$ but $[0]_2 \neq [3]_2$
$2n + 1 < 2^n$ for all $n \geq 1$	**F**	fails at $n = 1, 2$; true from $n = 3$

strong induction proves more statements than weak induction	**F**	1PI, 2PI and WOP are equivalent
every non-empty subset of $\mathbb{Z}$ has a least element	**F**	WOP is about $\mathbb{Z}_{\geq 0}$; take $\mathbb{Z}^-$
an empty sum is 0 and an empty product is 1	**T**	conventions (§31)
$\lvert\mathbb{Z}\rvert = \lvert 2\mathbb{Z}\rvert$	**T**	$n \mapsto 2n$ is a bijection
every infinite subset of $\mathbb{R}$ is uncountable	**F**	$\mathbb{Z}$, $\mathbb{Q}$
every subset of $\mathbb{Q}$ is countable	**T**	Theorem 7.4.3
a countable union of finite sets can be infinite	**T**	$\bigcup\{i\} = \mathbb{Z}^+$; still countable
$\mathcal{P}(\mathbb{N})$ is countable	**F**	diagonal argument (§38)
$P(n,r) = \binom{n}{r} \cdot r!$	**T**	choose a subset, then order it
the number of multisets of size 3 from 4 elements is $\binom{4}{3}$	**F**	$\binom{6}{3} = 20$
$\binom{n}{r} = \binom{n}{n-r}$	**T**	choose which to leave out
disjoint events are independent	**F**	if both have positive probability they are dependent
$E[X+Y] = E[X] + E[Y]$ needs independence	**F**	linearity always holds
$P(A \mid B) = P(B \mid A)$	**F**	equal only if $P(A) = P(B)$ or $P(A \cap B) = 0$
a graph can have exactly three vertices of odd degree	**F**	Proposition 10.1.3
all degrees even $\Rightarrow$ the graph has an Euler circuit	**F**	it must also be connected
every Eulerian graph is Hamiltonian	**F**	friendship graph F_2
a graph with n vertices and $n-1$ edges is a tree	**F**	only if connected (Theorem 10.5.4)
$K_{3,3}$ is planar	**F**	Kuratowski; K_5 is not planar either
equal degree lists $\Rightarrow$ isomorphic	**F**	C_6 and two disjoint triangles
pre-order and post-order determine a binary tree	**F**	a one-child vertex is ambiguous (§51)
a full binary tree has an odd number of vertices	**T**	$2k+1$
all spanning trees of a graph have the same number of edges	**T**	Proposition 10.7.1

61 · Symbol table

Symbol	Read as	Symbol	Read as	Symbol	Read as
$\sim p$	not p	$p \wedge q$	p and q	$p \vee q$	p or q
$p \to q$	if p then q	$p \leftrightarrow q$	p iff q	$p \oplus q$	exclusive or
$P \equiv Q$	logically equivalent	$\therefore$	therefore	**true**, **false**	tautology, contradiction (Epp: $\boldsymbol{t}$, $\boldsymbol{c}$)
$\forall$	for all	$\exists$	there exists	$\exists!$	there exists exactly one
$x \in A$	x is an element of A	$A \subseteq B$	A is a subset of B	$A \subsetneq B$	proper subset
$\varnothing$	empty set	$\lvert A\rvert$	cardinality	$\mathcal{P}(A)$	power set
$A \times B$	Cartesian product	$A \setminus B$	set difference	$\overline{A}$	complement
$A \oplus B$	symmetric difference	(a,b)	ordered pair	$\{x \in U : P(x)\}$	set-builder
$d \mid n$	d divides n	$d \nmid n$	d does not divide n	$a \equiv b \pmod n$	congruent mod n
xRy	$(x,y) \in R$	R^{-1}	inverse relation	$S \circ R$	R then S
R^r, R^s, R^t	closures	$[x]_\sim$	class of x	$A/\sim$	quotient: set of classes
$x \preccurlyeq y$	x precedes-or-equals y	$x \preccurlyeq^* y$	a linearization	$(A, \preccurlyeq)$	poset
$\mathbb{Z}^+, \mathbb{Z}_{\geq 0}$	positive / non-negative integers	$\mathbb{N}$	$\{0,1,2,\ldots\}$	$n \operatorname{div} d, n \bmod d$	quotient, remainder

62 · Symbols for Lectures 7–13

Symbol	Read as	Symbol	Read as	Symbol	Read as

$f : X \to Y$	function from X to Y	$x \mapsto y$	x maps to y	$f(A)$	setwise image
$f^{-1}(B)$	setwise preimage	f^{-1}	inverse function	$g \circ f$	f then g
id_X	identity on X	A^*	strings over A	ε	empty string
$\mathbb{Z}_n$	$\mathbb{Z}$ mod n (L7); $\{1, ..., n\}$ (L9)	$\lfloor x \rfloor, \lceil x \rceil$	floor, ceiling	$n!$	n factorial
$\sum_{k=m}^{n} a_k$	sum of $a_m, ..., a_n$	$\prod_{k=m}^{n} a_k$	product	F_n	Fibonacci number
$\aleph_0$	$\lvert\mathbb{Z}^+\rvert$, aleph-zero	$\lvert A\rvert = \lvert B\rvert$	same cardinality	$P(n, r)$	r-permutations
$\binom{n}{r}$	n choose r	$P(E)$	probability of E	$P(B \mid A)$	B given A
$E[X]$	expected value	$G = (V, E)$	graph	$\deg(v)$	degree
$\deg^-(v), \deg^+(v)$	indegree, outdegree	K_n	complete graph	$K_{m,n}$	complete bipartite graph
$\overline{G}$	complement graph	$G \cong G'$	isomorphic	A^n	n-th matrix power
$w(e), w(G)$	edge weight, total weight	C_n	cycle graph	I_n	identity matrix

63 · Topic locator

§	Definitions / methods	§	Definitions / methods
1–3	Statements, connectives, equivalence, conditional variants	13–14	Sets, notation, intervals, subset, equality, empty set
4	Arguments, validity, soundness, inference rules	15–16	Pairs, products, set operations, disjointness
5–7	Predicates, truth sets, quantifiers, negation, vacuity	17–18	Set proofs, partitions, power sets
8	Quantified inference rules and worked negations	19–20	Relations, domain/range, inverse, composition
9	Proof vocabulary, format, assumptions, biconditionals	21–22	Relation properties and closures
10	Even/odd, prime, rational, divisibility, absolute value	23	Equivalence, classes, quotient, congruence, representatives
11–12	Proof methods and number-theory examples	24–25	Posets, extrema, total/well orders, Hasse, Kahn
26–27	Functions, images, preimages, sequences, strings, equality	28–30	Injective, surjective, inverse, composition, $\mathbb{Z}_n$
31–32	Sums, products, mathematical induction	33–34	Strong induction, well-ordering, recursion, structural induction
35–36	Pigeonhole, finite sets, same cardinality	37–38	Countable and uncountable sets, diagonalization
39–41	Multiplication, addition, inclusion/exclusion, pigeonhole	42–43	Combinations, multisets, Pascal, binomial theorem
44–45	Probability axioms, expectation, Bayes, independence	46–49	Graphs, degree, Euler, Hamilton, matrices, isomorphism, planarity
50–52	Trees, binary trees, traversals, spanning trees, MST	53–58	Exam methods, counting, formula sheet, question locators
59–62	Self-tests and symbols	64–65	Appendix A; slide-numbered theorem locator

Part XII · Appendix A and Theorem Locator

64 · Appendix A · Properties of the Real Numbers

Source: Appendix A, cited in Lecture 1, slide 26; Lecture 3, slide 81; and Lecture 4, slide 4. The complete appendix is reproduced here as a statement reference, including its final least upper bound axiom. The symbols a, b, c, d below denote arbitrary **real numbers**, subject to each stated condition.

Appendix A · Field axioms F1–F6
Addition and multiplication are binary operations on $\mathbb{R}$: if $a, b \in \mathbb{R}$, then $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.

Appendix A label	Statement
Axiom F1 · Commutative laws	$a+b=b+a$; $ab=ba$.
Axiom F2 · Associative laws	$(a+b)+c=a+(b+c)$; $(ab)c=a(bc)$.
Axiom F3 · Distributive laws	$a(b+c)=ab+ac$; $(b+c)a=ba+ca$.
Axiom F4 · Identity elements	There are distinct $0,1\in\mathbb{R}$ such that $0+a=a+0=a$ and $1\cdot a=a\cdot 1=a$.
Axiom F5 · Additive inverses	For each a, there is $-a\in\mathbb{R}$ with $a+(-a)=(-a)+a=0$.
Axiom F6 · Reciprocals	For each $a\neq 0$, there is $a^{-1}=1/a\in\mathbb{R}$ with $a\cdot a^{-1}=a^{-1}\cdot a=1$.

Appendix A · Theorems T1–T16 (algebra)

Appendix A label	Statement
Theorem T1 · Addition cancellation	If $a+b=a+c$, then $b=c$. The additive identity 0 is unique.
Theorem T2 · Subtraction	For each a,b, exactly one x satisfies $a+x=b$. This x is denoted $b-a$; $0-a=-a$.
Theorem T3	$b-a=b+(-a)$.
Theorem T4	$-(-a)=a$.
Theorem T5	$a(b-c)=ab-ac$.
Theorem T6	$0\cdot a=a\cdot 0=0$.
Theorem T7 · Multiplication cancellation	If $ab=ac$ and $a\neq 0$, then $b=c$. The multiplicative identity 1 is unique.
Theorem T8 · Division	For $a\neq 0$, exactly one x satisfies $ax=b$. This x is denoted b/a; $1/a$ is the reciprocal of a.
Theorem T9	If $a\neq 0$, then $b/a=b\cdot a^{-1}$.
Theorem T10	If $a\neq 0$, then $\left(a^{-1}\right)^{-1}=a$.
Theorem T11 · Zero product	If $ab=0$, then $a=0$ or $b=0$.
Theorem T12 · Negative signs	$(-a)b=a(-b)=-(ab)$; $(-a)(-b)=ab$. If $b\neq 0$, $-(a/b)=(-a)/b=a/(-b)$.
Theorem T13 · Equivalent fractions	If $b\neq 0$ and $c\neq 0$, then $a/b=(ac)/(bc)$.
Theorem T14 · Addition of fractions	If $b\neq 0$ and $d\neq 0$, then $a/b+c/d=(ad+bc)/(bd)$.
Theorem T15 · Multiplication of fractions	If $b\neq 0$ and $d\neq 0$, then $(a/b)(c/d)=(ac)/(bd)$.
Theorem T16 · Division of fractions	If $b\neq 0$, $c\neq 0$ and $d\neq 0$, then $(a/b)/(c/d)=(ad)/(bc)$.

Appendix A · Order axioms Ord1–Ord3

Appendix A label	Statement
Axiom Ord1	If a and b are positive, then $a+b$ and ab are positive.
Axiom Ord2	If $a\neq 0$, exactly one of a and $-a$ is positive.
Axiom Ord3	0 is not positive.

Appendix A · Order definitions: $a<b$ means $b+(-a)$ is positive; $b>a$ means $a<b$. Also, $a\leq b$ means $a<b$ or $a=b$, and $b\geq a$ means $a\leq b$. A number a is **negative** if $a<0$, and **nonnegative** if $a\geq 0$.

Appendix A · Theorems T17–T27 (order)

Appendix A label	Statement
Theorem T17 · Trichotomy	Exactly one holds: $a<b$, $b<a$, or $a=b$.
Theorem T18 · Transitivity	If $a<b$ and $b<c$, then $a<c$.
Theorem T19	If $a<b$, then $a+c<b+c$.
Theorem T20	If $a<b$ and $c>0$, then $ac<bc$.

Theorem T21	If $a \neq 0$, then $a^2 > 0$.
Theorem T22	$1 > 0$.
Theorem T23	If $a < b$ and $c < 0$, then $ac > bc$.
Theorem T24	If $a < b$, then $-a > -b$. In particular, if $a < 0$, then $-a > 0$.
Theorem T25	If $ab > 0$, then a and b are both positive or both negative.
Theorem T26	If $a < c$ and $b < d$, then $a + b < c + d$.
Theorem T27	If $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.

AXIOM **Appendix A · Axiom LUB · Least upper bound**
Every nonempty subset S of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$. Explicitly, let $B = \{u \in \mathbb{R} : \forall s \in S, s \leq u\}$.
If $S \neq \varnothing$ and $B \neq \varnothing$, then there is $\ell \in B$ such that $\forall u \in B, \ell \leq u$. This ℓ is the **least upper bound** (supremum) of S. This axiom holds for $\mathbb{R}$; the analogous statement for subsets and bounds in $\mathbb{Q}$ fails.

Lecture 4, slide 4 · Additional assumptions: equality is reflexive, symmetric and transitive for all objects; no integer lies strictly between 0 and 1; and $\mathbb{Z}$ is closed under addition, subtraction and multiplication. Lecture 1 additionally permits the even/odd dichotomy (Assumption 1, slide 27) and reduction of every rational to lowest terms (Assumption 2, slide 37).

65 · Theorem locator · Slide numbering

Theorems are stated with their proofs or definitions in the listed sections. Labels match the slides. **L** = lecture; **s** = slide.

Slide label	Topic	Source	Section
Theorem 2.1.1	Logical equivalences	L2 s25–26	§2
Theorem 3.2.1; Theorem 3.2.2	Negating ∀; negating ∃	L3 s29–30	§6
Theorem 4.2.1	Integers are rational	L4 s19	§12
Theorem 4.2.2; Corollary 4.2.3	Sum; double of a rational	L4 s20–21	§12
Theorem 4.3.1	Positive divisor bound	L4 s24	§12
Theorem 4.3.2	Divisors of 1	L4 s25	§12
Theorem 4.3.3	Transitivity of divisibility	L4 s26	§12
Theorem 4.4.1	Quotient–remainder	L5 s32	§10
Lemma 4.4.4; Theorem 4.4.6	Absolute value bounds; triangle inequality	L1 s24	§12
Theorem 4.6.1	No greatest integer	L4 s29	§12
Proposition 4.6.4	Even square implies even integer	L1 s38; L4 s31–32	§9
Theorem 4.7.1	Irrationality of $\sqrt{2}$	L1 s36, 39; L4 s28	§12
Theorem 6.2.1	Subset relations	L5 s40	§16
Theorem 6.2.2	Set identities	L5 s42–43	§2
Theorem 6.2.4	Empty set	L5 s15	§14
Theorem 6.3.1	Power-set cardinality	L5 s34–36	§18
Theorem 8.3.1	Relation induced by a partition	L6 s39	§23
Lemma Rel.1	Equal or disjoint classes	L6 s47–49	§23
Theorem 8.3.4; Theorem Rel.2	Classes partition the underlying set	L6 s50, 57–59	§23
Propositions (unnumbered)	Composition: associativity; inverse	L6 s18	§20
Proposition (unnumbered)	Congruence is an equivalence relation	L6 s54	§23
Proposition (unnumbered)	Smallest implies minimal	L6 s83	§24
Theorem 7.1.1	Function equality	L7 s22	§27
Proposition (unnumbered); Theorem 7.2.3	Unique inverse; bijective iff invertible	L7 s34–36	§28
Theorems 7.3.1, 7.3.2	Composition with identity; with the inverse	L7 s44, 46	§29
Theorem (unnumbered); Theorems 7.3.3, 7.3.4	Associativity; composing injections, surjections	L7 s47, 50, 52	§29
Propositions (unnumbered)	$+$ and $\cdot$ on $\mathbb{Z}_n$ are well defined	L7 s57–58	§30
Theorem 5.1.1	Properties of sums and products	L8 s14	§31
Theorem 5.2.2; Theorem 5.2.3	Sum of first n integers; geometric sum	L8 s26, 28	§32

Proposition 5.3.1; Proposition 5.3.2	$3 \mid (2^{2n}-1)$; $2n+1 < 2^n$	L8 s29–30	§32
Well-Ordering Principle	Non-empty subsets of $\mathbb{Z}_{\geq 0}$ have a least element	L8 s43–44	§33
Pigeonhole; Dual Pigeonhole	Injections and surjections between finite sets	L9 s6–8	§35
Theorem Cardinality.1	Subsets of finite sets are finite	L9 s11	§35
Theorem 7.4.1	Same cardinality is an equivalence relation	L9 s14–16	§36
Proposition 9.1; Lemma 9.2	Countability via sequences	L9 s33–34	§37
Theorem 7.4.2	$(0,1)$ is uncountable	L9 s38–40	§38
Theorem 7.4.3; Corollary 7.4.4	Subsets of countable sets; uncountable subsets	L9 s41–43	§37
Proposition 9.3; Lemma 9.4	Countably infinite subset; union of two	L9 s44–45	§37
Theorem 9.1.1; Theorem 9.2.1	Elements of a list; multiplication rule	L10 s12, 21	§39
Theorem 9.2.2; Theorem 9.2.3	Permutations; r-permutations	L10 s34, 37	§39
Theorems 9.3.1, 9.3.2, 9.3.3	Addition, difference, inclusion/exclusion	L10 s40, 43, 50	§40
Generalized Pigeonhole Principle	Some value has at least $k+1$ preimages	L10 s62, 65	§41
Theorem 9.5.1; Theorem 9.5.2	Formula for $\binom{n}{r}$; repeated objects	L11 s10, 15	§42
Theorem 9.6.1	Multisets	L11 s21	§42
Theorem 9.7.1; Theorem 9.7.2	Pascal's formula; binomial theorem	L11 s26, 34	§43
Theorem 9.9.1	Bayes' theorem	L11 s66	§45
Theorem 10.1.1; Corollary 10.1.2; Proposition 10.1.3	Handshake theorem and consequences	L12 s25	§46
Lemma 10.2.1	Paths, circuits and connectedness	L12 s37	§47
Theorems 10.2.2–10.2.4; Corollary 10.2.5	Euler circuits and trails	L12 s41–43	§47
Proposition 10.2.6	Necessary conditions for a Hamiltonian circuit	L12 s52	§47
Theorem 10.3.2	Powers of the adjacency matrix count walks	L12 s74	§48
Theorem 10.4.1; Kuratowski; Euler's formula	Isomorphism; planarity	L12 s81, 83, 86	§49
Lemma 10.5.1; Theorem 10.5.2	Leaves; a tree has $n-1$ edges	L13 s7, 10	§50
Lemma 10.5.3; Theorem 10.5.4; Lemma 10.5.5	Removing a circuit edge; converse; two paths	L13 s14–15; T11	§50
Theorem 10.6.1; Theorem 10.6.2	Full binary trees; leaves and height	L13 s27, 29	§51
Proposition 10.7.1; Algorithms 10.7.1, 10.7.2	Spanning trees; Kruskal; Prim	L13 s45, 50, 55	§52